

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

**ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED
RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED
EVALUATION FOR THE COLLEGE OF ENGINEERING,
COMPUTER STUDIES, AND ARCHITECTURE AT
LYCEUM OF THE PHILIPPINES
UNIVERSITY - CAVITE**

An Undergraduate Thesis
Submitted to the Faculty of the
College of Engineering, Computer Studies and Architecture
Lyceum of the Philippines University-Cavite

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Computer Science
with specialization in Software Engineering

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July 2025

COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

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TITLE OF WORK: ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI - ASSISTED EVALUATION FOR THE COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

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**Bachelor of Science in Computer Science
with Specialization in Software Engineering**

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COMPUTER STUDIES, AND ARCHITECTURE
AT LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE**

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This thesis entitled **ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI - ASSISTED EVALUATION FOR THE COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES UNIVERSITY-CAVITE** prepared and submitted by **WINSTON L. AGUSTIN, NEIL S. VICEDO, JERALD RYAN D. GERONA, and IVAN KERWIN R. ILANO**, in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH SPECIALIZATION IN SOFTWARE ENGINEERING** is hereby accepted.

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Title of work: ATLAS: THESIS MANAGEMENT SYSTEM WITH
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We hereby declare that this thesis is our own work and that, to the best of our knowledge and belief, contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgment is made in the text.

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LIST OF ABBREVIATIONS	
LPU-C	Lyceum of the Philippines University Cavite
COECSA	College of Engineering, Computer Studies, and Architecture
LLM	Large Language Model
SQL	Structured Query Language
CMS	Content Management System
GA	Genetic Algorithm
ASS	Automated Scheduling System
DSS	Decision Support System
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
PHP	Hypertext Preprocessor
NLP	Natural Language Processing

LPU-C

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NLP

Natural Language Processing

ABSTRACT

This research presents the development and evaluation of ATLAS: Thesis Management System with Automated Research Defense Scheduling and AI-Assisted Evaluation, a web-based system tailored for the College of Engineering, Computer Studies, and Architecture at the Lyceum of Philippines University – Cavite. The System addresses common issues in thesis management, such as inefficient manual coordination, scheduling conflicts, and the lack of systematic evaluation process. ATLAS integrates modules for managing group proposals, real-time scheduling of defenses, and streamlined research evaluation with AI-generated feedback. It allows different functions to perform specific functions with role-based access control. The system was developed using PHP, MySQL, and other modern web technologies. System Testing was conducted with test cases for the system’s roles. The system’s performance was evaluated using modified ISO/IEC 25010 quality standards. The outcome of the evaluation suggests that ATLAS is deemed “Highly Acceptable”, this shows a high level of satisfaction, confirming the system’s effectiveness in enhancing thesis management processes. The research concludes that ATLAS is a viable solution for academic institutions seeking a centralized, automated, and intelligent platform for managing undergraduate thesis workflows. Future improvements may include enhanced AI features, better mobile compatibility, better layout, and broader institutional deployment.

Keywords: thesis defense management, web-based system

CHAPTER I

INTRODUCTION

Background and Rationale of the Study

Thesis management systems play a crucial role in higher education by enhancing academic performance and supporting the timely completion of research projects. Digitalization is an effective method for enhancing processes and improving work efficiency, and this principle extends to the education sector for educators as well (Kauppinen, 2023). These systems are pivotal in integrating technology into academic workflows, improving communication, and ensuring high-quality output. Recent studies further emphasize the importance of effective research management in academic institutions. For instance, (Bouasangthong, Thalangsy, and Leusinxay, 2025) explored the current state of research management at the National University of Laos. Their study revealed that while the system was generally perceived positively, several challenges remained. Key issues such as limited budget allocation, the absence of comprehensive research manuals, and the underutilization of research outputs were identified as areas needing improvement. Moreover, while research processes were frequently practiced, the feedback mechanisms lacked the necessary depth for driving meaningful enhancements. The study highlighted the critical need for university managers to secure better financial support, provide thorough research guidelines, and

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	maximize the application of research findings to foster a more robust academic environment. These insights underline the growing necessity for institutions to not only implement digital research management systems but also to continuously refine them to meet evolving educational demands. The integration of AI-driven tools into academic processes has become a critical solution for addressing the challenges faced by educational institutions, particularly in managing thesis-related tasks. Recent studies have highlighted key aspects of thesis management systems, such as scalability and maintainability, which enable institutions to handle growing student populations more efficiently (Khozaimi, 2023). Generative AI tools, such as Gemini, have been shown to streamline the drafting and evaluation of final-year projects, improving both submissions and processes and faculty assessments (Pereira, 2023). Additionally, academic writing tools that provide real-time feedback have proven effective in enhancing students' writing skills and increasing their chances of success (Storey, 2023). Furthermore, AI's capability to analyze student performance data and offer personalized insights for evaluators enrich the thesis evaluation process, fostering tailored learning experiences and better academic outcomes (Abbas et al., 2023). These advancements highlight the transformative potential of AI in improving the efficiency and quality of thesis management systems. Moreover, studies reveal the ethical considerations of AI in academia. Tindle addresses concerns about academic integrity, urging institutions to establish	
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	guidelines for ethical AI use in assessments and writing (Tindle, 2023). Widodo examines AI's application in lecturer performance evaluations, illustrating its capacity to process large datasets on teaching effectiveness, research, and feedback for informed decision-making Widodo (2023). Together, these studies underscore AI's transformative role in higher education, from improving student outcomes and academic processes to enhancing institutional decision-making, while emphasizing the need for ongoing research and ethical frameworks to ensure AI's effective implementation. The College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University-Cavite (LPU-C) is committed to cultivating students' practical expertise, promoting lifelong learning, and developing globally competent leaders. COECSA holds distinguished accreditations, such as Level 3 PACUCOA accreditation for the BS Information Technology program and Level 2 for its engineering programs. Furthermore, the college ensures industry readiness by offering professional certifications like MTA and CCNA, while producing graduates who exemplify the core values of truth, fortitude, and service (LPU, n.d.). Despite its academic excellence, LPU-C faces challenges in managing research presentations due to its growing student population and reliance on manual system. Scheduling defense, coordinating the availability of students, advisers, and panelists, and organizing documents are time-consuming tasks that often lead to conflicts, inefficiencies, and delays. Manual processes further	
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complicate thesis management, as they are prone to errors in grading this hinders the tracking of schedules and submissions, and create administrative burdens, such as duplicate titles and unsecured data storage (Cagana & Istiono, 2024; Nugraha, 2022). These inefficiencies not only slow down communication and thesis progress but also highlight the need for a more streamlined and automated approach. As Kauppinen et al. (2023) emphasizes, digitalization in education can address these challenges by providing tools that enhance efficiency, improve organization, and generate valuable data for decision-making, ensuring a smoother thesis management process. The study proposes the development of ATLAS: Thesis Management System with Automated Research Defense Scheduling and AI-Assisted Evaluation, a comprehensive solution tailored for LPU-C. ATLAS integrates a suite of AI-driven tools to address key challenges in thesis management. The Thesis Topic Decision Tool helps students select impactful research topics by analyzing current trends.

The Research Title Acceptance Tool, powered by Gemini Pro LLM for its advanced capabilities in multi-turn dialogues, context-aware interactions, and robust performance, making it an ideal choice for developing reliable, high-quality applications that integrate seamlessly (MindsDB, 2024). ensures originality by assessing the uniqueness of proposed titles. The Scheduling System, utilizing a Genetic Algorithm, automates defense coordination, optimizing availability and reducing conflicts. Additionally, the Content Management System (CMS) centralizes academic resources, streamlining administrative tasks. Moreover,

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ATLAS is a flexible system that can be easily adapted by other colleges through CMS. The Requirement Checker Tool tracks document submissions and sends reminders, while the Pass Recommendation System employs AI-driven sentiment analysis to provide consistent feedback during evaluations. Together, these tools aim to improve efficiency, accuracy, and transparency in thesis management, ultimately enhancing the academic experience for students and faculty alike.

Objectives of the Study

The general objective of this study is to develop ATLAS for Lyceum of the Philippines University – Cavite.

Specifically, the project will:

1. Develop ATLAS for LPU-Cavite with the use of the following features.
 - a) Thesis Topic Decision Tool
 - b) Research Title Acceptance Tool
 - c) Scheduling System
 - d) Content Management System
 - e) Requirement Checker Tool
 - f) Pass Recommendation System
2. Design a user-friendly, interactive, and scalable system with an intuitive interface for seamless functionality and improved user experience.
3. Evaluate the system using ISO/IEC 25010 standards.

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	For Students, ATLAS helps students manage their time better, track progress clearly, and collaborate more effectively. It also encourages accountability by automating reminders and providing feedback based on their research work. Scope and Limitation This study is centered on the development and implementation of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation, specifically customized for the Lyceum of the Philippines University-Cavite (LPU-C). The system includes the following key features such as Scheduling System which automates the scheduling of thesis defenses to minimize conflicts and improve efficiency, Pass Recommendation Tool that utilizes sentiment analysis via Gemini, to analyze feedback and suggest pass/fail outcomes, Research Title Acceptance that detects similarities between proposed and existing research titles to prevent plagiarism and ensure originality, Requirement Checker Tool that tracks thesis chapter submissions and ensures compliance with deadlines and academic requirements, and lastly Thesis Topic Decision Tool that provides students with topic suggestions based on trends in computing and AI. The research will take place at LPU-C and will concentrate on overcoming the academic challenges faced by LPU-C, with data collection being carried out over one semester. The system will be applicable only within LPU-C, with its features specifically designed to meet the needs of the department. These features	
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	may require modifications for use in other departments. Additionally, the system offers a CMS feature, other colleges can also benefit from it. The study will include key stakeholders such as students, research advisers, and panelists. It will not address broader administrative activities, plagiarism checks beyond title comparisons, or other academic programs at the university. Although AI-driven tools like the Gemini Pro LLM will aid in the decision-making process, human input is important for final grading and evaluation to maintain fairness and accuracy. The system necessitates an internet connection and compatible devices, and it is built for web-based access. ATLAS requires a reliable internet connection, and it requires a Windows 10 operating system version 10.0.19045 Build 19045. For CPU a minimum of Intel i3 3rd Gen or Ryzen 3 3000 series or 4000 series APU and a minimum of 4GB RAM. As for the resolution, a minimum of resolution of 720x565 (also with a 16:9 aspect ratio) and 720 x480 (for 4:3). For browsers, Google Chrome version 131 or higher, Opera version LVL6 with core version 114 or higher, Microsoft Edge version 131 or higher, and Firefox version 131 or higher. For mobile users, the minimum Android Version is Android 10, for iOS is iOS 15. As for the resolution for mobile devices, the system requires a minimum resolution of 565x720 (with a 16:9 aspect ratio) and 720x480 (for 4:3) to ensure optimal functionality and a fully responsive design. These specifications allow users to effectively use all features, with a layout designed for both mobile and desktop environments.	
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CHAPTER II

REVIEW OF RELATED LITERATURE

The proponents of this project collected relevant literature and studies to facilitate the research. This collection aims to assist readers in understanding the study's content. The literature review clarifies terms closely linked to the research, while the review of related studies delves into findings and recommendations important to this project. Also, a synthesis is provided at the end of each topic to show the connection between literature and the researcher's work.

Genetic Algorithm

Genetic Algorithms are powerful optimization techniques inspired by the principles of natural selection and genetics. Since their inception by Holland in the 1970s, GAs have evolved into versatile tools widely applied across various domains, including engineering, artificial intelligence, and computational biology. The fundamental mechanism of Genetic Algorithm involves iteratively searching for optimal solutions through processes analogous to biological evolution, such as selection, crossover, and mutation (Chen and Okle, 2024). Genetic Algorithms have garnered significant attention in recent years due to their versatility and effectiveness in solving complex optimization problems across various fields. Their

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	application ranges from machine learning to biomedical research, showcasing their adaptability and robustness in diverse scenarios. One prominent application of Genetic Algorithm is in the optimization of machine learning models. For instance, a study developed an automatic grading system for English compositions that integrates a Genetic Algorithm with a Convolutional Neural Network (CNN) . This approach enhances grading accuracy and optimizes the evaluation process by continuously refining the model through genetic operations such as selection, crossover, and mutation (Wang ,2023). Scheduling problems are often challenging and difficult to tackle, it is often solved using optimization algorithms. Genetic Algorithm stands out stands out due to its ability to provide better solutions over generations. The steps and process are simple where it does not require finding derivatives making it effective for reaching global optima (Gurusinga, et al. 2025). Genetics Algorithm are used within ATLAS for optimization tasks, particularly in enhancing scheduling and resource management. Inspired by natural selection, genetic algorithms iteratively search for the best solutions to complex problems. In ATLAS, they can be applied to optimize schedules, ensuring that defense presentations are arranged with minimal conflicts by efficiently allocating time slots based on the availability of students, advisers, and panelists. This not only reduces scheduling conflicts but also improves the overall efficiency of the system, making it more adaptable to the growing number of students and faculty involved in research.	
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Large Language Model (LLM)

Large Language Models (LLMs) have emerged as transformative tools in various fields, particularly in Natural Language Processing (NLP). Their architecture, characterized by a vast number of parameters, allows them to capture the complexities of human language, enabling applications ranging from text generation to complex problem-solving in scientific domains. For instance, recent studies have highlighted the application of LLMs in gene-related tasks, for instance, recent studies have highlighted the application of large language models in gene-related tasks, where these models are used to enhance gene-focused analysis through retrieval-augmented generation techniques, showcasing their versatility beyond traditional text processing (Lin, 2024).

The computational demands of LLMs, however, pose significant challenges. As noted in recent research, while these models achieve unprecedented performance levels, they require substantial computational resources for both training and inference, which can limit their accessibility and practical applications (Huangpu, 2024). This has led to ongoing efforts in model compression and knowledge distillation, aimed at maintaining performance while reducing the computational burden (Huangpu, 2024). Moreover, the scalability of large language models, as demonstrated by the increasing size of models like PaLM, highlights the need to optimize their architectures to improve their practical effectiveness across various applications (Cai, 2023).

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	The study highlights the ethical and academic challenges of students using Large Language Models (LLMs) to write theses, including concerns over academic integrity, lack of authentic learning, and the risk of passing off AI-generated work as their own. While guidelines were proposed to address these issues, the project improves on this by integrating AI responsibly. Through tools like the Thesis Topic Decision Tool and Research Title Acceptance Tool, the system supports students in selecting relevant and unique research topics while ensuring originality. Additionally, the Pass Recommendation System ensures that AI aids decision-making without replacing critical human evaluation, safeguarding both academic integrity and learning outcomes. (Leary ,2023) LLM such as the Gemini Pro LLM used in ATLAS, adds a layer of AI-driven intelligence for thesis evaluations and feedback. LLMs can analyze text data, such as research titles or student-submitted documents, to provide relevant feedback, detect potential issues, and suggest improvements. For instance, the LLM helps in assessing the uniqueness of thesis titles by comparing them with existing works and generating recommendations to avoid redundancy or plagiarism. The use of LLM in ATLAS ensures that evaluations are more consistent and precise, as it supports the decision-making process with AI-driven insights, while human reviewers still maintain control over the final evaluations.	
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	Automated Scheduling Systems is their ability to optimize class schedules based on various constraints such as classroom availability, instructor schedules, and student preferences. For instance, the development of an automated scheduling system using the 1C:Enterprise platform has demonstrated significant improvements in managing educational processes within higher education institutions. This system not only reduces the time required for scheduling but also facilitates easier modifications to the timetable, ensuring that the educational process runs smoothly (Rymar & Voytko, 2021). Similarly, research has shown that employing genetic algorithms can effectively address complex scheduling problems in universities, allowing for the generation of optimal course schedules that meet diverse constraints (Lei et al, 2024). In addressing the challenges of scheduling in universities, (Chen et al., 2021) focused on optimizing class schedules through automated conflict resolution and efficient resource management. Their work highlights the importance of minimizing scheduling conflicts to enhance the educational experience for both students and faculty. While their study employed traditional software algorithms to achieve these goals, ATLAS applies AI-driven scheduling algorithms to further enhance the accuracy and efficiency of the scheduling process. This approach not only streamlines the scheduling of research presentations but also optimizes panelist and student availability, offering a more flexible and responsive solution to scheduling challenges.	
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	One notable approach to automated scheduling is the use of mixed-integer programming (MIP) for university course timetabling. Rappos et al. present a comprehensive MIP model that addresses the complexities of scheduling courses while considering various constraints such as classroom availability and instructor assignments. Their findings indicate that MIP can effectively minimize conflicts and optimize resource allocation, making it a suitable choice for educational institutions facing intricate scheduling challenges (Rappos et al., 2022). Additionally, the integration of genetic algorithms (GAs) into scheduling systems has been explored by Dogan and Yurtsal, who developed a decision support system for exam scheduling. Their system utilizes GAs to efficiently assign courses to time slots, demonstrating the algorithm's capability to handle large datasets and complex constraints (Dogan & Yurtsal, 2021). Another innovative approach is highlighted in the work of Uma et al., who developed an automatic timetable generation system that incorporates artificial intelligence to optimize resource utilization. This system takes inputs such as subjects, teachers, and workloads to generate timetables that adhere to predefined constraints, significantly reducing the manual effort required in traditional scheduling processes (Uma et al., 2023). The architecture of this system emphasizes adaptability and efficiency, showcasing how AI can enhance automated scheduling. Automated Scheduling System is another critical component that simplifies the logic of research presentations and defenses. By leveraging AI and algorithms, this system automatically aligns the schedules of students, advisers, and panelists,	
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to proactively address issues. This aligns with the growing use of DSS in academic institutions, similar to how AI-driven systems like ATLAS streamline research management and decision-making in educational environments. (Adedapo et al., 2023).

Decision Support System is essential in guiding students and faculty members throughout the thesis process by providing structured recommendations based on analyzed data. In the context of ATLAS, the DSS helps students select thesis topics that align with current academic trends and their strengths. By evaluating available data and research trends, the DSS supports more informed decision-making, ensuring that students work on relevant and impactful topics. The system uses this decision support mechanism to provide objective recommendations, which helps streamline the thesis selection and development process, improving both the quality and relevance of the research produced.

Structured Query Language

Structured Query Language (SQL) remains a cornerstone in the management and retrieval of data within relational database management systems (RDBMS). Its declarative nature allows users to specify what data they wish to retrieve without detailing how to obtain it, which simplifies the interaction with complex databases (Ares et al., 2023). SQL employs relational algebra and calculus, making it a powerful tool for querying and manipulating structured data efficiently. This characteristic is particularly beneficial in environments where data

integrity and consistency are paramount, such as in financial and healthcare applications (Ares et al., 2023).

SQL is critical for managing the backend database of ATLAS, as it enables efficient storage, retrieval, and manipulation of data. Given that the system will need to manage a variety of datasets, including student records, research topics, schedules, and grades, SQL ensures that all this data can be organized, accessed, and updated seamlessly. The system will rely on SQL to perform complex queries, such as matching schedules between students and panelists, tracking progress on research projects, and ensuring data integrity. Without SQL, it would be challenging to handle the large volume of data that ATLAS will process, especially when maintaining consistency and preventing data duplication.

JavaScript

JavaScript has undergone significant evolution since its inception, transitioning from a simple client-side scripting language to a robust full-stack programming language. This transformation has been largely influenced by the standardization efforts surrounding ECMAScript, particularly with the introduction of ES6, which enhanced JavaScript's capabilities and maintainability ("Modern JavaScript Frameworks and JavaScript's Future as a FullStack Programming Language", 2023). The adoption of modern JavaScript frameworks has also played a crucial role in this evolution, as they provide developers with tools that streamline the development process and improve application performance (Ferreira et al.,

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	The LPU-C-DCS faced challenges with manually managing the scheduling, grading, and administration of capstone and research presentations due to a growing student population and faculty workload. To address this, (Leaño et al, 2013) created an online system that automated scheduling, grading, and report generation, reducing errors and saving time. Our system improves on this by incorporating AI-driven scheduling, LLM-powered feedback analysis, automated grading support, and a more robust content management system. These enhancements will further streamline the process, reduce conflicts, and provide intelligent assistance for both faculty and students CMS will serve as a crucial tool for managing academic content within ATLAS. It provides a flexible platform for faculty members to easily update rubrics, research titles, team lists, and panelist assignments without needing technical expertise. The CMS ensures that all important documents and data related to thesis projects are well-organized and easily accessible. By centralizing content management, the CMS will help streamline academic processes, reduce manual errors, and enhance the overall efficiency of handling research presentations and evaluations. Generative Artificial Intelligence Artificial Intelligence (AI) has emerged as a transformative force across various sectors, influencing methodologies, operational efficiencies, and ethical considerations. In the realm of education, AI's potential to enhance teaching and	
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	The application of Artificial Intelligence (AI) in research has gained significant traction in recent years, particularly in the context of academic writing, data analysis, and educational methodologies. This literature review synthesizes recent studies that highlight the transformative potential of AI across various research domains, emphasizing its role in enhancing academic performance, improving research methodologies, and addressing ethical considerations. AI's integration into academic writing has been a focal point of recent research. For instance, this discusses how AI is reshaping academic research, particularly in personalized learning and intelligent tutoring systems, which bridge the gap between academic results and industry needs (William, 2024). This sentiment matches with (Pigola, 2023), who notes that AI's influence extends beyond mere data processing to fundamentally altering the landscape of academic research itself. The potential of AI tools, such as Gemini, to assist in literature reviews and data analysis has been recognized as an asset for researchers, facilitating more efficient and thorough investigations (Burger et al., 2023). Furthermore, the exploration of AI's role in enhancing students' educational performance underscores its capability to tailor educational content based on individual learning patterns, thereby improving engagement and knowledge retention (Abbas et al., 2023). The integration of AI in education presents a multifaceted opportunity to enhance learning experiences and outcomes. As educational institutions continue to adopt AI technologies, ongoing research and collaboration will be essential to navigate the complexities and maximize the benefits of AI in educational contexts.	
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Evaluating Academic Systems Using ISO/IEC 25010 Standards

The evaluation of academic systems using the ISO/IEC 25010 standards has gained prominence in recent years, as educational institutions increasingly rely on software solutions to enhance their operational efficiency and educational outcomes. ISO/IEC 25010, part of the Software Quality Requirements and Evaluation (SQuaRE) framework, provides a comprehensive model for assessing software quality through eight key characteristics: functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety (Ani et al., 2022).

The importance of ISO/IEC 25010 in assessing the quality of academic information systems. The authors argue that the standard's comprehensive nature allows for a nuanced evaluation of software performance in educational settings, ensuring that systems meet the diverse needs of users, including students and faculty (Mulyawan et al., 2021). Who demonstrated that integrating ISO/IEC 25010 into the "PSG Balita" application improved the quality of nutritional status data by streamlining data management processes, thereby enhancing the overall educational experience which aligns with findings (Rahmad, 2023),. Such improvements underscore the potential of ISO/IEC 25010 to facilitate better data handling and reporting in academic contexts.

The ISO/IEC 25010 standards serve as a robust framework for evaluating academic systems, offering a structured approach to assess software quality in educational environments. The integration of these standards not only enhances the

Conceptual Framework of the Study

The proponents undertaking the project must acquire knowledge in HTML, CSS, JS, SQL, and PHP, as these skills are essential for the development of Decision Support System. The framework shows the system's components and their relationship, focusing on how an AI-driven website processes can optimize scheduling and assessment within the academic environment.

As illustrated in Figure 1 below, the Scrum model was chosen for this study because it provides an agile framework that enables iterative development, continuous feedback, and adaptability, ensuring the efficient implementation of ATLAS's complex, user-driven features.

Figure 1.

Conceptual Framework of ATLAS.

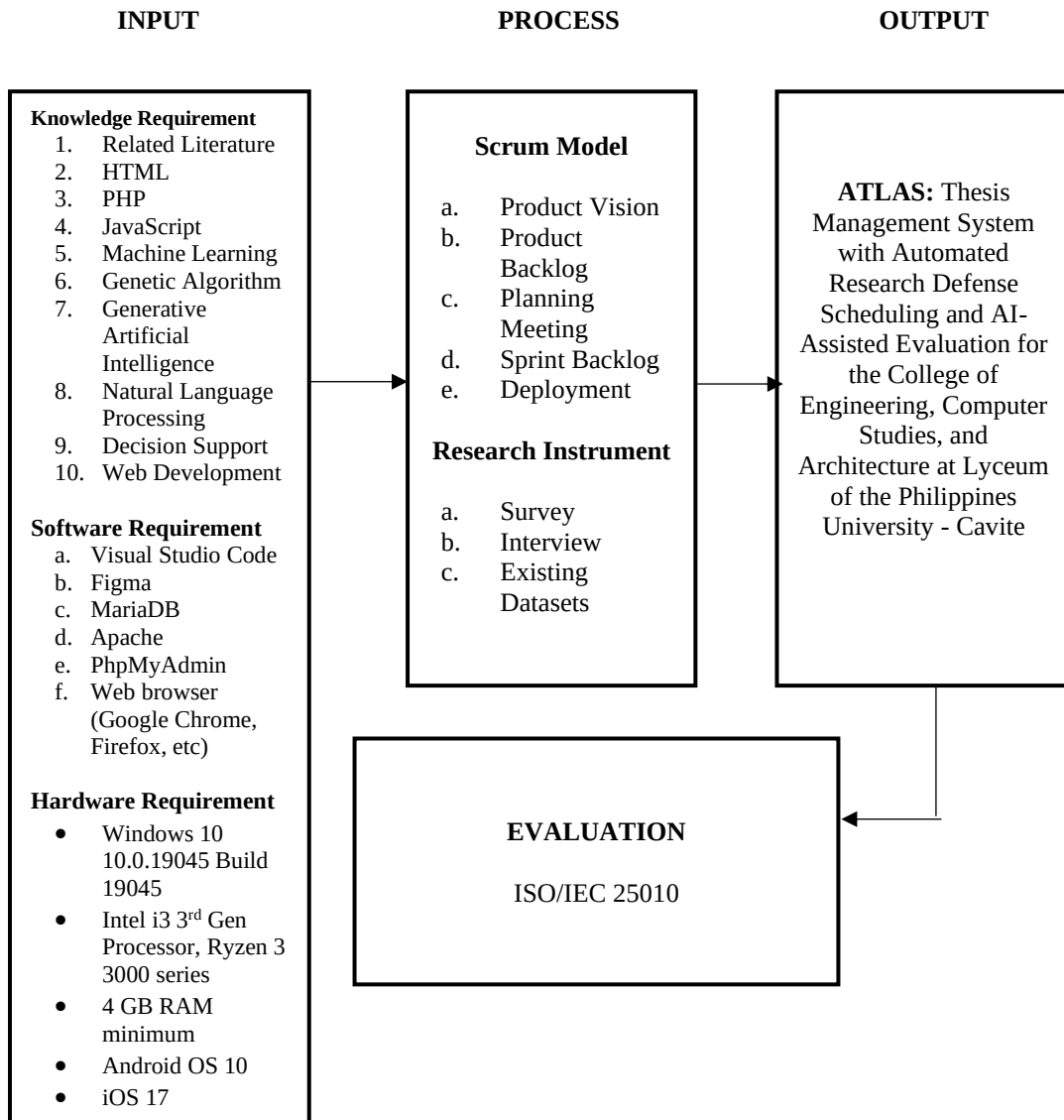


Figure 1 illustrates the project's framework representing the Input, Process, and Output phases of the study. The Input section shows the required Knowledge requirements, Software requirement, and Hardware requirements which are

Content Management System. A digital platform that helps users create, manage, and publish content for websites and apps. In ATLAS it streamlines document management, ensures transparency, reduces redundancy, and minimizes errors, providing students and faculty with easy access to essential materials for a cohesive academic experience.

Gemini. A Large Language Model (LLM) developed by Google AI. In ATLAS, Gemini is used to empowers the Thesis Topic Decision Tool, Research Title Acceptance, and Pass Recommendation features.

Large Language Model. A type of AI program that uses deep learning to process and generate human language. LLM allows the features in ATLAS to work.

Genetic Algorithm. A computational method that uses natural selection and genetic operators to solve problems by evolving a population of candidate solutions. In ATLAS, GA optimizes scheduling by efficiently assigning research presentations to available slots.

MariaDB. A free, open-source relational database management system that is used for storing data in a tabular format. It is used in ATLAS as a server database.

PHP. A server-side scripting language embedded in HTML in its simplest form. PHP is used where the server-side is coded.

Apache. A free, open-source web server that delivers web content over the internet. It is used in ATLAS as the web server.

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	Visual Studio Code. A source code editor that helps developers build, debug, and test code. VS Code is used as the IDE for ATLAS Decision Support System (DSS). An AI-based system that aids in academic decision-making by analyzing data to recommend thesis topics for students and assess work quality for advisers. It ensures informed, unbiased, and consistent recommendations. Scheduling System. Uses a Genetic Algorithm to optimize thesis defense schedules by aligning participant availability and institutional constraints. It minimizes conflicts, reduces manual effort, and improves efficiency. Thesis Topic Decision Tool. An AI-powered tool that analyzes research trends to suggest unique, relevant, and impactful thesis topics aligned with academic and industry standards. Requirement Checker Tool. Tracks mandatory thesis defense documents, alerting students to incomplete or missing items to ensure timely completion and a smooth defense process. Pass Recommendation System. Uses Gemini Pro LLM to analyze panelists' feedback and scores, providing preliminary AI-based pass/revision recommendations. It reduces grading inconsistencies and errors while maintaining academic rigor. Research Title Acceptance Tool. Assesses title originality by comparing proposed titles with online research databases to prevent duplication and ensure unique contributions.	
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CHAPTER III

METHODOLOGY

Research Design

In this study, the proponents used a quantitative survey research approach to evaluate and see the potential effectiveness of the Atlas: Thesis Management System with Automated Research Defense Scheduling And AI-Assisted Evaluation for The College of Engineering, Computer Studies, And Architecture at Lyceum of the Philippines. This approach was chosen to get valuable data from students, advisers, and program chairs. The data will be analyzed to see and understand the relevance of the proposed system.

To optimize scheduling, the Genetic Algorithm was chosen as the main algorithm for ATLAS. Genetic Algorithm is a robust optimization technique inspired by the process of natural selection. It is used in ATLAS to efficiently allocate limited slots for research presentations and to avoid conflicts, ensuring that all users of ATLAS have an optimized and organized schedule.

Additionally, Gemini, a Large Language Model (LLM), is utilized within ATLAS to perform tasks related to natural language processing (NLP). Gemini aids in the text analysis of ATLAS, especially in sentiment analysis of feedback from

that, compared to other algorithms, GA yields better average fitness values and computational efficiency, making it an ideal choice for addressing the multi-objective demands of course scheduling. (Xu., 2021)

Gemini, with its dual encoder model and memory-augmented mechanism, is uniquely positioned to excel in thesis management compared to other AI models. Unlike traditional algorithms, which may lack the ability to handle complex, multi-turn dialogues, Gemini’s architecture is specifically designed to understand and engage in sequential, context-rich interactions. This enables it to navigate the nuanced and evolving steps in thesis tasks, such as suggesting relevant topics, validating research titles, and providing pass recommendations. The model’s memory-augmented mechanism allows it to retain context across multiple dialogue turns, making it ideal for tasks that require consistency and continuity, such as following up on a student’s thesis development over time. Additionally, Gemini’s fine-grained dialogue state tracking ensures accurate and relevant responses by keeping track of the evolving context, which is essential in academic environments where clarity and adherence to specific guidelines are critical. While other Large Language Models (LLM) offer strengths in open-ended conversation and task-specific support, Gemini’s tailored approach to multi-step dialogue and task tracking makes it a superior choice for the specialized requirements of thesis management. (Krify, 2024)

The proponents chose Gemini over GPT because it meets the specific technical requirements of our project, particularly in handling multi-turn dialogues

and providing context-aware interactions tailored for academic processes. Google’s Gemini is highlighted as one of the best models for developing robust applications due to its combination of high performance, versatility, and cutting-edge features.

As a proprietary model from one of the leading AI companies, Gemini benefits from Google's extensive resources, data, and expertise in natural language processing and machine learning. It excels in tasks requiring deep reasoning and a broad general knowledge base, making it suitable for a wide range of applications. Additionally, Gemini's integration with Google's ecosystem and tools enhances its utility for developers, offering seamless compatibility with cloud services, APIs, and advanced deployment options. For applications needing top-tier performance out of the box, Gemini is presented as a strong choice, alongside other models particularly when quality and reliability are top priorities. (MindsDB, 2024) Also, in an article by (Jackson, 2023) it is stated that Google AI is one of the ethical companies that practices AI. Gemini is identified as the most adversarially robust AI model on the leaderboard due to its superior performance in resisting adversarial attacks designed to exploit vulnerabilities in AI systems. This robustness demonstrates its advanced architecture and training methodology, enabling it to maintain accuracy and reliability under challenging conditions (Scale, 2024).

Stratified random sampling will be employed to gather responses from Professors, Research Advisers, Research Panelists, Department Chairs, and Students through an online survey. This approach involves dividing the population into distinct subgroups (strata) based on their roles within the thesis process. A

random sample will then be drawn from each subgroup, ensuring a proportionate and representative selection of participants. By capturing input from all relevant groups, this method ensures comprehensive feedback on the college's thesis system and the potential benefits of an AI-driven decision support tool, while minimizing bias and enhancing the validity of the research findings.

Participants of the Study

The participant of this study includes three key groups from LPU-C: students, thesis advisers/professors, and the program chair. All the mentioned participants are vital in giving valuable feedback on potential improvements of ATLAS.

- Student Researcher/Proponent will provide insights on how the system supports their scheduling, performance tracking and research title management.
- Research advisor monitoring the students' progress and improving the evaluation of research presentations.
- Research Panel will assess the student's performance during defense.
- The Program Chair will offer its insights into the system's overall impact on the department.

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	Research Locale This study will be conducted at Lyceum of the Philippines University – Cavite (LPU-C), located in General Trias, Cavite. LPU-C is known for its commitment to academic excellence and fostering innovation, making it an ideal setting for this study. The institution has been chosen as it houses more complex research presentations and academic assessments. The institution is home to a diverse group of faculty members and students who actively engage in academic research, making them the main beneficiary of the proposed system. By focusing on LPU-C, the study aims to address real-world challenges in managing research defenses and enhance the efficiency of academic assessments. Research Instrument To gather data for this research, an online survey questionnaire using Microsoft Forms. The survey will target the key stakeholders including Professors, Research Advisers, Research Panelists, Department Chairs, and Students. The objective is to gain valuable insights into the current thesis system used within the college and to explore the potential of implementing an AI-driven decision support tool.	
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Data Gathering Procedure

The proponents relied on online surveys with the selected respondents. The survey was sent through email and can be accessed by using their LPU account.

The questions for the survey were based on the research objectives. The survey has categories depending on the respondents if either they are students, thesis professors/advisers and program chair. The respondents' privacy and confidentiality were strictly maintained throughout the data gathering process.

Multiple Constraints Analysis

The ATLAS project is influenced by several constraints that shape its development and deployment. These constraints are primarily technical, user-related, and regulatory. While other areas, such as environmental and health-related constraints, have no relevance to this system.

First technical constraints include the system's dependency on hardware such as internet access, mobile devices and computers, which are critical for accessing and operating the system. Limited access to these resources, especially in areas with poor connectivity or insufficient hardware availability, could impact on the system's functionality and user experience. A stable internet connection is essential, as the system cannot function or be accessed without it, potentially delaying research processes.

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	Regarding data privacy constraints, the system handles sensitive information such as student, adviser, and administrative accounts, necessitating strict measures to protect this data. All personal information will be anonymized to ensure confidentiality, and access will be restricted to authorized personnel only, reducing the risk of unauthorized access or data breaches. These measures align with privacy laws and reinforce user trust in the system's integrity and security. As for the constraints that are not applicable to this project, environmental constraints are not relevant since ATLAS is a software-based project, it has no direct interaction with physical environments or ecosystems. It uses existing digital infrastructure, resulting in no significant ecological impact. Similarly, health and safety constraints are not relevant, since the project is entirely digital, it does not involve physical risks to users. There are no requirements for emergency preparedness measures or occupational health precautions. Lastly, sustainability constraints, while sustainability in terms of long-term system usability is a consideration, there are no concerns regarding physical resource management or ecological sustainability, as the system does not rely on consumables	
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System Development Process

The proponents utilized the Scrum model for the development of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation for LPU-C. The Scrum model has 5 components that the proponents utilized. This includes Product Vision, Product Backlog, Planning Meeting, Sprint Backlog, and Deployment.

Figure 2.

Scrum Model

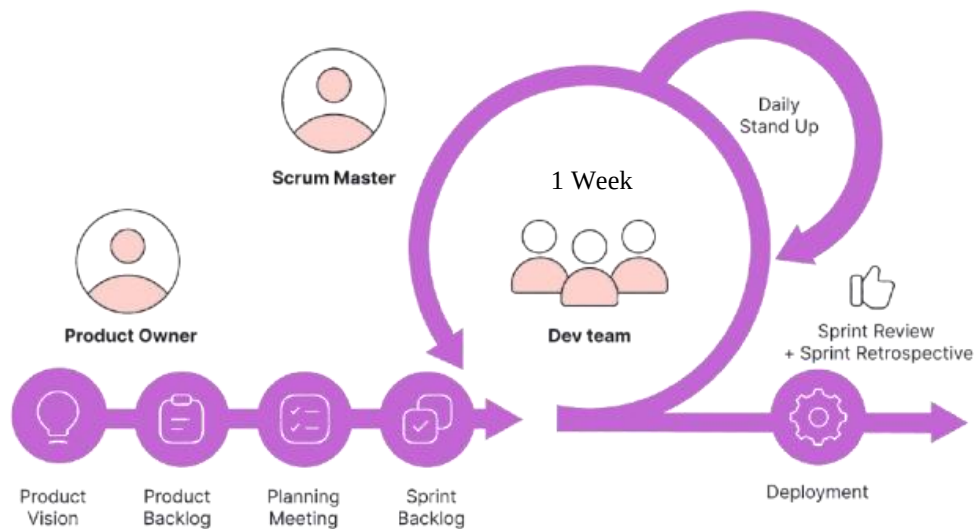


Figure 2 illustrates the process of Scrum Model. Below discusses each step pertaining to the actions of the life cycle:

Scrum Framework Overview

The Scrum model operates in short sprints typically lasting between 1-4 weeks. But for this study the researchers intend to operate weekly. Each sprint consists of planning and development. Key roles in Scrum includes Product Owner, Scrum Master, and Development Team, each roles plays a crucial role ensuring the success of the project.

Roles:

Product Owner: Defines the product vision, prioritizes the product backlog, and ensures value delivery.

Scrum Master: Facilitates Scrum processes, removes blockers, and drives team improvement.

Development Team: Builds, tests, and documents the product increment collaboratively.

Artifacts and Events:

Product Vision: The goal and purpose of the product.

Product Backlog: A prioritized list of all tasks, features, and requirements for the product.

Sprint Planning Meeting: A session to define the sprint goal and select backlog items for the sprint.

Sprint Backlog: A focused subset of the product backlog for the sprint.

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Daily Standup: A quick meeting to align team progress and identify blockers.

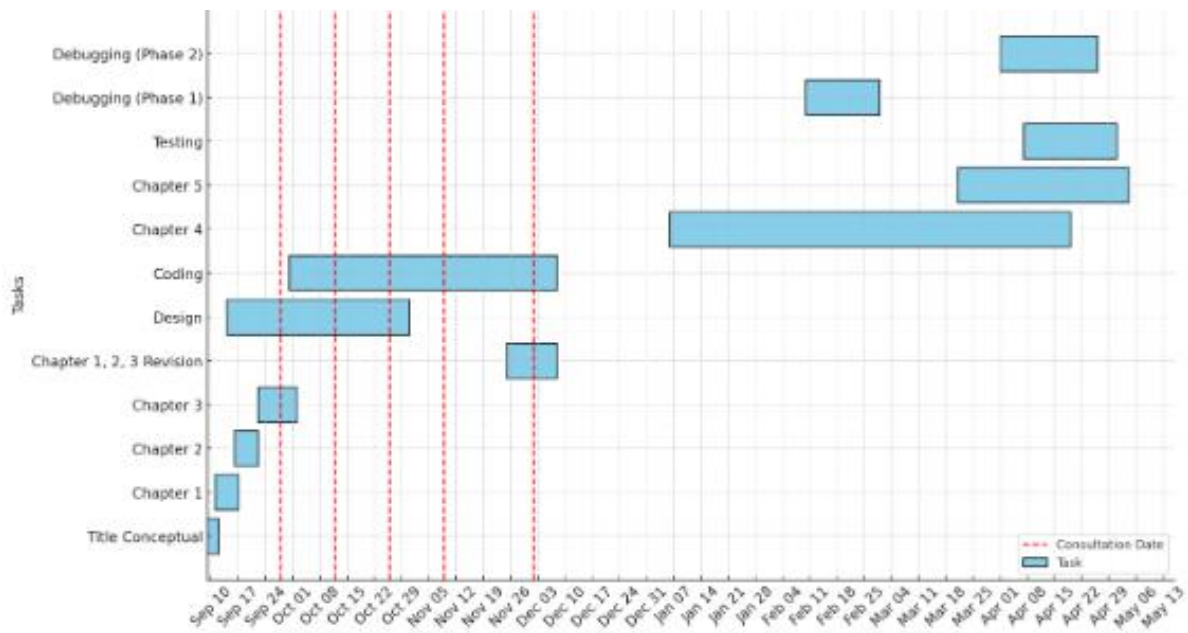
Sprint Review: A presentation of completed work for stakeholder feedback.

Sprint Retrospective: A reflection meeting to identify and plan process improvements.

Deployment: The release of the product increment to stakeholders or production.

Figure 3.

Gantt Chart of ATLAS

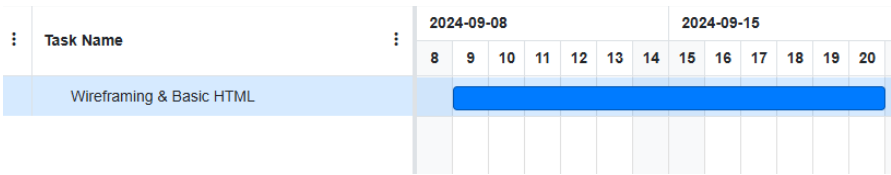


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	The Gantt Chart of ATLAS outlines the timeline for the various tasks involved in the project’s development, providing a clear breakdown of activities, durations, and consultation dates. The tasks are organized from conceptualization to revision phases, starting on September 10 and ending on May 3. Key activities include Title Conceptualization, Chapter 1–5 Writing, Design, Coding, Testing, and Debugging, arranged sequentially based on dependencies and priority. Chapters 1, 2, and 3 were completed progressively between mid-September and early October. The Design phase spanned from mid-September to late October, overlapping with the beginning of the Coding process, which ran from October 1 to December 10. Chapter 4 was developed from January 6 to April 18, followed by Chapter 5, conducted between March 21 and May 3. The Debugging process occurred in two phases: the first from February 10 to February 28, and the second from April 1 to April 25, ensuring ongoing system refinement. Meanwhile, Testing was carried out from April 7 to April 30, validating system performance and functionality. The Gantt Chart also highlights specific Consultation Dates (marked in red dashed lines) on October 1, October 29, November 5, and November 26. These consultations align with major milestones, such as the completion of design and key stages in coding, ensuring timely feedback during critical development phases. The Chapter 1–3 Revisions took place between late November and early December, providing a structured opportunity for reflection and improvement. This	
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comprehensive and well-sequenced timeline ensures a balanced workflow of development, feedback, and refinement, crucial for the successful completion of ATLAS.

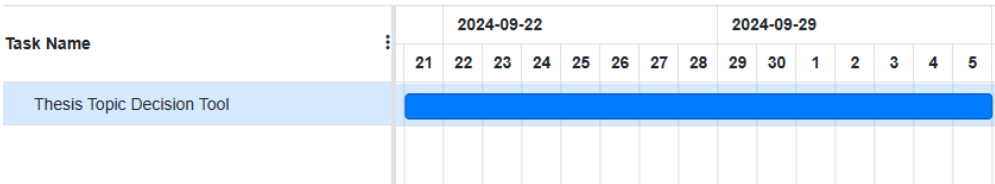
Sprint 1: Wireframing and Basic HTML Lay-outing

The initial sprint laid the groundwork for the system’s interface and user experience. Wireframes were created to visualize the overall system structure, and simple HTML lay-outing was implemented to build the skeletal framework of the user interface. This phase emphasized intuitive design and navigational clarity.



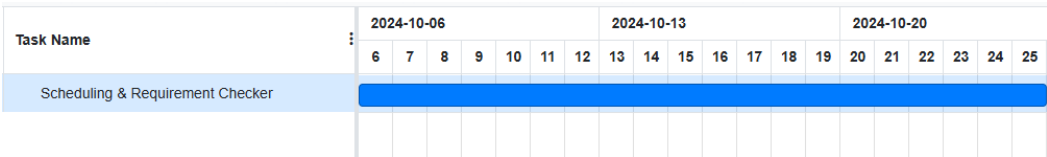
Sprint 2: Module 1 Development – Thesis Topic Decision Tool

The second sprint focused on developing the Thesis Topic Decision Tool. This module was independently designed, coded, and tested to ensure it functioned as a standalone feature, setting the precedent for modular development across the system.



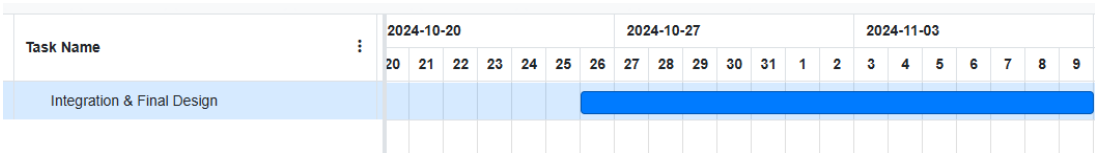
Sprint 3: Module 2 and 3 Development – Scheduling System & Requirement Checker Tool

In the third sprint, the Scheduling System and Requirement Checker Tool were developed. These modules were constructed concurrently, each undergoing individual testing to verify their functionality before integration. Isolating these features ensured modular stability and reduced the risk of systemic errors later on.



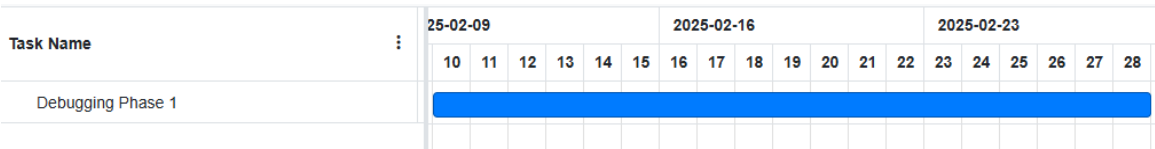
Sprint 4: System Integration and Design Finalization

During this sprint, all developed modules were integrated into a unified system. Interoperability was a key focus, ensuring seamless communication between features. Final UI/UX designs were also implemented to provide a consistent and visually appealing interface.



Sprint 5: Debugging Phase 1

The fifth sprint addressed early-stage bug identification and resolution. Occurring between February 10 and 28, this phase involved initial system testing and refinement, targeting functional inconsistencies and performance bottlenecks ahead of full testing.



Sprint 6: Testing and Debugging Phase 2

The final sprint covered full-scale system testing (April 7–30) and final debugging (April 1–25). Comprehensive evaluations were conducted to detect remaining issues, ensuring the system met all intended requirements and operated smoothly before project completion.

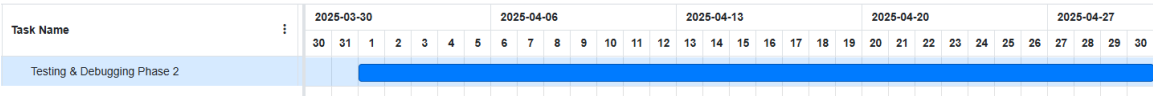
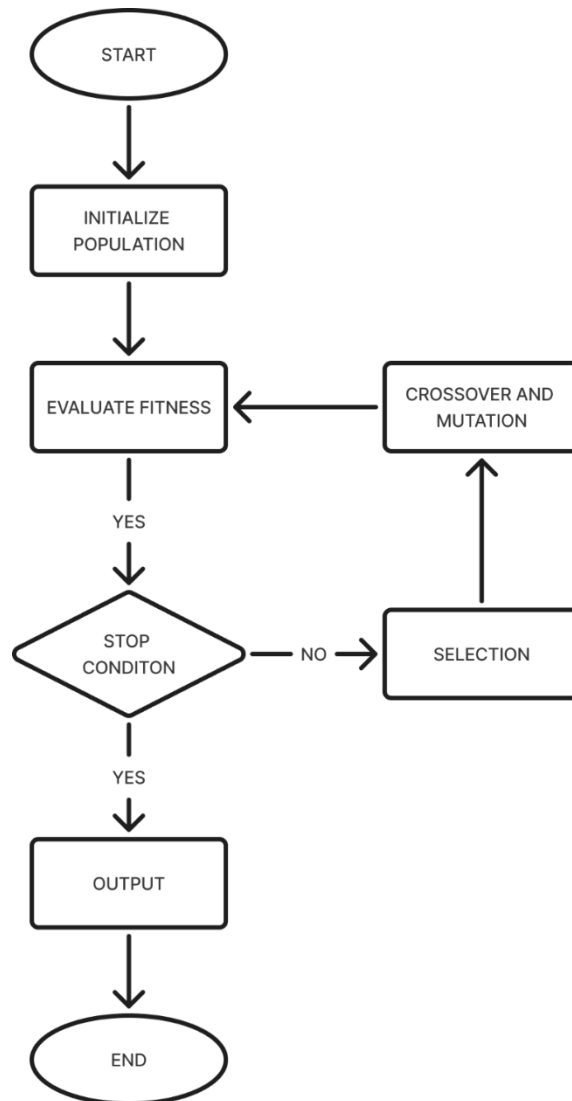


Figure 4.

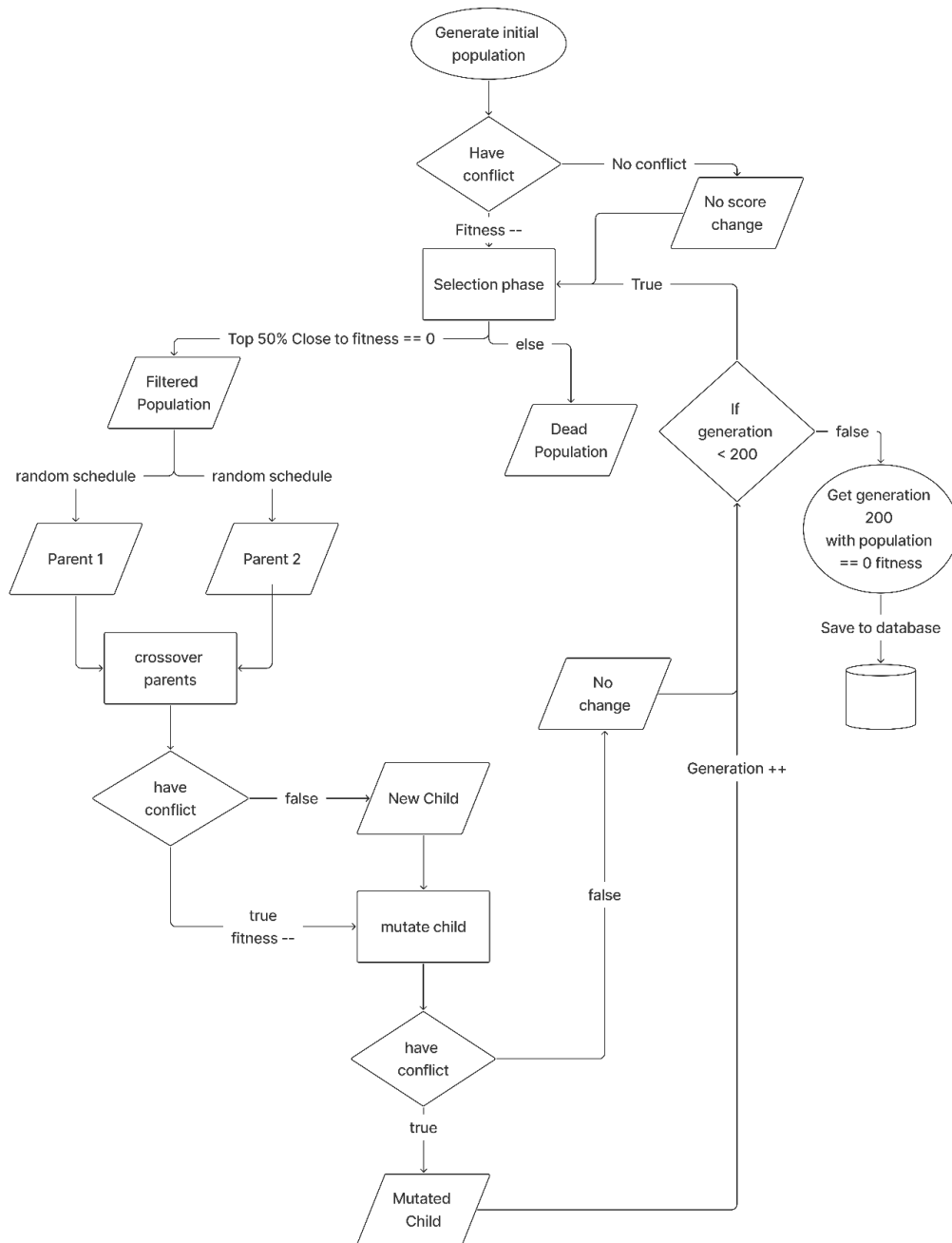
Base Genetic Algorithm



Setup

Figure 5.

Genetic Algorithm Simulation



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Parameters:

- **Teams:** Team A and Team B
- **Panelists:** P1, P2, P3, and P4
- **Rooms:** Room 1, Room 2
- **Time Slots:** 9:00 AM, 10:00 AM
- **Days:** December 11, 2024

Initial Population:

- **Population Size:** 2
- **Generations:** 2 (example)
- **Mutation Rate:** 10% (0.1)

Initial Population

Table 1.

Individual 1 (Schedule)

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P2, P3, P4	Room 1	9:00 AM	December 11, 2024

The table showed both teams having a schedule in the same room with panelists P2 and P3 having overlapping schedules reducing fitness score by 20, 10 for each conflict.

Table 2.

Individual 2 (Schedule)

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P4	Room 2	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 1	10:00 AM	December 11, 2024

The table shows a conflict panelist P4 is double-booked between the two teams A and B which deducts its fitness score by 10 compared to Individual 1, and there is a crossover between Parent 1 Individual 1 and Parent 2 Individual 2.

Table 3.

Child (New Schedule):

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 1	10:00 AM	December 11, 2024

Table 3 shows that panelist P3 has overlapping schedule and Room 1 has been booked twice where and that there is a mutation chance of 10% to change the team's schedule. Team B changes its room as part of the Mutation.

Table 4.

Mutated Child

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 2	10:00 AM	December 11, 2024

Table 4 shows no conflict after the mutation where the child of the 1st Generation has a fitness of 0 which dominates and is better compared to Individual 2 with a fitness of -10. Another crossover may occur, but results will favor conflict-free schedules.

It will iterate depending on the set number of generation, the tuning of 100 population, 200 generations, and the 0.1 mutation rate returned significant changes with least errors.

The input values will be sent as POST data to the run_scheduler.php file, which will also retrieve data from the database, including information about teams, users (panelists), and user schedules.

The algorithm will generate a randomized initial population of 100 solutions and iterate through 200 generations, using a learning rate of 0.1. Each team will be limited to a single schedule.

For each generation:

1. The algorithm computes the fitness score and counts conflicts for each individual in the population.

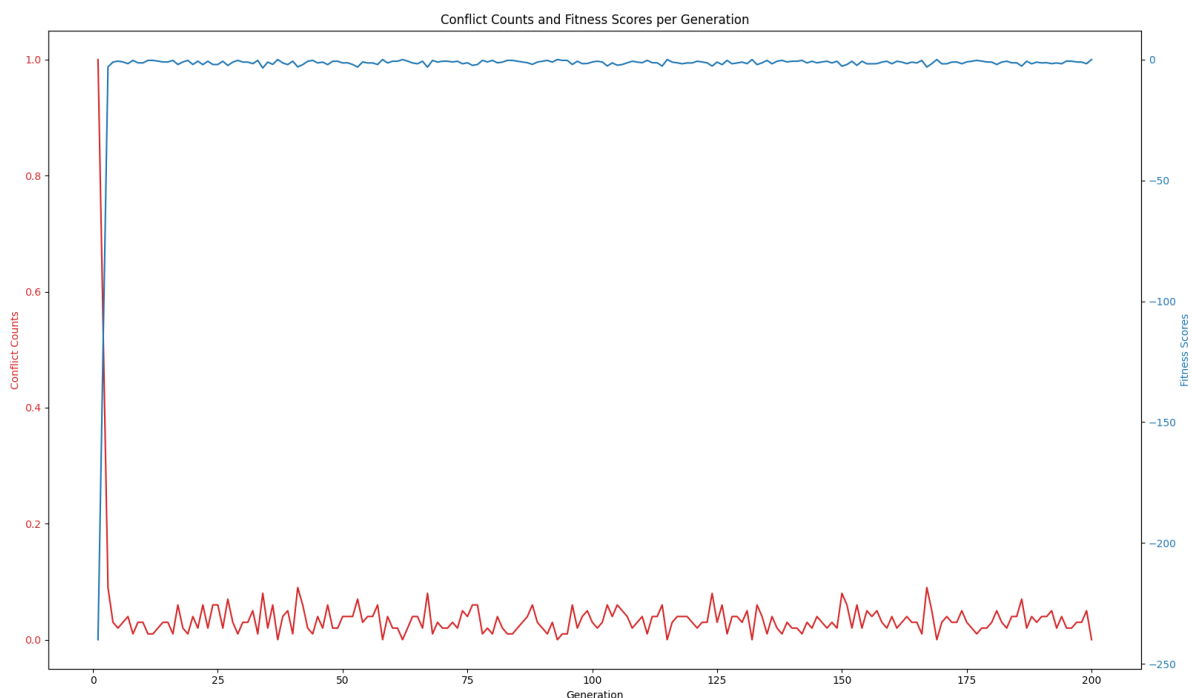
2. It calculates the average fitness and conflict count for the generation.
3. The solution with the least conflicts is selected, and genetic crossover is performed on its attributes (e.g., time slots, days, and rooms). These attributes are pooled and passed down to the next population.

The genes are then randomly assigned to the new population. If a newly generated gene assignment introduces more conflicts, the program reverts the changes and moves on to the next population member.

Finally, the population is sorted from the best to worst fitness scores, with the top individual representing the current best solution. Below is an example run.

Figure 6.

Conflict Counts and Fitness Scores per Generation



The graph shows the performance of a genetic algorithm (GA) over generations, with two metrics plotted:

1. **Conflict Counts (left y-axis):** This likely represents the number of violations or conflicts (e.g., constraints in a scheduling problem) in the solution at each generation.
2. **Fitness Scores (right y-axis):** This represents how optimal or "fit" the solutions are (lower fitness scores could indicate better performance if minimizing is the goal).

Here's why the graph looks like this:

Initial Generations

- **Sharp drop in conflict counts:** In the early generations, the algorithm explores the search space and rapidly improves the population by eliminating many conflicts. This is typical as random solutions with high conflicts are refined through crossover, mutation, and selection.
- **Corresponding fitness improvement:** Fitness scores also improve quickly during these initial iterations, reflecting better solutions with fewer conflicts.

Stabilization Phase

- **Conflict counts oscillation:** After the initial rapid improvement, the red line stabilizes around a low value with slight fluctuations. This suggests the algorithm has reduced most conflicts but is struggling with local optimization and minor adjustments.

- **Fitness scores plateau:** Similarly, the blue line plateaus, indicating the population is converging to solutions that are near-optimal.

Reason for Oscillations

- **Genetic diversity:** Fluctuations in both metrics may result from mutations or crossovers introducing diversity into the population. This can momentarily worsen solutions before re-converging to better solutions.
- **Local optima:** The GA might be oscillating around a local optimum if global exploration is limited.

Observations

- The fitness scores and conflict counts don't seem to improve significantly after a certain point, which may indicate:
 - The algorithm has converged to a near-optimal solution.
 - Further refinement may require a hybrid approach, such as local search methods post-GA.

Mathematical Model of Genetic Algorithm

The mathematical model of Genetic Algorithm consists of several components that outline its functioning for solving optimization problems.

1. Population Representation

The population at generation t is denoted as $P(t) = \{x_1, x_2, \dots, x_n\}$, where x_i represents an individual solution (a potential candidate). Each individual is encoded, often as binary strings, real values, or other representations depending on the problem.

2. Fitness Function

The fitness function evaluates the quality of each solution X in the population. Solutions with higher fitness values are considered better candidates for the problem

$$f: X \rightarrow \mathbb{R}$$

3. Selection Probability

Individuals are selected based on their fitness. The selection probability of an individual x_i is calculated as:

$$p(x_i) = f(x_i) / \sum_j f(x_j)$$

4. Crossover Operator

Crossover combines the genetic material of two parent solutions x_p^1 and x_p^2 to produce an offspring x_o :

$$x_o = \text{Crossover}(x_p^1, x_p^2)$$

Common techniques include single-point, multi-point, and uniform crossover.

5. Mutation Operator

Mutation introduces randomness into the offspring to avoid local optima and maintain genetic diversity. It modifies an individual x_i with a small probability, producing x'_i :

$$x'_i = M(x_i)$$

For binary encoding, this might involve flipping bits; for real-valued encoding adding small perturbations.

6. Replacement Strategy

The next generation $P(t+1)$ is created by replacing some or all the old population $P(t)$ with offspring $O(t)$. The replacement can be generational (entire population replaced) or elitist (best individuals are retained).

$$P(t+1) = \text{Replacement}(P(t), O(t))$$

7. Termination Condition

The algorithm stops when one of the following conditions is met:

- A maximum number of generations is reached $t \geq T$.
- A satisfactory fitness value is achieved:

$$\exists x \in P(t) \text{ such that } f(x) \geq f_threshold$$

The Genetic Algorithm mathematically models an evolutionary process inspired by natural selection. It iteratively evolves a population of candidate solutions through selection, crossover, mutation, and replacement until an optimal or satisfactory solution is found. Each component ensures balance between exploration (diversity) and exploitation (using the best solutions).

Algorithm Comparison

The researchers conducted a comparative analysis of the Genetic Algorithm (GA) and Memetic Algorithm (MA) based on the study by Nico et al. (2020) and their own findings. According to Nico et al., the Memetic Algorithm is efficient for solving simpler problems, leveraging its hybrid approach of global search and local optimization. However, the researchers observed different results when applying both algorithms to the ATLAS system, which involves a more intricate set of variables and constraints.

In their experiments with a population of 100 and 200 generations, the Genetic Algorithm outperformed the Memetic Algorithm. The GA satisfied all constraints without violations across 20 trials, including key constraints such as no consecutive panelists and adviser cannot be panelists. Conversely, the Memetic Algorithm violated two constraints under the same conditions.

While the Memetic Algorithm may offer advantages in less complex scenarios, the Genetic Algorithm proved more effective for ATLAS, providing more robust and optimal solutions. Its ability to explore diverse possibilities and adapt to dynamic constraints makes it better suited for handling the complexities of ATLAS.

Figure 7.

Memetic Algorithm Scores

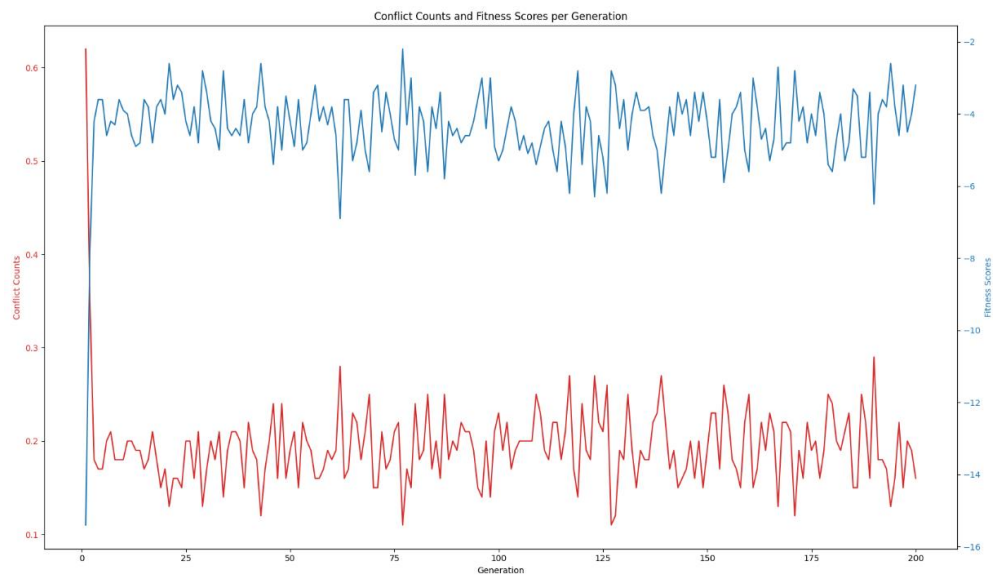
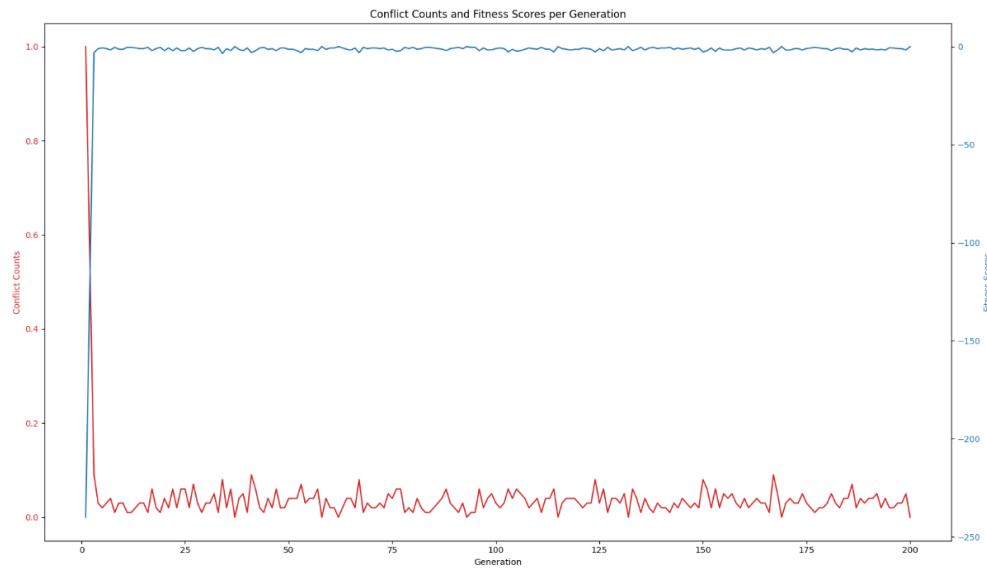


Figure 8.

Genetic Algorithm Scores



System Architecture

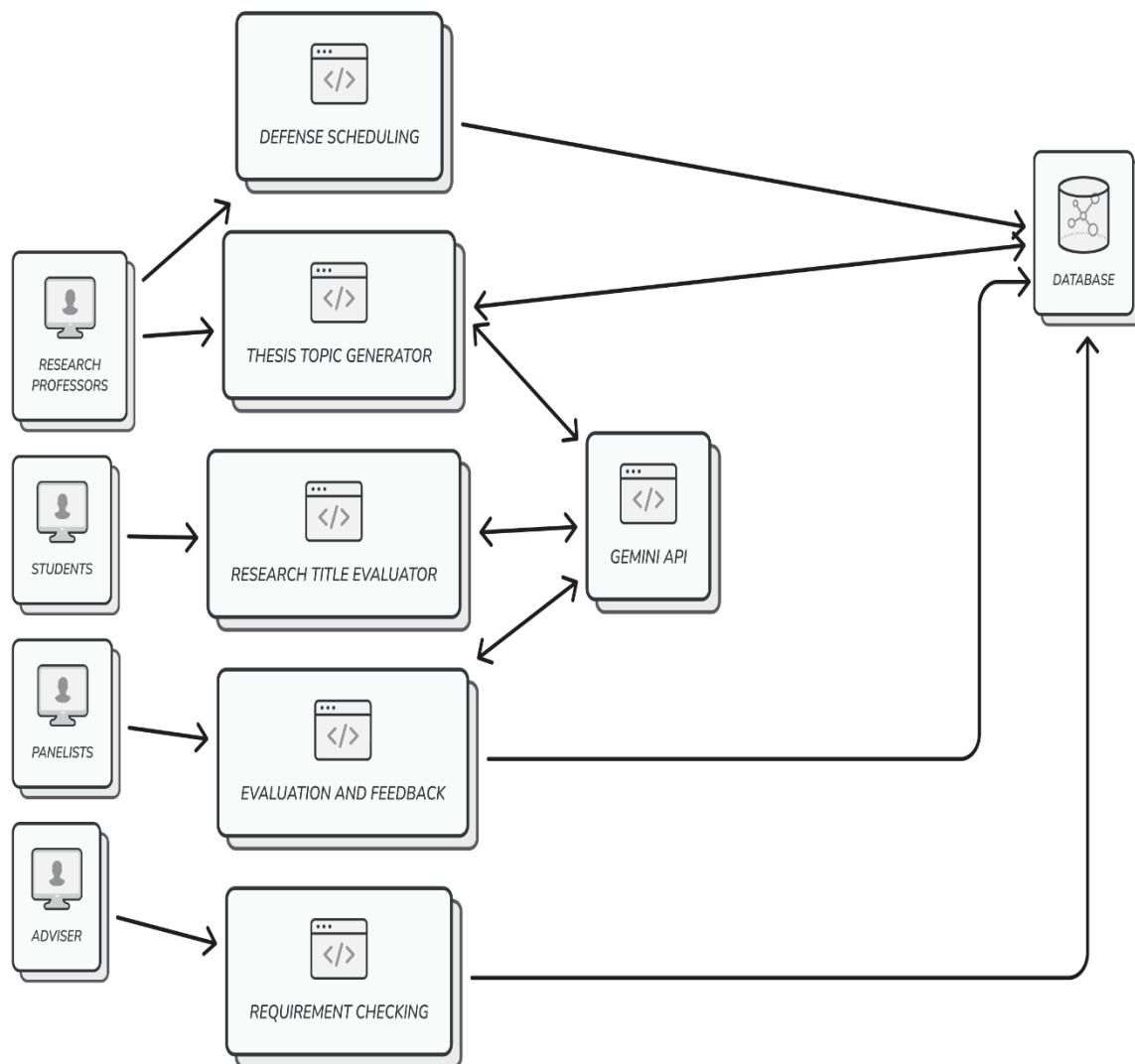
The users log in to access personalized dashboards based on their roles. Students input their interests and grades into the Thesis Topic Decision Tool, which leverages AI analysis to recommend relevant research topics. Upon proposing a title, the Research Title Acceptance Tool checks its uniqueness against an existing database. Students then the student will wait for the time of their defense, and the Scheduling System aligns schedules by querying availability to confirm a defense time. The Requirement Checker will be manual input by the prof/admin and the document submissions, alerting students to any pending requirements. Following

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the defense, Professor panelist provides feedback that Pass Recommendation System analyzes to classify student performance as "Pass" or "Fail." Lastly, admins utilize the Content Management System (CMS) to manage rubrics and panel assignments, ensuring all data remains updated and accessible.

Figure 9.

Systems Architecture



Object Modelling

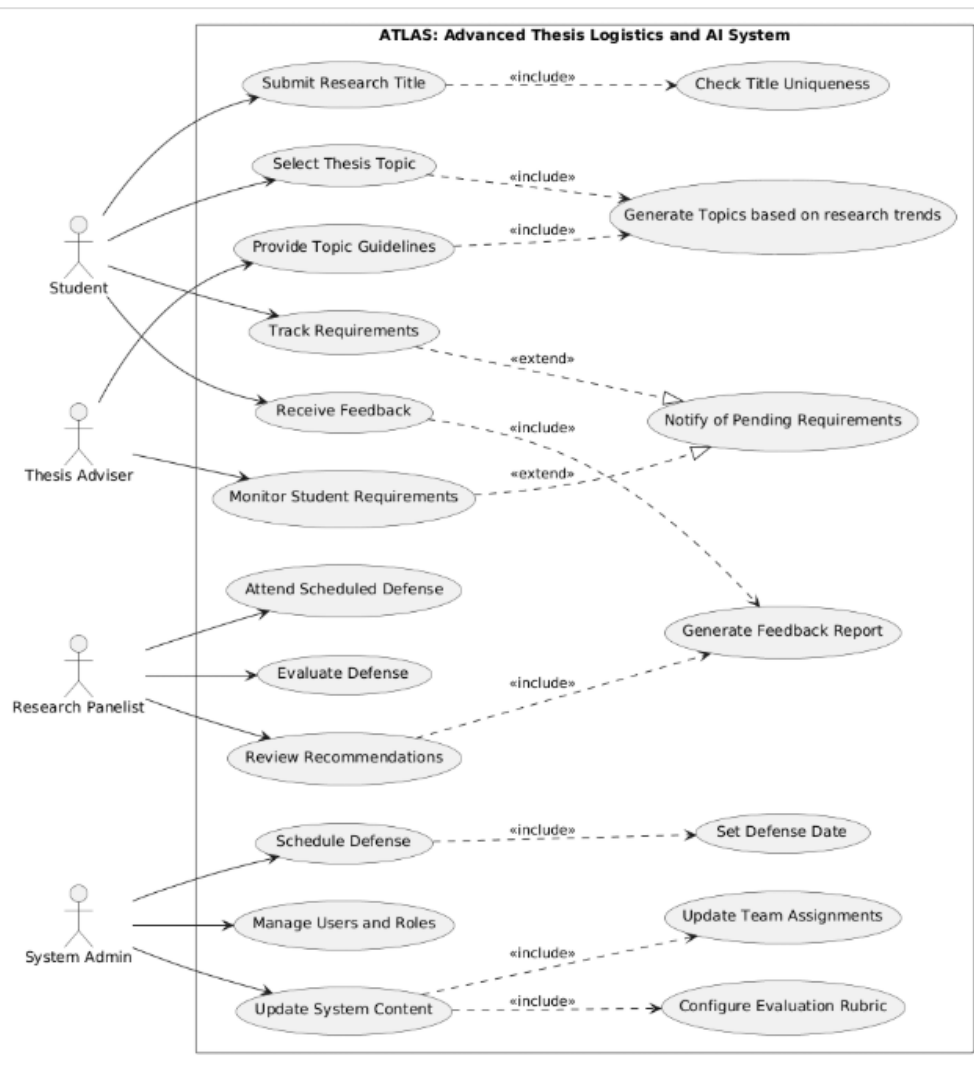
This portion of the design of the application demonstrates the expected client of the application and the capacities that were accessible for the clients to utilize.

Use Case Diagram

The ATLAS system streamlines research topics by enabling students to select topics, submit titles, and track requirements, while advisers and panelists oversee evaluations and feedback. The system admin manages scheduling, user roles, and system content, ensuring an efficient and transparent thesis process.

Figure 10.

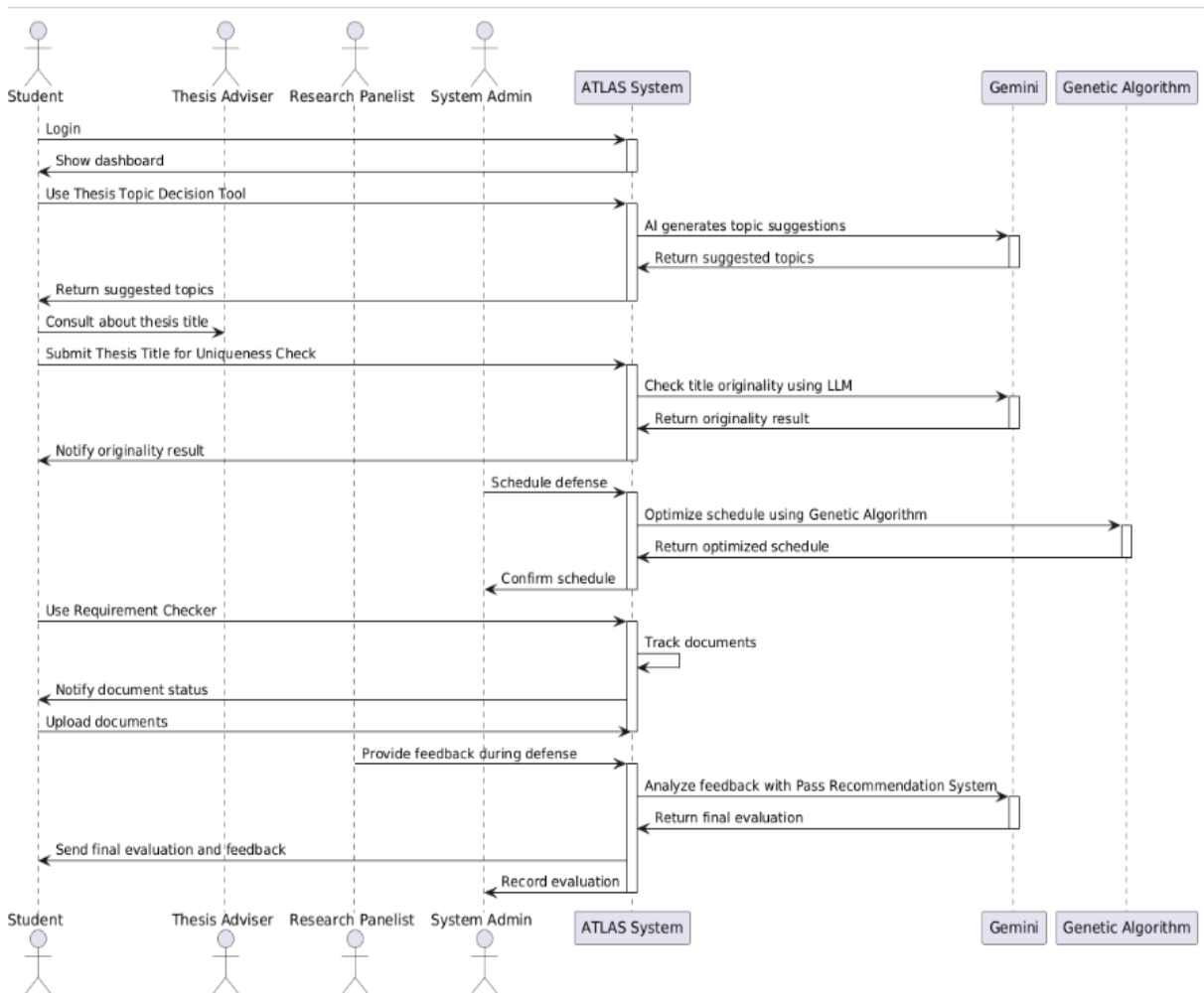
Use Case Diagram



Students initiate the process by selecting a topic with assistance from the Suggest Relevant Topics feature, ensuring relevance. They then submit a research title, which is verified for originality using Check Title Uniqueness. Students also track their document submissions through Track Requirements, with optional

Figure 11.

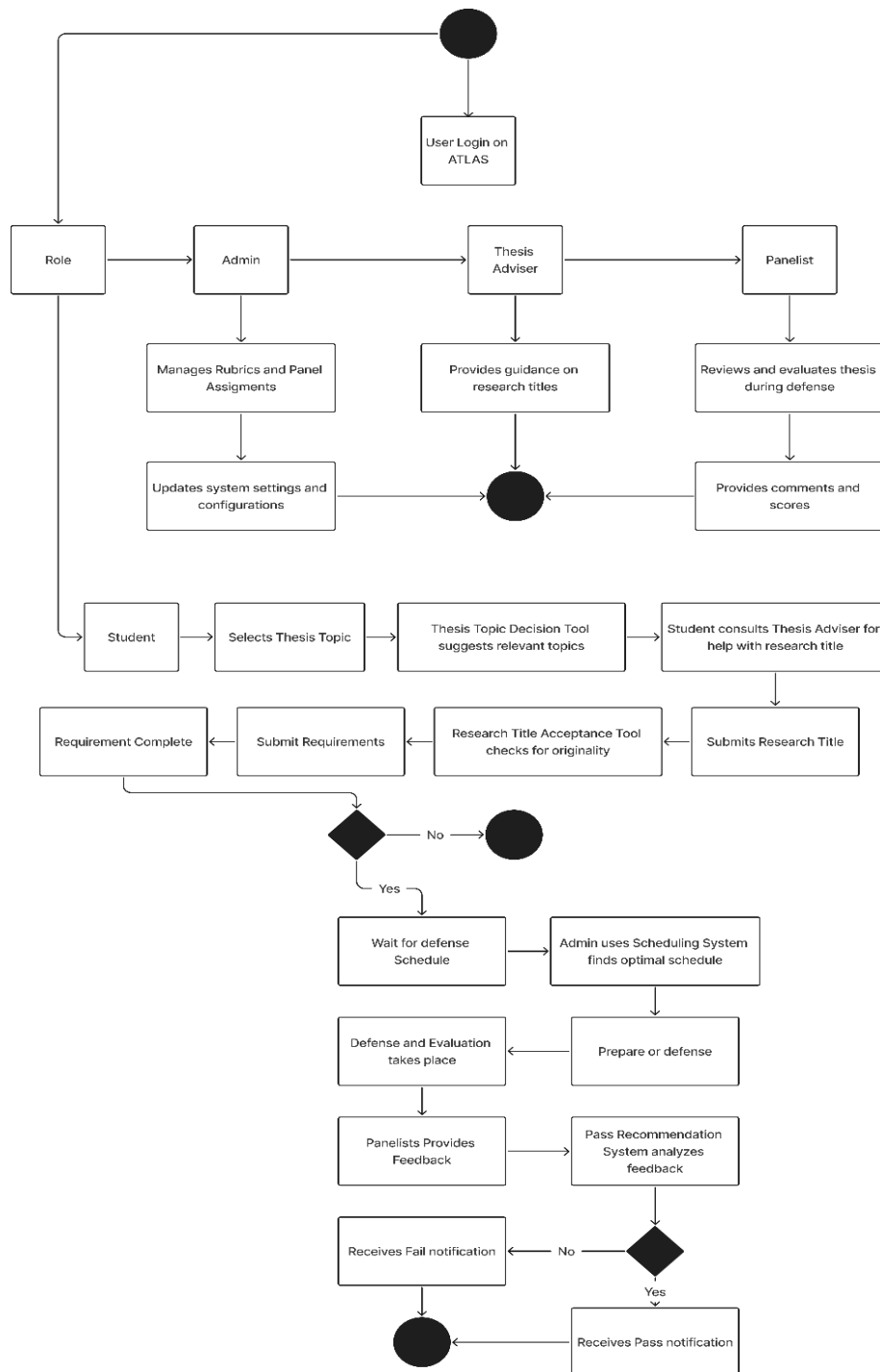
Sequence Diagram



The process starts with the student logging in to use the Thesis Topic Decision Tool, where AI generates tailored topic suggestions. After consulting their Thesis Adviser, the student submits the chosen title to the system for an AI-powered originality check. The system evaluates the title and notifies the student of the

Figure 12.

Activity Diagram



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	Students begin by selecting a thesis topic and using the Thesis Topic Decision Tool for suggestions. They consult their thesis adviser for guidance on their research title before submitting it for originality verification via the Research Title Acceptance Tool. Afterward, students complete and submit all requirements. Once all requirements are fulfilled, students request a defense schedule, and the system's scheduling tool assigns an optimal time slot. During the defense, panelists review and evaluate the thesis, providing feedback and scores. The Pass Recommendation System analyzes this feedback to determine the final result, notifying the student of a pass or fail outcome. Admins manage rubrics, panel assignments, and system configurations, while thesis advisers guide students on research and provide feedback during the defense. Panelists focus on thesis evaluation and scoring. This process ensures efficient collaboration and streamlines thesis management.	
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Data Analysis

20 students identify the key challenges faced in the research process. A notable 65% of them have difficulty getting their research titles approved, and 70% struggle with selecting a suitable research topic. This emphasizes the necessity for AI tools such as the Research Title Acceptance Tool and Thesis Topic Decision Tool to improve these areas. What's more, 70% of students encounter issues with timely submissions of requirements, while 65% face confusion or delays regarding defense schedules. AI-driven solutions like a requirement tracker and an automated scheduling system could effectively resolve these challenges. When it comes to choosing research topics, 70% favor AI-powered suggestion tools, and 80% use online resources such as Google Scholar. To ensure the uniqueness of titles, 55% prefer using AI tools, although 50% also appreciate feedback from peers or mentors, indicating that a hybrid approach may be advantageous. In managing document submissions, 45% opt for a manual checklist, while 40% prefer an AI-driven requirement tracker and guidance from faculty. This suggests that integrating AI with traditional methods could offer the most flexibility. Regarding evaluation tools, 55% find an AI-powered pass recommendation system quite useful, and 75% view a centralized system for managing defense requirements as highly beneficial. Opinions on AI-generated performance evaluations are varied; 45% find them very useful, while 25% still prefer traditional methods, indicating that AI should complement human evaluation rather than replace it.

Ethical Considerations

In the development of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation for LPU-Cavite, the following considerations are needed to ensure the protection of all stakeholders and the responsible use of AI technology.

1. Data Privacy and Confidentiality

- The proponents ensures that all the data/information gathered throughout the duration of this project will be handled with strict confidentiality. The proponents ensures that unauthorized users will not have access to this data. All participants will stay anonymous.

2. Informed Consent

- Before participating in this study, all participants are informed about the objectives and procedures involved. The proponents ensure that the participants fully understand, their rights and the voluntary nature of their involvement. Consent will be obtained through signed letters by the Dean of the department.

3. Intellectual Property and Plagiarism

- The system will respect and protect the intellectual property of both participants and the institution. Research titles, theses, and other academic content submitted by the institution and students will be

safeguarded within the system. The plagiarism detector will ensure originality and have no similar existing titles.

5. Cybersecurity and Protection from Harm

- ATLAS will be developed with strong cybersecurity protocols to prevent unauthorized access, data breaches, or system manipulation. Protection from harm also includes ensuring that the system does not expose participants to any form academic, emotional, or reputational harm through errors or misuse.

6. Bias and Fairness

- To address potential algorithmic bias in ATLAS, specific attention was given to ensure fairness in the Genetic Algorithm used for scheduling and Gemini LLM used in Thesis Topic Decision Tool, Research Title Acceptance Tool, and Pass Recommendation Tool. Actions were made to prevent any biases that might favor groups or lead to unfair outcomes.
- Transparency is also a guiding principle in this study. A clear documentation was provided on how the proponents gathered data, used the data in this study, and analyzed the data within ATLAS.

7. Accountability and Integrity

- The researchers and developers of ATLAS will maintain accountability throughout the system's design, deployment and operation. The research team also ensures that ATLAS operates with the highest level of academic integrity, supporting fair evaluation practices.

8. Collaboration Ethics

- ATLAS encourages collaboration between students, faculty, and the administrative staff while upholding ethical standards in all interactions. Contributions made by all participants during the systems development will be acknowledged. The researchers and developers of ATLAS ensures that the system will be used as a tool for academic growth while promoting responsible conduct in research and academic evaluation.

These ethical considerations ensures that the thesis project is conducted responsibly, safeguarding the interests of individuals, society, and the broader tech community.

7. Document defects or issues identified during testing for further review.
8. Prioritize and address system defects, resolving critical issues before retesting.
9. Retest the affected functionality to verify that previous issues have been fixed.
10. Evaluate the overall testing process by summarizing test results, assessing performance metrics and identifying areas for improvement.
11. Prepare and distribute test summary reports to relevant stakeholders to finalize system adjustments.

Evaluation

Software evaluation ensures that ATLAS meets its intended requirements, operates efficiently, and provides a seamless user experience. Following ISO/IEC 25010 standards, the system is assessed for functionality, usability, performance, security, and reliability. Key tests include verifying core features, evaluating user-friendliness, measuring system speed and stability, ensuring data security, and checking for continuous operation. The evaluation aims to identify areas for improvement, optimize performance, and enhance ATLAS's effectiveness in efficient thesis management at LPU-C.

Evaluation Instruments

ATLAS will be evaluated using the ISO/IEC 25010 standard to ensure software quality. The assessment will focus on eight key characteristics: functional suitability, reliability, performance efficiency, usability, security, compatibility, maintainability, and portability. These criteria will measure the system's effectiveness, efficiency, security, and adaptability. This evaluation guarantees that ATLAS meets academic and technical requirements, providing a seamless and reliable thesis management experience for LPU-C.

Evaluation Procedure

The ISO/IEC 25010 software quality framework will be used to assess ATLAS, ensuring its effectiveness, reliability, and usability. The evaluation will involve advisers, research panelists, IT experts, and student, representing a significant portion of the system's end-users. The results will determine whether ATLAS meets academic and technical standards while identifying areas for improvement.

The evaluation will be conducted online, following this process:

1. Define evaluation requirements, including hardware, software, and network configurations.
2. Invite software evaluators to assess the system.
3. Discuss the ISO/IEC 25010 quality characteristics and explain the evaluation purpose.
4. Distribute evaluation forms to respondents for feedback.

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5. Compare the results against software quality criteria to measure compliance.
6. Prepare a detailed evaluation report, summarizing strengths and areas for improvement.
7. Document recommendations to enhance system performance and usability.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the actual output of the system, analysis obtained from evaluation forms given to respondents of Lyceum of the Philippines University – Cavite. The research findings are discussed in this section.

Project Design

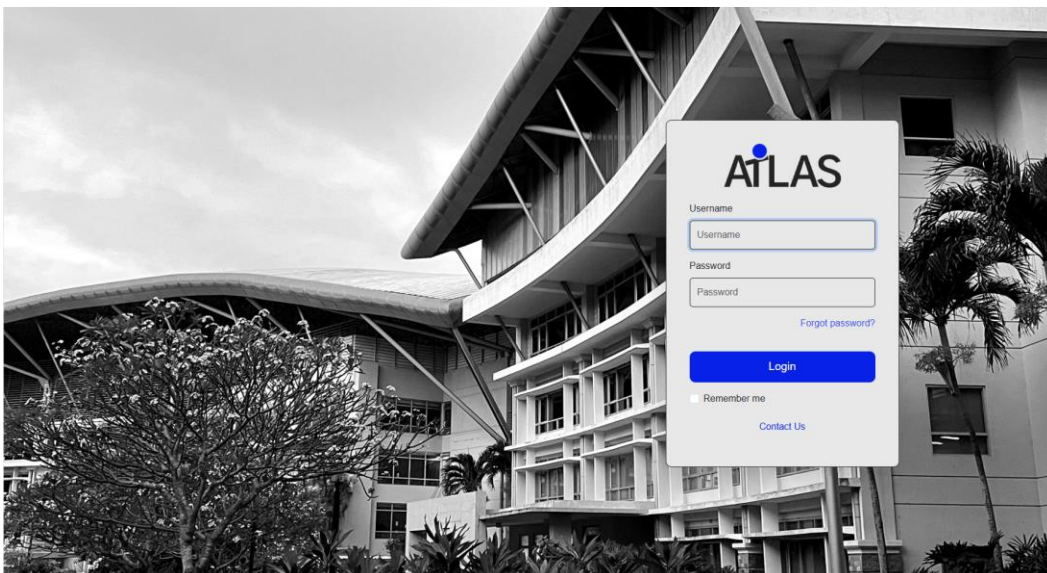
The software project's look and feel are the focus of the user interface design. This also addresses the project's visual appeal, usability, and simplicity.

Module 1: Login Page

The page allows users to login to the system.

Figure 13

Log in module of ATLAS



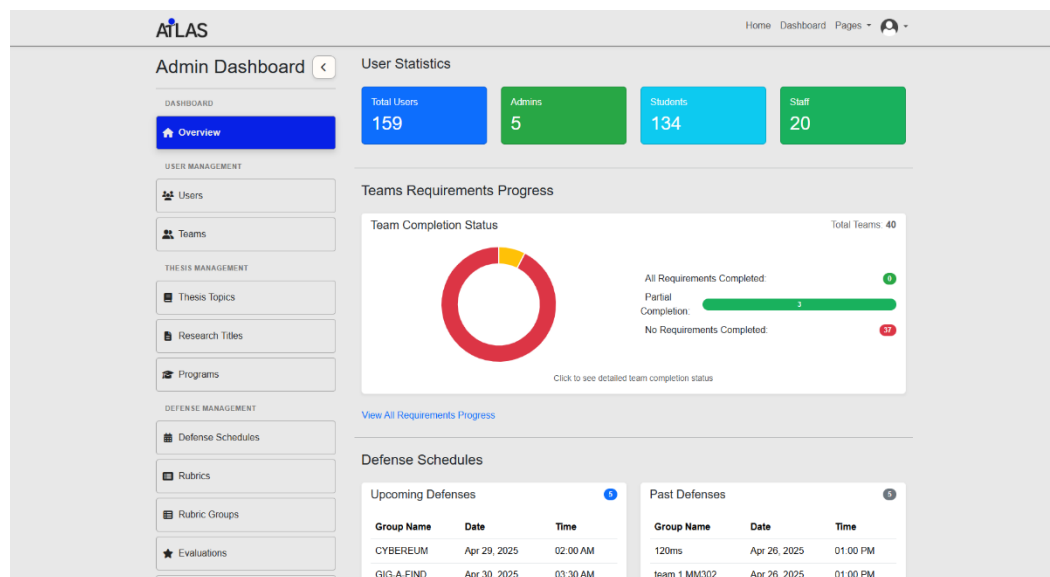
There are two (2) input fields which are the username and password. The system verifies the input of the user if the user exists or if the password is wrong and will tell via a message under each of the forms. Once verified, the system redirects the users to their respective dashboards based on their user type such as: admin, staff, and students. Users who forgot their account password may click the link to have the system email a password reset link to reset their password anytime.

Module 2: Admin Overview

This tab allows administrators to view information about user statistics, team requirements progress, and defense schedules, for the administrators monitoring.

Figure 14

Admin Overview Preview Module of the Admin Dashboard



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The Overview Tab displays details about the users, teams, and defense schedules. The data shown for the Users Statistics are its total count and per user type count. The Teams Requirements Progress displays details of teams which are their total count, the amount of progress of the Teams in a chart and numerical form in its requirement compliance, and the progress of teams per requirement. The Defense Schedules section displays a quick overview of a team's defense schedule with the team's name, team's adviser, and date and time of the scheduled defense for the teams.

Module 3: Admin Users

This tab display allows the administrators to manage the users of the system.

Figure 15

Admin Users Tab Module of the Admin Dashboard

Username	Email	First Name	Last Name	User Type	Action
2021-2-01881	2021-2-01881@lpunetwork.edu.ph	VINCE LAWRENCE N/A	GARGALLO	Student	Edit Delete
2020-2-02151	2020-2-02151@lpunetwork.edu.ph	DANE ISAAH NOVA	EYAYA	Student	Edit Delete
2022-2-02621	2022-2-02621@lpunetwork.edu.ph	LINDSAY BORRES	ESCALA	Student	Edit Delete
2022-2-02842	2022-2-02842@lpunetwork.edu.ph	SHANE ANGELIC ARAYAS	DIAZ	Student	Edit Delete
2021-2-01946	2021-2-01946@lpunetwork.edu.ph	MICHAEL ONEAL TABIANTE	DESPI	Student	Edit Delete
2021-2-01472	2021-2-01472@lpunetwork.edu.ph	CEDRIC JAMES ROXAS	DELOS REYES	Student	Edit Delete
2021-2-02577	2021-2-02577@lpunetwork.edu.ph	ROLANDO BI MACAJANGAY	CORONADO	Student	Edit Delete
2017-2-02314	2017-2-02314@lpunetwork.edu.ph	RAFAEL CARLO SUGUI	CAMPO	Student	Edit Delete
2021-2-00326	2021-2-00326@lpunetwork.edu.ph	MILES ANDREI LUPISAN	CAJAZAN	Student	Edit Delete
2019-2-02303	2019-2-02303@lpunetwork.edu.ph	SHANEY LOUISSE	BAUZON	Student	Edit Delete

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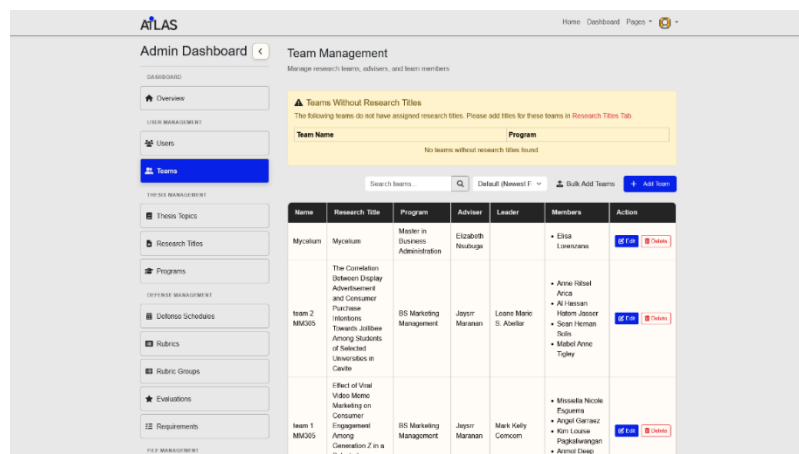
The Users Tab displays all the created users categorized in tabs for each type of user. The admin can create users individually and add students in bulk using a csv file, pasting csv data, or inputting multiple students at a time. The admin can edit the data of the users such as their user type, username, email, first and last name, gender, headline, bio, and their programs. Additionally for the editing of staff details, the admin can edit the area of expertise and employment type. The admin can also delete users if necessary. The admin can also sort the table of users according to its date created, username, first name, and last name accordingly in both ascending and descending formats. There is also a search function to find the users needed.

Module 4: Admin Teams

This tab display allows the administrators to manage the teams of the system.

Figure 16

Admin Teams Tab Module of the Admin Dashboard



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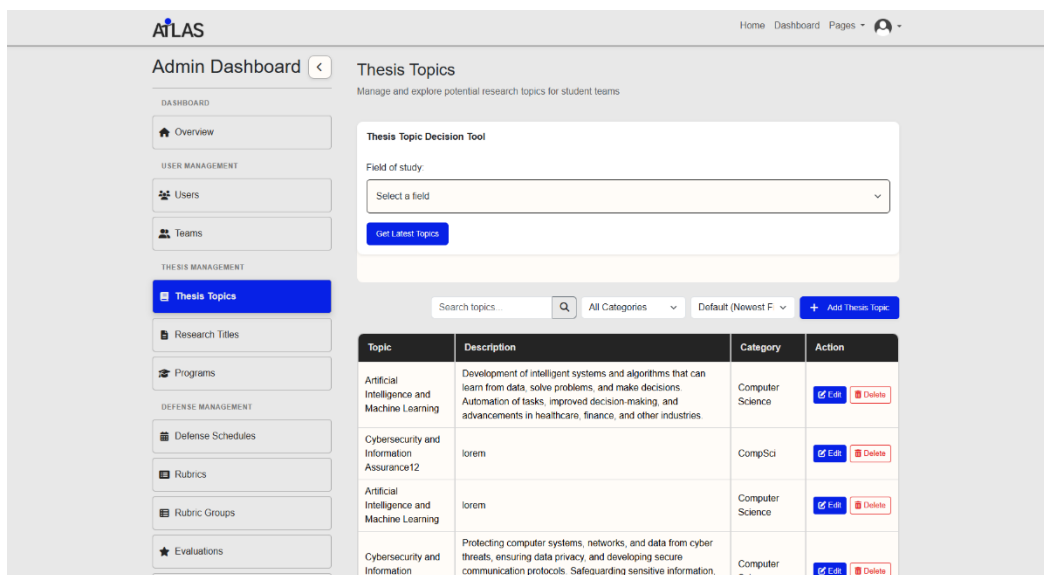
This tab displays all the teams created. The admin can create teams individually and teams in bulk using a csv file, pasting csv data, or inputting multiple teams at a time. The admin can edit the teams data such as the team name, research title, area of expertise, program, and team members and their roles. The admin can also delete teams if necessary. The admin can also sort the table of teams according to its date created, and team name accordingly in both ascending and descending formats. There is also a search function to find the teams needed.

Module 5: Admin Thesis Topics

Admin Thesis Topics Module. This tab allows the administrators to view and manage the topic ideas in creating a thesis.

Figure 17

Admin Thesis Topics Tab Module of the Admin Dashboard



LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

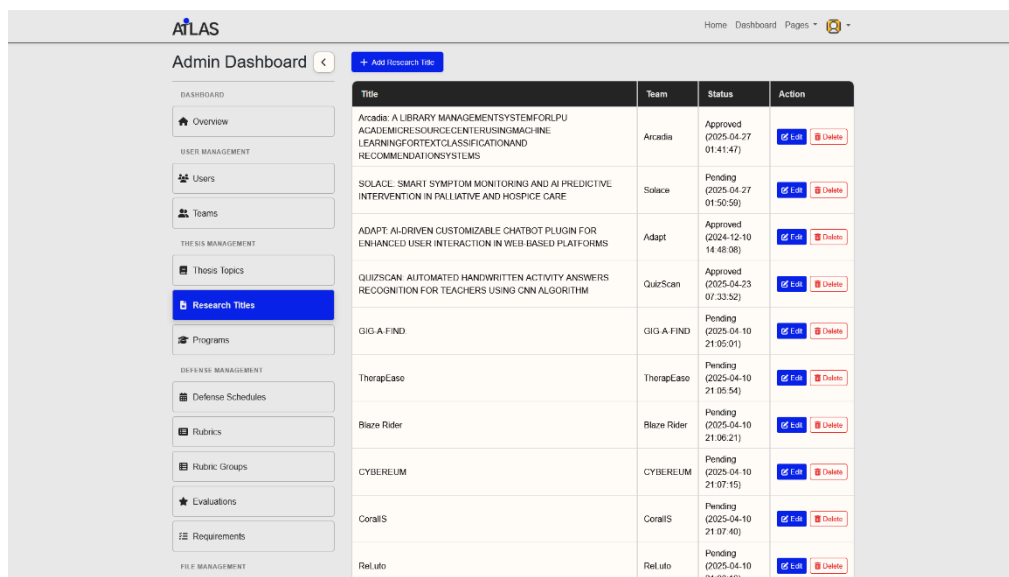
This tab displays all the approved topics for ideas for the student's thesis creation. The admin can select a field of a topic, and the system will get topics from academic sources like google scholar and gives the admin an option to add the topics given or not. The admin can also edit the contents of topic added such as its topic, description, and category. The admin can also delete the topics if necessary. Topics can also be sorted per category, date added, topic name alphabetically, or its category alphabetically, both in ascending and descending format. There is also a search function to find the topics needed.

Module 6: Admin Research Titles

Admin Research Titles Module. This tab allows the administrator to view and manage the research titles.

Figure 18

Admin Research Titles Tab Module of the Admin Dashboard



Title	Team	Status	Action
Arcade: A LIBRARY MANAGEMENT SYSTEM FOR LPU ACADEMIC RESOURCE CENTER USING MACHINE LEARNING FOR TEXT CLASSIFICATION AND RECOMMENDATION SYSTEMS	Arcadia	Approved (2025-04-27 01:41:47)	Edit Delete
SOLACE: SMART SYMPTOM MONITORING AND AI PREDICTIVE INTERVENTION IN PALLIATIVE AND HOSPICE CARE	Solace	Pending (2025-04-27 01:50:50)	Edit Delete
ADAPT: AI-DRIVEN CUSTOMIZABLE CHATBOT PLUGIN FOR ENHANCED USER INTERACTION IN WEB-BASED PLATFORMS	Adapt	Approved (2024-12-10 14:48:08)	Edit Delete
QUIZSCAN: AUTOMATED HANDWRITTEN ACTIVITY ANSWERS RECOGNITION FOR TEACHERS USING CNN ALGORITHM	QuizScan	Approved (2025-04-23 07:33:52)	Edit Delete
GIG A FIND	GIG A FIND	Pending (2025-04-10 21:05:01)	Edit Delete
TherapEaso	TherapEaso	Pending (2025-04-10 21:05:54)	Edit Delete
Blaze Rider	Blaze Rider	Pending (2025-04-10 21:06:21)	Edit Delete
CYBERELUM	CYBERELUM	Pending (2025-04-10 21:07:15)	Edit Delete
CoralS	CoralS	Pending (2025-04-10 21:07:40)	Edit Delete
Reluto	Reluto	Pending (2025-04-10 21:08:01)	Edit Delete

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

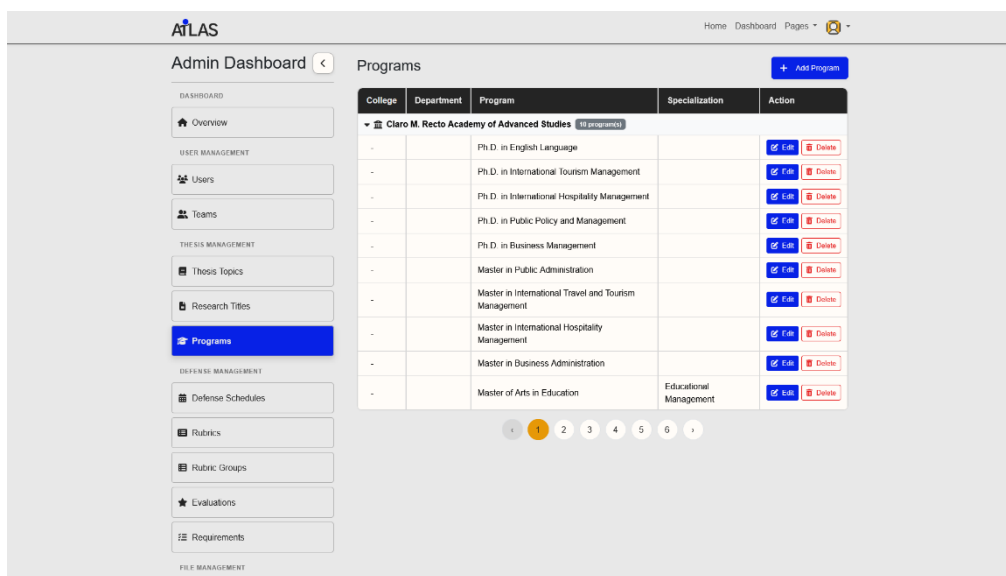
This tab lists the titles for each team along with its status of either approved or pending. The admin can also edit the title of the paper for corrections and change the status of the paper for approval. The admin can also delete the research titles if necessary.

Module 7: Admin Programs

At this tab allows administrators to view and manage the programs of the colleges.

Figure 19

Admin Programs Tab Module of the Admin Dashboard



This tab allows admins to view the list of programs available and manage the programs. The admins can add a program by filling up information such as the

college, department, program name, and parent program ID for sub programs. The admins can delete programs if necessary.

Module 8: Admin Defense Schedules

This tab allows administrators to create and manage defense schedules.

Figure 20

Admin Defense Schedules Tab Module of the Admin Dashboard

Date & Time	Team	Adviser	Thesis Title	Panelist 1	Panelist 2	Panelist 3	Room
Apr 29, 2025 7:00 AM - 8:00 AM	QuizScan	Raymund Constante	QUIZSCAN: AUTOMATED HANDWRITTEN ACTIVITY ANSWERS RECOGNITION FOR TEACHERS USING CNN ALGORITHM	Leamie Carlos	Joven Cagges	Elizabeth Neubuge	bob
Apr 30, 2025 7:00 PM - 8:30 PM	CYBERELUM	Gerson Mendoza	CYBERELUM	Sean Charleston Gono	Toni Granada	Jerian Peren	#
Apr 27, 2025 4:00 AM - 5:00 AM	Team 1 MM302	Vincent Cortinas	The Influence of Corporate Social Responsibility Programs of a Legacy Brand in Tagaytay City Towards Customer Loyalty	Sean Charleston Gono	Toni Granada	Jerian Peren	#
Apr 28, 2025 2:00 PM - 6:00 PM	120ms	Sean Charleston Gono	aaaaaaaaaaaa	Sean Charleston Gono	N/A	N/A	N/A
Apr 28, 2025 2:00 PM - 5:00 PM	120ms	Sean Charleston Gono	aaaaaaaaaaaa	Leah Santos	N/A	N/A	N/A
			The Influence of Corporate				

This tab allows admins to create defense schedules by entering the desired settings from: the rooms to be used, duration of each defense, start and end time in a day, day/s for the defense, and choose for what programs the defense is. There is also an option to manually add a defense schedule to the existing schedule and the system will validate if the manually added can fit to the existing schedule set. The admins can also edit the schedule such as the schedule data, start and end time, the

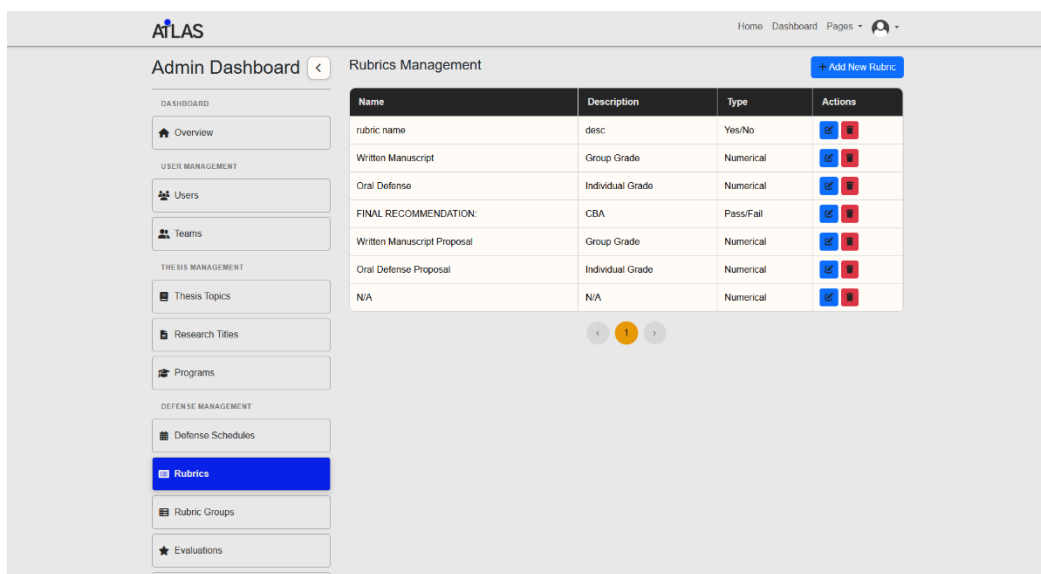
room the defense will be conducted, the team, and the panelists. The admin can also delete schedules if necessary.

Module 9: Admin Rubrics

Admin Rubrics Module. This tab allows administrators to manage rubrics for the different types of defenses and rubric types.

Figure 21

Admin Rubrics Tab Module of the Admin Dashboard



This tab allows admins to create rubrics based on the defense type: Title Defense, Proposal Defense, Final Defense, and Re-defense, and also the rubrics type: numerical scoresheet, rubrics with yes/no option, and rubrics with pass/fail option. Admins can also configure the name of the rubrics, its description, the applicable programs for the rubrics, and the number of rows and the respective content for each row for each criterion added. The edit function allows admins to

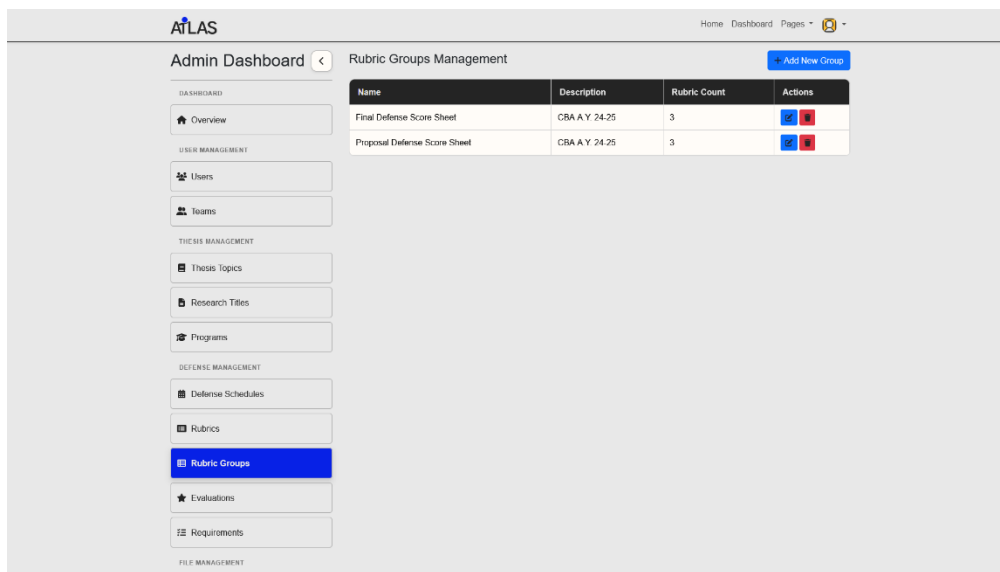
edit existing data in the same manner as adding it, but with the created data for the rubrics type. The admins can also delete rubrics if necessary.

Module 10: Admin Rubrics Group

This tab allows administrators to manage rubrics and group them accordingly to the type of defense needed.

Figure 22

Admin Rubrics Group Tab Module of the Admin Dashboard



This tab allows admins to create a group of rubrics from the created rubrics in the Rubrics Tab. This allows flexibility in creating scoresheets for the type of rubrics a college or departments needs. The admin can freely edit the rubrics group accordingly such as its name, description, and what type of rubrics are in the group. The admin can also delete rubrics group if necessary.

Module 11: Admin Evaluations

This tab allows administrators to view the details about the evaluations of the panelists.

Figure 23

Admin Evaluations Tab Module of the Admin Dashboard

Team Name	Evaluator	Student	Group Score	Individual Score	Total Score	Comments
120ms	Sean Charlston Oloro	van Kerwin Ilano	60	30.6667	90.6667	
120ms	Sean Charlston Oloro	Jerald Ryan Geronia	60	40	100	
120ms	Sean Charlston Oloro	Winstonini Pagenini	60	40	100	

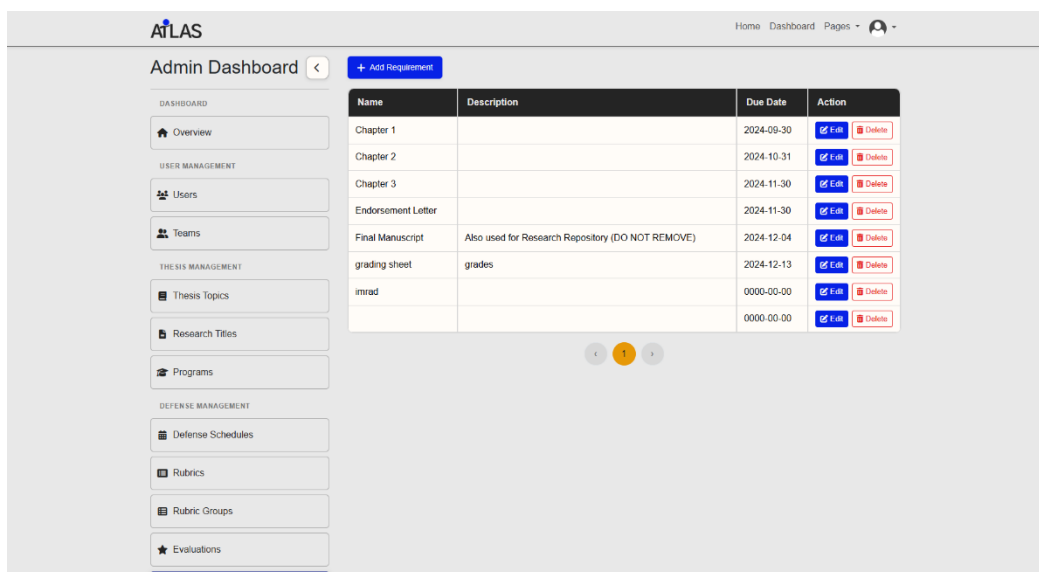
This tab allows admins to view the evaluations made throughout the defenses. An option to change view to be a detailed view with all information displayed or a simplified view which focuses on a group's scores.

Module 12: Admin Requirements

This tab allows administrators to manage the requirements of students for the thesis and the defense.

Figure 24

Admin Requirements Tab Module of the Admin Dashboard



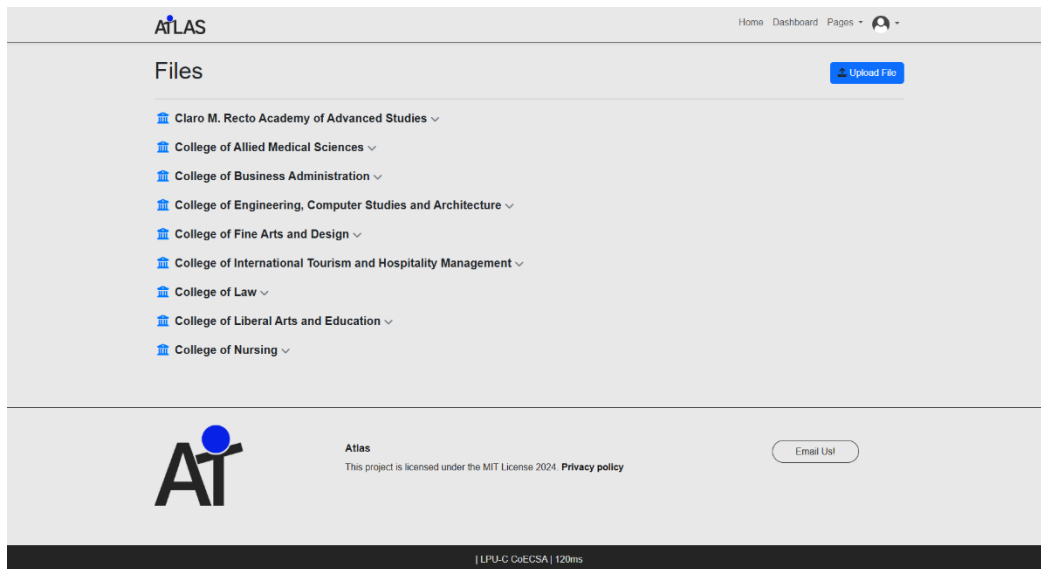
This tab allows admins to create the requirements needed the students need to complete for the subject and as a defense requirement and set due dates and descriptions or instructions for it. Editing the requirements allows admins to freely edit its name, description, and due dates. The admins can also delete the requirements if necessary.

Module 13: Admin Files

This tab allows administrators to manage the research files for each college.

Figure 25

Admin Files Tab Module of the Admin Dashboard



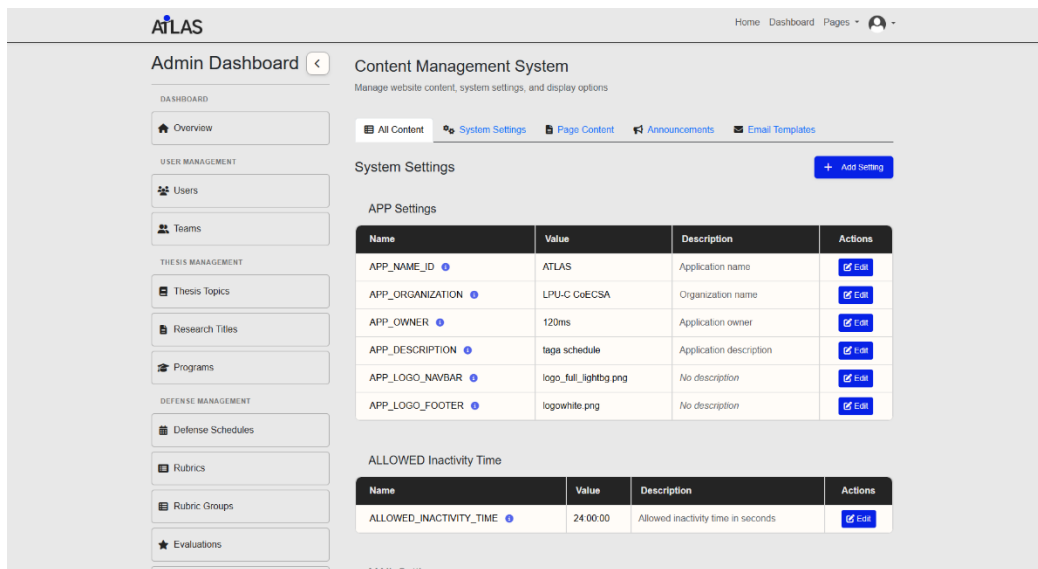
This page allows admins to upload research papers published for each college and assign them to a specific program and give it a description. It functions as an online repository for the published research papers.

Module 14: Admin Content Management

This allows administrators to manage the website details.

Figure 26

Admin Content Management Tab Module of the Admin Dashboard



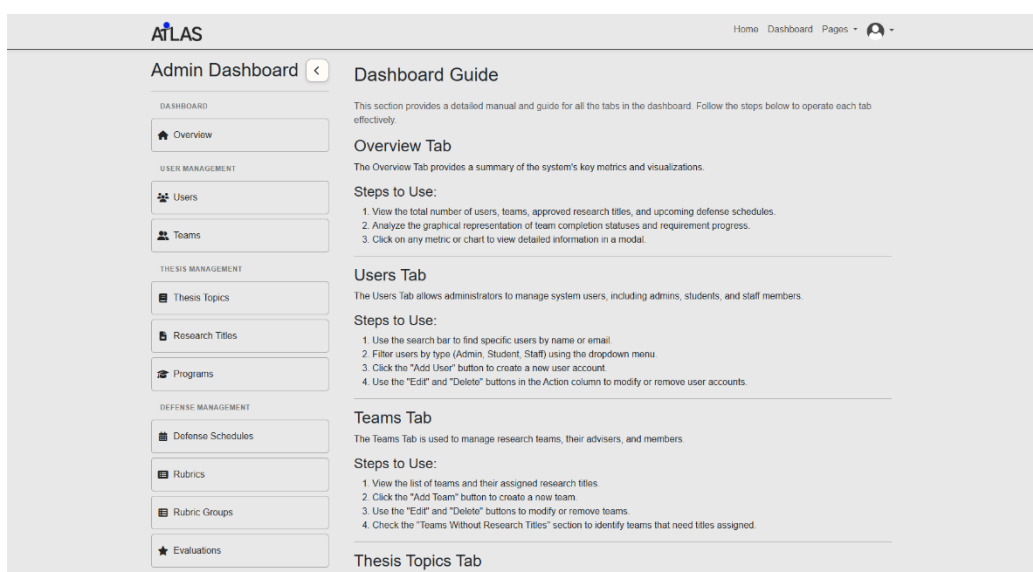
This tab allows admins to manage the website settings as well as the pages and its content. Admins can add a system setting by inputting the desired key, value, and description for the website. Additionally, the admin can create and publish pages with a WYSIWYG text editor for customization of the page content. The admin can delete the pages created in the page content tab if necessary.

Module 15: Admin Guide

This tab provides administrators with a simple guide on how to operate each tab in the dashboard.

Figure 27

Admin Guide Tab Module of the Admin Dashboard



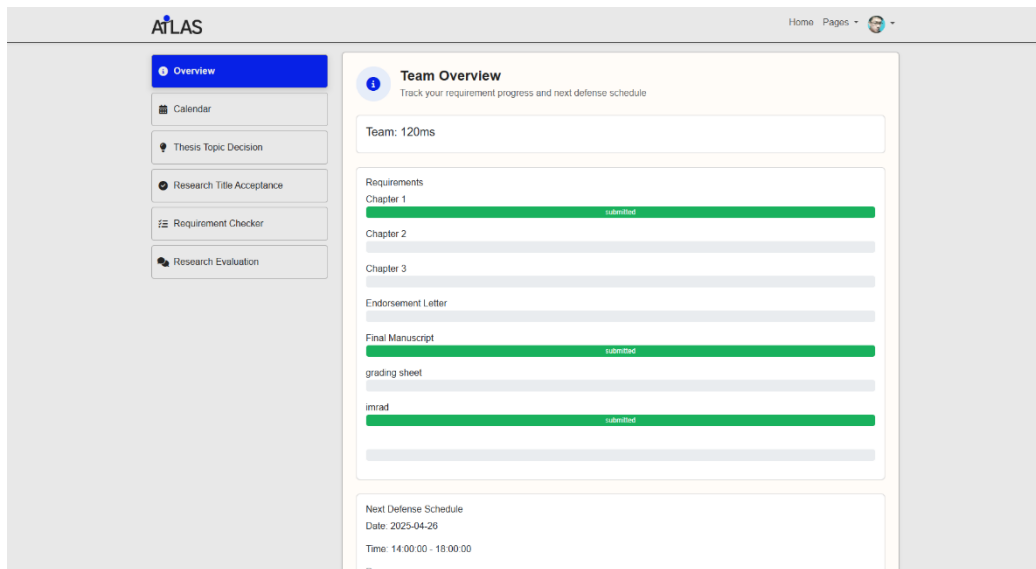
The Guide Tab provides instructions and descriptions for the admins who may fail to recall some details on how to operate the functions for each tab of the admin dashboard.

Module 16: Staff Overview

Staff Overview Module. This tab of the staff dashboard allows research advisers to view and monitor their team progress in the requirements and also the next defense schedule for their team.

Figure 28

Staff Overview Tab Module of the Staff Dashboard



This tab provides research advisers with a quick display to view and monitor their team's requirements, they can also view other teams requirements if they are assigned to multiple teams. There is also a small section which displays the next defense schedule for their team with its date and time, and the room where they are assigned.

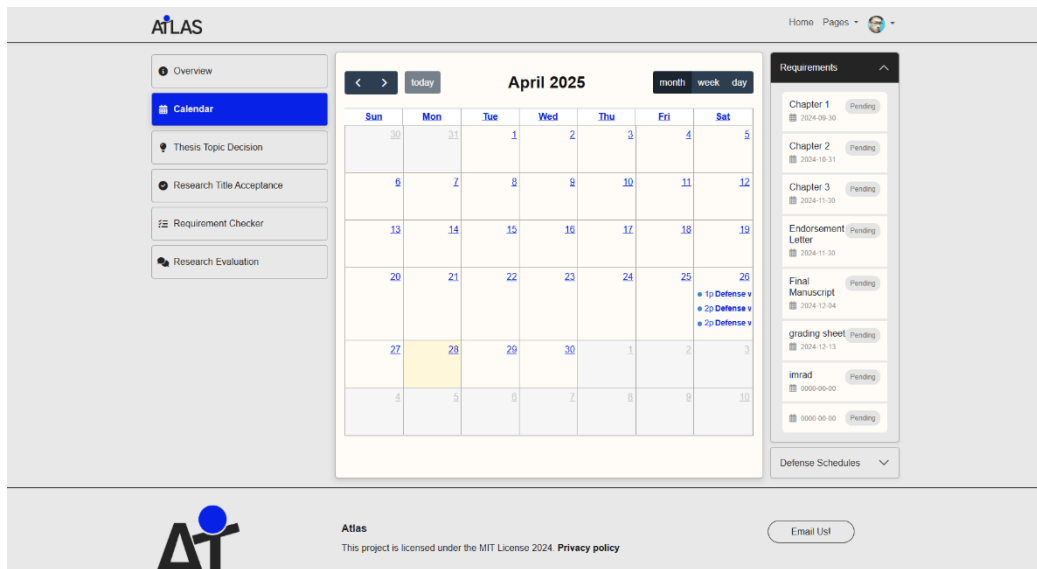
Module 17: Staff Calendar

This tab allows a quick display of the defense schedule and the requirements.

Figure 29

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

Staff Calendar Tab Module of the Staff Dashboard



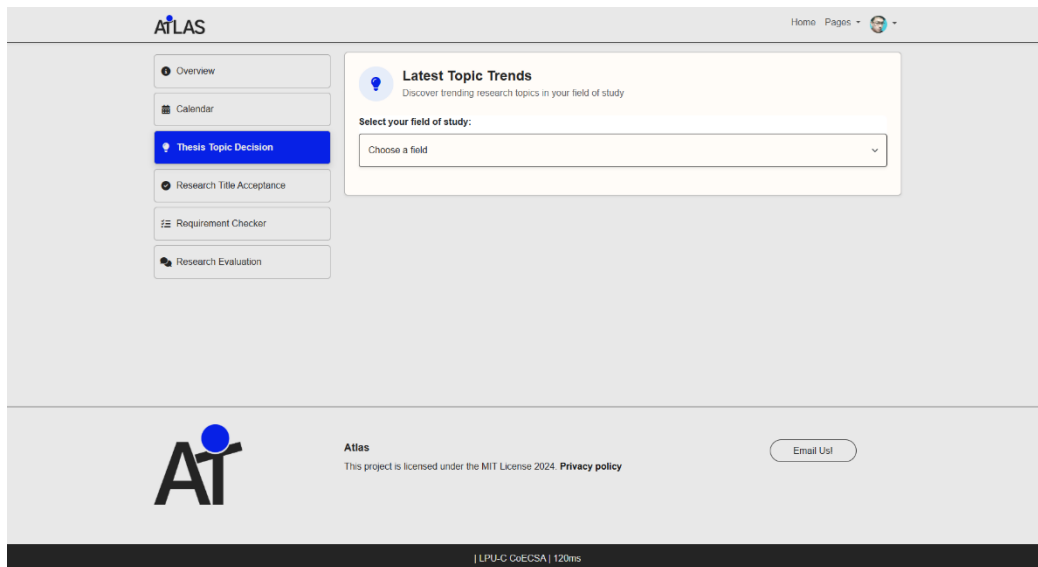
This tab displays a calendar view which also displays the defense schedules scheduled for them. It comes with the view of their team requirements and their status. Defense schedules are also displayed in the same way for an easier view.

Module 18: Staff Thesis Topic Decision

This tab allows research advisors to check for the latest approved thesis topic ideas by the administrators.

Figure 30

Staff Thesis Topic Decision Tab Module of the Staff Dashboard



This tab allows research advisors to view the different latest topic trends and ideas for thesis creation for their group which are approved by the admins. The dropdown menu are options for different fields of interest such as Architecture, Information Technology, Computer Science, Aeronautical Engineering, etc.

Module 19: Staff Research Title Acceptance

This tab allows research advisors to check their team's title and get Artificial Intelligence (AI) insights about it.

Figure 31

Staff Research Title Acceptance Tab Module of the Staff Dashboard

The screenshot displays the ATLAS Staff Research Title Acceptance Tool interface. On the left, a sidebar menu includes options: Overview, Calendar, Thesis Topic Decision, **Research Title Acceptance** (highlighted in blue), Requirement Checker, and Research Evaluation. The main content area is titled "Research Title Acceptance Tool" with a sub-header "Check the uniqueness of your research title and get AI-powered suggestions". It features a "Proposed Research Title" input field, a "Research Field" dropdown menu (showing "e.g., Computer Science"), and a "Problem Statement" text area (with the placeholder "Brief description of the problem"). A blue "Check Title" button is positioned below the input fields. The footer of the interface includes the ATLAS logo, the text "Atlas This project is licensed under the MIT License 2024. Privacy policy", an "Email Us!" button, and a footer bar with the text "[LPU-C CoECSA] 120ms".

This tab allows research advisors to check their teams' current titles, get AI insights. It also checks if the paper has similar topics by browsing online repository of the system and from research sites like Google Scholar. The input fields are to give AI about the context of the research project, users can enter on all the input or even just the title and another either of the input and the system will be able to provide AI feedback and insights about its analysis of the title or even give suggestions on alternatives for the title.

Module 20: Staff Requirement Checker

This tab allows research advisors to check for their teams' requirement progress and submissions.

Figure 32

Staff Requirement Checker Tab Module of the Staff Dashboard

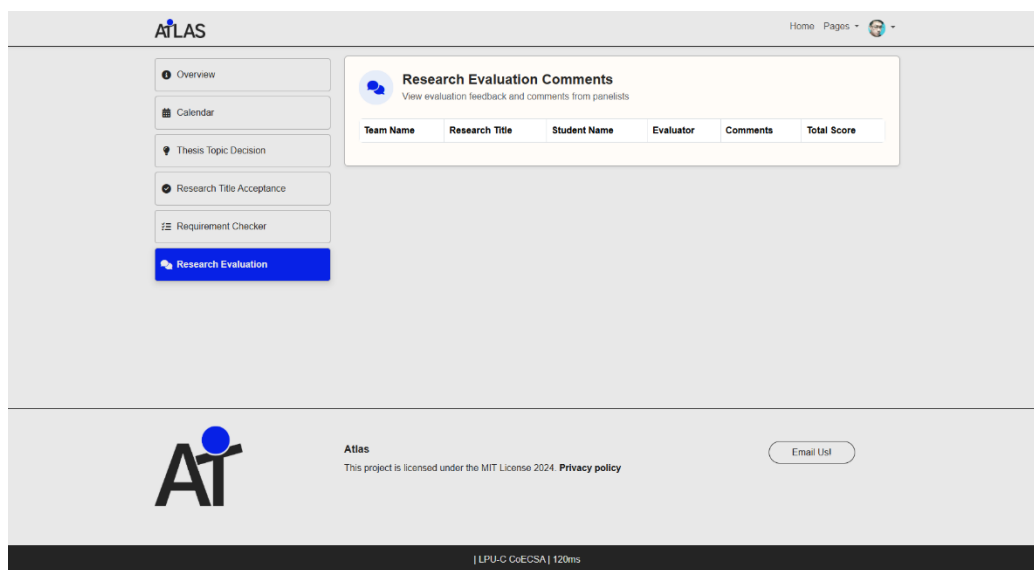
This tab allows research advisors to check their teams' requirements progress and submissions and manage them. Advisors can change the status of the requirements, download or view the submitted file, and provide annotated file for feedback for its students.

Module 21: Staff Research Evaluation

This tab allows research advisors to view the feedback from the defense for their team.

Figure 33

Staff Research Evaluation Tab Module of the Staff Dashboard



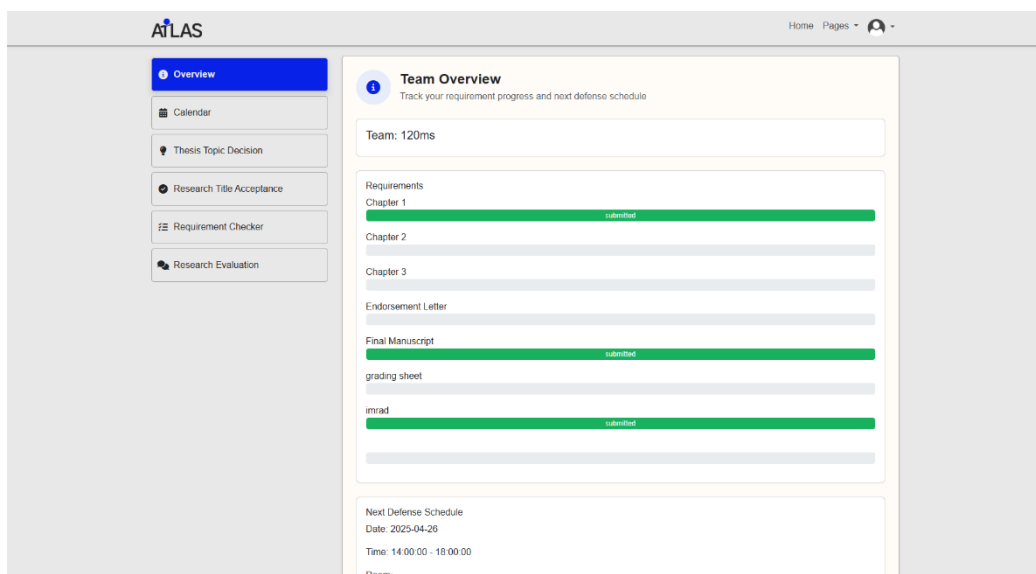
This tab allows research advisors to see the evaluations made during the defense for their team. It provides a table view with necessary information such as team name, the research title, members of the group, the evaluators, comments, and the total score for their team.

Module 22: Student Overview

This tab of the student dashboard allows students to view and monitor their team progress in the requirements and the next defense schedule for their team.

Figure 34

Students Overview Tab Module of the Students Dashboard



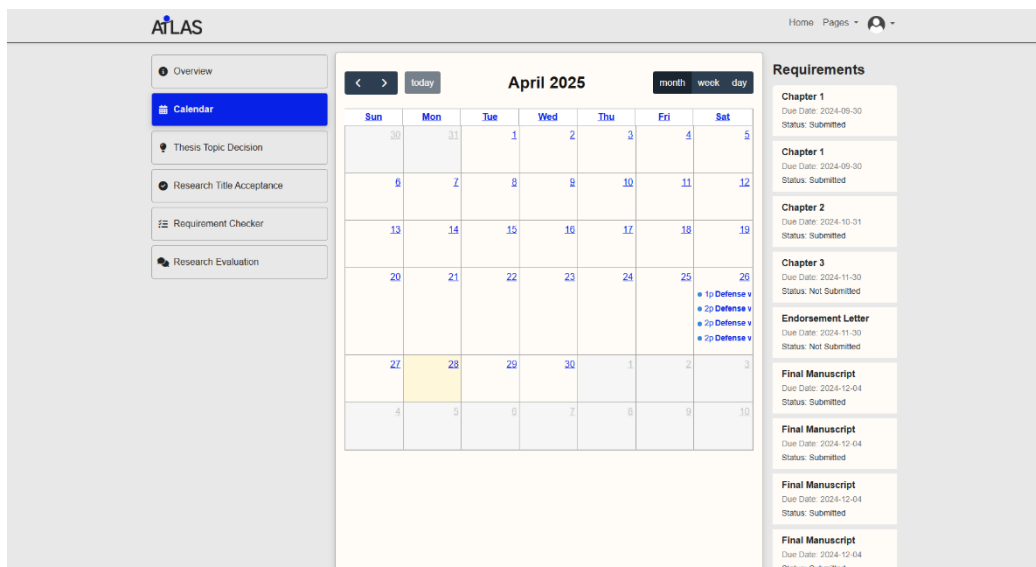
This tab provides students with a quick display to view and monitor their team's requirements, they can also view other team's requirements if they are assigned to multiple teams. There is also a small section which displays the next defense schedule for their team with its date and time, and the room where they are assigned.

Module 23: Student Calendar

Student Calendar Module. This tab allows a quick display of the defense schedule and the requirements.

Figure 35

Student Calendar Tab Module of the Student Dashboard



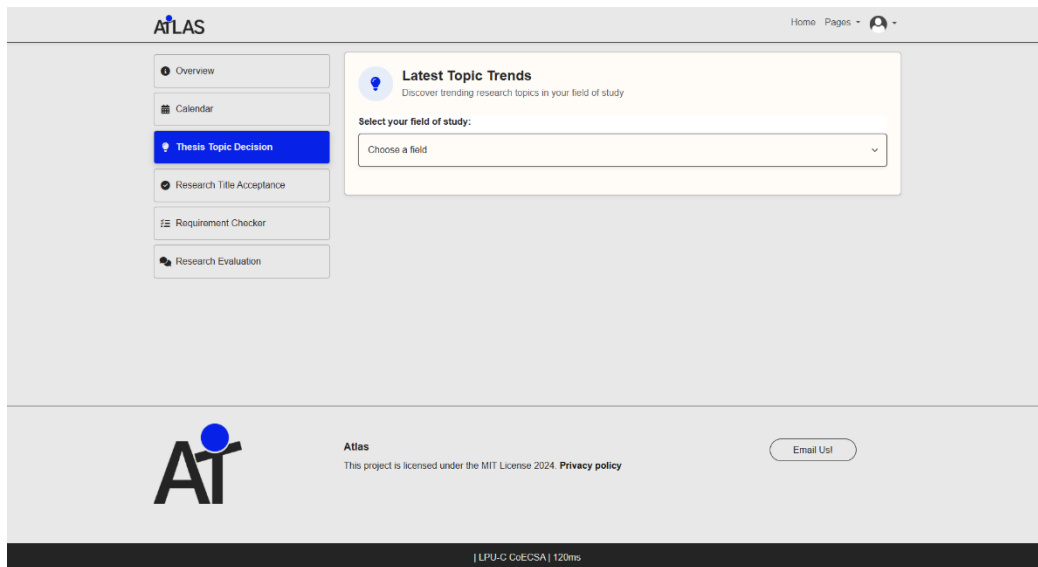
This tab displays a calendar view which also displays the defense schedules scheduled for them. It comes with a view of their team requirements and their status and due dates.

Module 24: Student Thesis Topic Decision

This tab allows students to check for the latest approved thesis topic ideas by the administrators.

Figure 36

Student Thesis Topic Decision Tab Module of the Student Dashboard



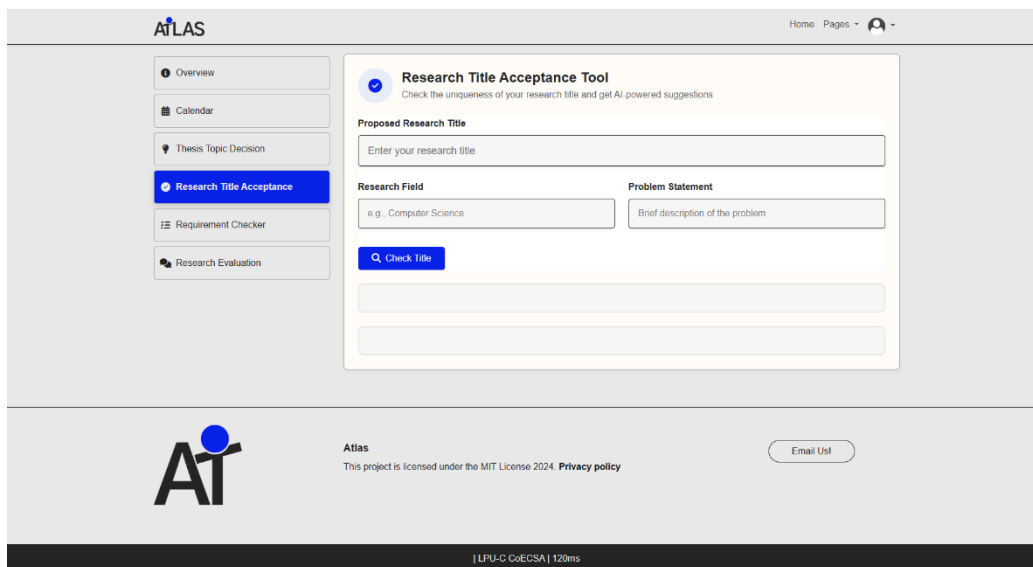
This tab allows students to view the different latest topic trends and ideas for thesis creation for their group which are approved by the admins. The dropdown menu options are for different fields of interest such as Architecture, Information Technology, Computer Science, Aeronautical Engineering, etc.

Module 25: Student Research Title Acceptance

This tab allows students to check their team's title and get Artificial Intelligence (AI) insights about the topic.

Figure 37

Student Research Title Acceptance Tab Module of the Student Dashboard



The screenshot displays the 'Research Title Acceptance Tool' interface within the ATLAS system. On the left, a sidebar menu includes options like Overview, Calendar, Thesis Topic Decision, Research Title Acceptance (highlighted), Requirement Checker, and Research Evaluation. The main content area features a form with a 'Proposed Research Title' input field, a 'Research Field' dropdown (with 'e.g., Computer Science' as an example), and a 'Problem Statement' text area. A 'Check Title' button is positioned below the input fields. The footer contains the ATLAS logo, a license notice ('This project is licensed under the MIT License 2024'), a privacy policy link, and an email list button.

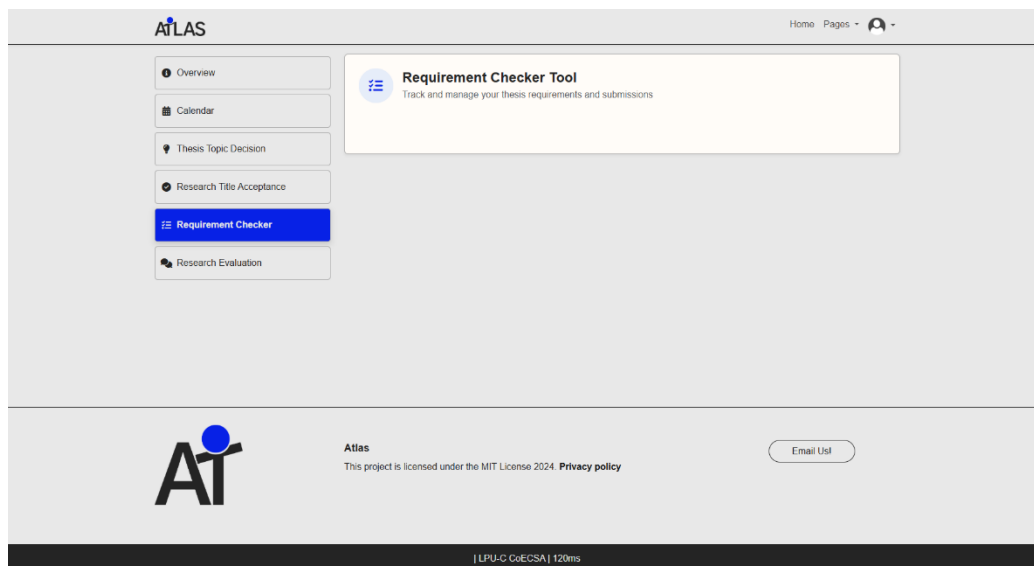
This tab allows students to check their teams' current titles, get AI insights. It also checks if the paper has similar topics by browsing online repository of the system and from research sites like Google Scholar. The input fields are to give AI about the context of the research project, users can enter on all the input or even just the title and another either of the input and the system will be able to provide AI feedback and insights about its analysis of the title or even give suggestions on alternatives for the title.

Module 26: Student Requirement Checker

This tab allows students to check for their teams' progress and submissions.

Figure 38

Student Requirement Checker Tab Module of the Student Dashboard



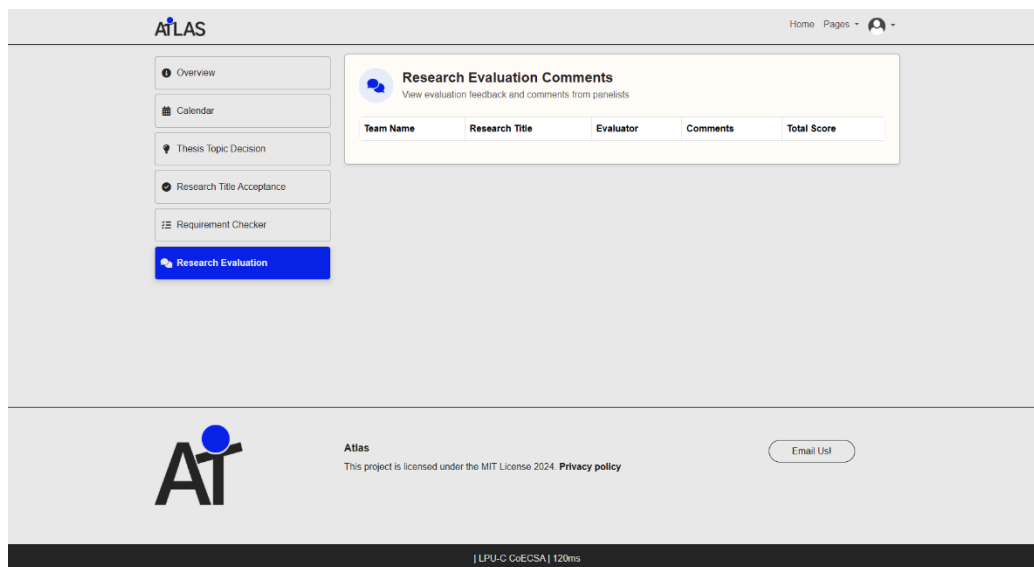
This tab allows students to check their team's requirements and submissions. Team leaders of the team are able to submit files, while all members can view the status and feedback of the requirements.

Module 27: Student Research Evaluation

This tab allows students to view the feedback from the defense for their team.

Figure 39

Student Research Evaluation Tab Module of the Student Dashboard



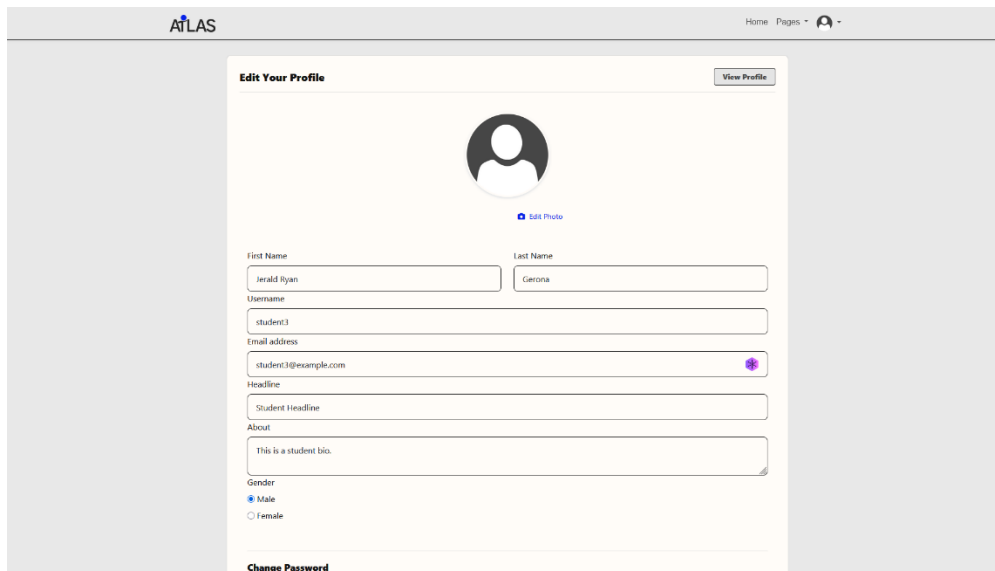
This tab allows students to see the evaluations made during the defense for their team. It provides a table view with necessary information such as team name, the research title, members of the group, the evaluators, comments, and the total score for their team.

Module 28: User Profile

This page allows users to manage their profile details.

Figure 40

User Profile Page Module



The screenshot displays the 'Edit Your Profile' interface within the ATLAS system. The page features a header with the ATLAS logo and navigation links. The main content area is titled 'Edit Your Profile' and includes a 'View Profile' button. A profile picture placeholder is shown with an 'Edit Photo' link. Below this, there are input fields for 'First Name' (Jerald Ryan), 'Last Name' (Gerona), 'Username' (student3), 'Email address' (student3@example.com), 'Headline' (Student Headline), 'About' (This is a student bio.), and 'Gender' (Male). A 'Change Password' link is located at the bottom of the form.

Figure 40.1

User Profile Page

The screenshot displays a user profile page with a light gray background. The main content area is a white card with rounded corners. At the top, there's a 'Student Headline' input field. Below it is an 'About' section with a text area containing 'This is a student bio.' and a small edit icon. The 'Gender' section has two radio buttons: 'Male' (selected) and 'Female'. The 'Change Password' section includes four input fields: 'Current Password', 'New Password', 'Confirm Password', and another 'Confirm Password' field. A blue 'Save Changes' button is at the bottom of the card. The footer features the 'Atlas' logo, a license notice 'This project is licensed under the MIT License 2024. Privacy policy', an 'Email Us!' button, and a status bar 'LPU-C CoECSA | 120ms'.

This page allows the users: admin, staff, and students to modify their details freely, input fields are provided to edit their existing details. Users can also change their password by entering the current, the new password, and the confirmation of the new password.

Table 5. Demographic Profile of Respondents

Profile	Sub-variables	Frequency (n=364)	Percent (100%)
Classification	IT Experts	10	2.7
	Subject-Matter Experts	11	3
	Students	343	94.2
Age	22 and below	177	48.6
	23 and above	187	51.4
Sex	Female	181	49.7
	Male	183	50.3
Participant's Category	Admin	8	2.2
	End-User	346	95.1
	IT Expert	10	2.7

The majority of the respondents were students (94.2%), with IT and subject-matter experts comprising a small percentage. The respondents were balanced in terms of sex. This distribution indicates that the evaluation results mainly reflect the perspective of actual users of the system.

Table 6. Level of Satisfaction of the respondents about the Different Features of the Software in terms of User Interface

Indicator	Respondent Group	Mean	Standard Deviation	Description
Navigation	IT- Experts	3.70	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.55	0.53	Highly Acceptable
	Overall	3.69	0.47	Highly Acceptable
Visual Design	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.45	0.82	Highly Acceptable
	Students	3.52	0.55	Highly Acceptable
	Overall	3.46	0.63	Highly Acceptable
Labeling and Icons	IT- Experts	3.50	0.53	Highly Acceptable
	Subject-Matter Experts	3.64	0.67	Highly Acceptable
	Students	3.58	0.54	Highly Acceptable
	Overall	3.57	0.58	Highly Acceptable
Layout Consistency	IT- Experts	3.40	0.70	Highly Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.53	0.54	Highly Acceptable
	Overall	3.52	0.58	Highly Acceptable
Responsiveness	IT- Experts	3.20	0.63	Acceptable
	Subject-Matter Experts	3.73	0.47	Highly Acceptable
	Students	3.53	0.56	Highly Acceptable
	Overall	3.49	0.55	Highly Acceptable

All respondent categories rated the system's interface as "Highly Acceptable", especially in terms of ease of navigation, clarity of design and overall responsiveness. This suggests that the system delivers a user-friendly experience across divers user groups.

Table 7. Level of Satisfaction of the Respondents about the Different Features of the Software in terms of Bug Frequency

Indicator	Respondent Group	Mean	Standard Deviation	Description
Bug-Free Operation	IT- Experts	2.90	0.74	Acceptable
	Subject-Matter Experts	3.09	0.70	Acceptable
	Students	3.45	0.67	Highly Acceptable
	Overall	3.15	0.70	Acceptable
Stability	IT- Experts	3.10	0.57	Acceptable
	Subject-Matter Experts	3.36	0.67	Highly Acceptable
	Students	3.53	0.55	Highly Acceptable
	Overall	3.33	0.60	Highly Acceptable

While Students rated this aspect as "Highly Acceptable", IT experts and subject-matter experts gave it an "Acceptable" score, indicating occasional system issues or minor bugs. Although still positive, this shows there is room for improvement in terms of stability and bug handling.

Table 8. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Functional Stability

Indicator	Respondent Group Mean Standard Deviation Description			
Functional Completeness	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.36	0.50	Highly Acceptable
	Students	3.57	0.52	Highly Acceptable
	Overall	3.41	0.50	Highly Acceptable
Functional Correctness	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.62	0.53	Highly Acceptable
	Overall	3.58	0.47	Highly Acceptable
Functional Appropriateness	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.60	0.62	Highly Acceptable
	Overall	3.57	0.50	Highly Acceptable

Respondents from all categories rated the system as “Highly Acceptable” in its ability to support essential academic operations. This reflects that ATLAS performs its core tasks reliably and meets user expectations functionality.

Table 9. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Performance Efficiency

Indicator	Respondent Group	Mean	Standard Deviation	Description
Time Behavior	IT- Experts	3.00	0.67	Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.56	0.59	Highly Acceptable
	Overall	3.40	0.59	Highly Acceptable
Resource Utilization	IT- Experts	3.10	0.32	Acceptable
	Subject-Matter Experts	3.73	0.47	Highly Acceptable
	Students	3.55	0.56	Highly Acceptable
	Overall	3.46	0.45	Highly Acceptable
Capacity	IT- Experts	3.00	1.05	Acceptable
	Subject-Matter Experts	3.73	0.47	Highly Acceptable
	Students	3.63	0.50	Highly Acceptable
	Overall	3.45	0.67	Highly Acceptable

The results show that students and subject-matter experts find the system efficient, while IT experts rated it slightly lower. Overall, the system was still evaluated as “Highly Acceptable”, especially in its ability to handle performance tasks effectively under normal load conditions.

Table 10. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Compatibility

Indicator	Respondent Group	Mean	Standard Dviation	Description
Co-existence	IT- Experts	3.20	0.63	Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.62	0.51	Highly Acceptable
	Overall	3.49	0.55	Highly Acceptable
Interoperability	IT- Experts	3.20	0.42	Acceptable
	Subject-Matter Experts	3.73	0.47	Highly Acceptable
	Students	3.60	0.56	Highly Acceptable
	Overall	3.51	0.48	Highly Acceptable

The systems were rated “Highly Acceptable” across most groups, showing that it can operate well with other tools and perform data transfer reliably. However, IT experts gave a lower score, possibly due to deeper technical expectations.

Table 11. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Usability

Indicator	Respondent Group	Mean	Standard Deviation	Description
Appropriateness Recognizability	IT- Experts	3.00	0.47	Acceptable
	Subject-Matter Experts	3.55	0.52	Highly Acceptable
	Students	3.61	0.58	Highly Acceptable
	Overall	3.39	0.52	Highly Acceptable
Learnability	IT- Experts	2.90	0.74	Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.56	0.59	Highly Acceptable
	Overall	3.36	0.61	Highly Acceptable
Operability	IT- Experts	3.10	0.32	Acceptable
	Subject-Matter Experts	3.45	0.52	Highly Acceptable
	Students	3.61	0.57	Highly Acceptable
	Overall	3.39	0.47	Highly Acceptable
Error Protection	IT- Experts	2.70	0.67	Acceptable
	Subject-Matter Experts	3.55	0.52	Highly Acceptable
	Students	3.63	0.57	Highly Acceptable
	Overall	3.29	0.59	Highly Acceptable
User-Interface Aesthetics	IT- Experts	3.10	0.57	Acceptable
	Subject-Matter Experts	3.36	0.67	Highly Acceptable
	Students	3.58	0.56	Highly Acceptable
	Overall	3.35	0.60	Highly Acceptable
Accessibility	IT- Experts	3.20	0.42	Acceptable
	Subject-Matter Experts	3.55	0.69	Highly Acceptable
	Students	3.61	0.54	Highly Acceptable
	Overall	3.45	0.55	Highly Acceptable

This aspect received consistently “Highly Acceptable” ratings from students and experts, especially in terms of learnability, ease of navigation, and design. The slight differences in ratings from IT experts may suggest usability improvements for more technical use cases.

Table 12. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Reliability

Indicator	Respondent Group	Mean	Standard Deviation	Description
Maturity	IT- Experts	2.90	0.32	Acceptable
	Subject-Matter Experts	3.45	0.52	Highly Acceptable
	Students	3.59	0.56	Highly Acceptable
	Overall	3.32	0.47	Highly Acceptable
Availability	IT- Experts	3.60	0.52	Highly Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.63	0.52	Highly Acceptable
	Overall	3.62	0.51	Highly Acceptable
Fault Tolerance	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.36	0.50	Highly Acceptable
	Students	3.55	0.58	Highly Acceptable
	Overall	3.44	0.53	Highly Acceptable
Recoverability	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.45	0.82	Highly Acceptable
	Students	3.63	0.52	Highly Acceptable
	Overall	3.50	0.62	Highly Acceptable

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The system scored “Highly Acceptable” in this category, indicating that users experienced minimal downtime and found the system stable even during failures. This reflects positively on ATLAS’s ability to handle recovery and consistent operations.

Table 13. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Security

Indicator	Respondent Group	Mean	Standard Deviation	Description
Confidentiality	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.45	0.52	Highly Acceptable
	Students	3.66	0.49	Highly Acceptable
	Overall	3.50	0.51	Highly Acceptable
Integrity	IT- Experts	3.50	0.53	Highly Acceptable
	Subject-Matter Experts	3.36	0.50	Highly Acceptable
	Students	3.65	0.52	Highly Acceptable
	Overall	3.50	0.52	Highly Acceptable
Non-Repudiation	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.55	0.52	Highly Acceptable
	Students	3.64	0.51	Highly Acceptable
	Overall	3.53	0.52	Highly Acceptable
Accountability	IT- Experts	3.00	0.82	Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.55	0.55	Highly Acceptable
	Overall	3.40	0.62	Highly Acceptable
Authenticity	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.64	0.50	Highly Acceptable
	Overall	3.59	0.46	Highly Acceptable

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All groups rated the system's security features as "Highly Acceptable", especially in terms of data confidentiality, access control, and authentication. This supports the claim that the system is reliable for handling sensitive academic data.

Table 14. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Maintainability

Indicator	Respondent Group	Mean	Standard Deviation	Description
Modularity	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.64	0.50	Highly Acceptable
	Overall	3.62	0.47	Highly Acceptable
Reusability	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.59	0.57	Highly Acceptable
	Overall	3.57	0.49	Highly Acceptable
Analyzability	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.64	0.50	Highly Acceptable
	Students	3.61	0.51	Highly Acceptable
	Overall	3.52	0.50	Highly Acceptable
Modifiability	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.58	0.54	Highly Acceptable
	Overall	3.57	0.48	Highly Acceptable
Testability	IT- Experts	3.30	0.48	Highly Acceptable
	Subject-Matter Experts	3.82	0.40	Highly Acceptable
	Students	3.55	0.53	Highly Acceptable
	Overall	3.56	0.47	Highly Acceptable

The system's maintainability was evaluated as "Highly Acceptable" across all groups. Respondents found that components were easy to modify, update, and test, making the system suitable for continuous development and upgrades.

Table 15. Level of Satisfaction of the Respondents about the Different Features of the Software Based on ISO/IEC 25010 in terms of Portability

Indicator	Respondent Group	Mean	Standard Deviation	Description
Adaptability	IT- Experts	3.40	0.52	Highly Acceptable
	Subject-Matter Experts	3.55	0.52	Highly Acceptable
	Students	3.60	0.54	Highly Acceptable
	Overall	3.51	0.53	Highly Acceptable
Installability	IT- Experts	3.20	0.79	Acceptable
	Subject-Matter Experts	3.55	0.69	Highly Acceptable
	Students	3.62	0.50	Highly Acceptable
	Overall	3.45	0.66	Highly Acceptable
Replaceability	IT- Experts	3.10	0.57	Acceptable
	Subject-Matter Experts	3.27	0.65	Highly Acceptable
	Students	3.62	0.56	Highly Acceptable
	Overall	3.33	0.59	Highly Acceptable

Portability was also deemed "Highly Acceptable", particularly by students and subject-matter experts. Although IT experts were more reserved, the overall rating implies that ATLAS can be installed, adapted, and integrated with other systems when needed.

Table 16. Functionality test results

Tester	Total Test Cases	Failed Cases	Pass Rate (%)	Fail Rate (%)
Tester 1	151	6	96.03%	3.97%
Tester 2	151	0	100%	0%
Tester 3	151	0	100%	0%
Tester 4	151	0	100%	0%
Tester 5	151	0	100%	0%
Tester 6	151	0	100%	0%
TOTAL	906 (151 × 6)	6	99.27%	0.73%

A total of six (6) testers executed 137 identical test cases each, totaling 822 test executions. The first tester reported six (6) failed cases (4.38% failure rate), which were immediately resolved through system updates. Following these corrections, the subsequent five (5) testers achieved a 100% pass rate with no reported failures.

The overall pass rate across all testers was 99.27%, demonstrating the system's stability and readiness for deployment after corrective actions were applied.

Table 17. Compatibility test results

Tester	Total Test Cases	Passed Cases	Failed Cases	Pass Rate (%)	Fail Rate (%)
Tester 1	14	14	0	100%	0%
Tester 2	14	14	0	100%	0%
Tester 3	14	14	0	100%	0%
Tester 4	14	14	0	100%	0%
Tester 5	14	14	0	100%	0%
Tester 6	14	14	0	100%	0%
TOTAL	84 (14 × 6)	84	0	100%	0%

A total of six (6) testers conducted compatibility testing across 14 distinct test cases covering different browsers, screen resolutions, and device types. All testers achieved a 100% pass rate with no recorded failures.

Testing confirmed that the system is fully functional and accessible across major browsers (Google Chrome, Mozilla Firefox, Opera, Microsoft Edge), as well as on varying screen sizes and mobile devices (Android and Apple). Additionally, the AI evaluation feature performed consistently across all platforms.

These results demonstrate that the system is highly compatible, responsive, and stable across diverse environments, ensuring a seamless user experience for all intended users.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

Summary of Findings

This study was conducted to design and develop ATLAS (Automated Thesis and Learning Assessment System), a comprehensive platform aimed at enhancing the management of thesis defenses within the university specifically College of Engineering, Computer Studies, and Architecture (COECSA) and College of Business Administration (CBA). The system sought to address common challenges such as scheduling conflicts, inefficient panel evaluation processes, heavy administrative workload, and excessive paper trail by automating research procedures through a cloud-based system.

ATLAS was developed using modern web technologies, integrating a Genetic Algorithm to optimize defense scheduling and Google's Gemini LLM to power AI-driven functionalities such as research title validation, abstract analysis, and requirements checking. Quantitative survey methods were employed to gather feedback from IT experts, subject-matter experts, and student users, evaluating the system based on the ISO/IEC 25010 Software Quality Model, which covers critical aspects such as functionality, usability, security, maintainability, and portability.

The findings showed that across all respondent groups, ATLAS consistently achieved ratings of "Highly Acceptable" in all major quality categories. The

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	system’s usability, performance efficiency, and functional suitability received strong positive feedback, confirming its effectiveness and practicality. Early evaluations, however, revealed opportunities for improvement, particularly in areas like user interface design, input validation, instructional clarity, and error handling. System hiccups related to inactivity monitoring, freezing during typing, and technical error messages were initially observed but were successfully resolved through further optimization and refinement, significantly improving system stability and user-friendliness. Feedback from academic leaders including the IT Program Chair, the CBA Research Coordinator, a CITHM professor, the CFAD Research Coordinator, and the Center for Research and Development (CRD) affirmed that ATLAS’s defense scheduling module was well-aligned with institutional policies and guidelines, making it fully acceptable for official academic use. Furthermore, another CITHM professor, along with three CBA panelists, a CBA research coordinator, a CAMS research coordinator, and CRD representatives, highlighted the significant benefits of the student-side modules. They emphasized that tools like the research title acceptance feature and curated research topic suggestions would make the thesis process easier for students, greatly enhancing their productivity and allowing for more efficient research output. Overall, the results suggest that ATLAS has successfully fulfilled its objectives: providing an intuitive, AI-assisted, and efficient management system	
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for academic research defenses. It not only streamlined the thesis defense process and reduced administrative burdens but also demonstrated the practical potential of integrating optimization algorithms and large language models into educational administrative tasks. This study contributes an innovative, scalable solution that can significantly improve academic workflows, decision-making quality, and sustainable practices in higher education institutions.

Conclusion

In conclusion, the ATLAS system successfully met its primary objectives, offering an AI-assisted and efficient platform that significantly improved the management of academic research defenses within the College of Engineering, Computer Studies, and Architecture (COECSA). By integrating cutting-edge technologies such as the Genetic Algorithm and Google’s Gemini LLM, ATLAS optimized the scheduling process and provided AI-driven assistance for the various stages of thesis defense, making it a powerful tool for both students and faculty. The system not only streamlined the thesis defense process, improving communication and reducing administrative workload, but also demonstrated the potential of utilizing artificial intelligence and optimization algorithms to enhance decision-making in educational administrative tasks.

The results from the evaluation suggest that ATLAS is a highly effective solution for addressing the challenges faced in managing thesis defenses, including scheduling conflicts, evaluation inefficiencies, and paperwork overload. Moreover, the system's ability to adapt and evolve with AI technologies positions it as a

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5. Conducting user training for students and faculty on how to best use the system.
6. Suggest periodic security audits to protect sensitive academic data.
7. Study the integration of more advanced AI (like predictive analytics for student performance).
8. Explore multi-campus integration using cloud deployment.

LITERATURE CITED

Abbas, N., Imran, A., Manzoor, R., Hussain, T., & Hussain, M. (2023). Role of artificial intelligence tools in enhancing students' educational performance at higher levels. *Journal of Artificial Intelligence Machine Learning and Neural Network*, (35), 36-49.

<https://journal.hmjournals.com/index.php/JAIMLNN/article/view/2577>

Abdul-kareem, A., Fayed, Z., Rady, S., Amin, S., & Nema, B. (2023). Advances in Decision Support Systems' design aspects: architecture, applications, and methods. *International Journal of Intelligent Computing and Information Sciences*, 23(2), 74–104. https://ijicis.journals.ekb.eg/article_305267.html

Amber Jackson (2023). Top 10 Companies with ethical AI practices
<https://aimagazine.com/ai-strategy/top-10-companies-with-ethical-ai-practices>

Ares, G., Fernández, C. C., & Diego, I. (2023). Ultimate pit-limit optimization algorithm enhancement using structured query language. *Minerals*, 13(7), 966.
<https://www.mdpi.com/2075-163X/13/7/966>

Bauer, K., Sogade, O., Gage, B., Ruoff, B., & Lewis, L. (2020). Improving follow-up attendance for discharged emergency care patients using automated phone system to self-schedule: a randomized controlled trial. *Academic Emergency Medicine*, 28(2), 197-205.

<https://onlinelibrary.wiley.com/doi/10.1111/acem.14080>

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

Begum, I. (2024). Role of artificial intelligence in higher education- an empirical investigation. *Int Res J Adv Engg Mgt*, 2(03), 49-53.

<https://goldncloudpublications.com/index.php/irjaem/article/view/22>

Bouasangthong, V., Thalangsy, K., & Leusinxay, I. (2025). The Current State of Implementation of the Research Management System of the National University of Laos. *Journal of General Education and Humanities*, 4(1), 35-46.

Bioframe: operations on genomic intervals in pandas dataframes. *Bioinformatics*, 40(2). <https://academic.oup.com/bioinformatics/article/40/2/btae088/7613967>

Brown, Z. C., Robinson, N., Wingate, D., & Fulda, N. (2020). Towards neural programming interfaces. <https://arxiv.org/abs/2012.05983>

Burger, B., Kanbach, D., Kraus, S., Breier, M., & Corvello, V. (2023). On the use of ai-based tools to support management research. *European Journal of Innovation Management*, 26(7), 233-241.

<https://www.emerald.com/insight/content/doi/10.1108/ejim-02-2023-0156/full/html>

Cagananta and Wirawan Istiono (2024). Efficient Thesis Management: A Study of Universitas Multimedia Nusantara's Application Development Using Extreme Programming Principles. <https://www.journal-isi.org/index.php/isi/article/view/701/349>

Chen, J. and Okle, R. A. N. (2024). A study on the application of response surface methodology and bayesian optimization in parameter tuning of genetic algorithms. Fourth International Conference on Applied Mathematics, Modelling, and Intelligent Computing (CAMMIC 2024), 1, 187.

<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/13219/3036921/A-study-on-the-application-of-response-surface-methodology-and/10.1117/12.3036921.short>

Chika Yinka-Banjo, Ajao Kamal Abayomi, Precious Ifeanyi Ohalet (2021). Application of Genetic Algorithm to The Job Assignment Problem with Dynamic Constraints.

<https://journals.covenantuniversity.edu.ng/index.php/cjict/article/view/2441>

Dewi Nasien, Andi Andi (2022). Optimization of Genetic Algorithm in Courses Scheduling. <https://journal.uir.ac.id/index.php/ITJRD/article/view/7896>

Doğan, A. and Yurtsal, A. (2021). Developing a decision support system for exam scheduling problem using genetic algorithm. Eskişehir Technical University Journal of Science and Technology a - Applied Sciences and Engineering, 22(3), 274-289. <https://dergipark.org.tr/en/pub/estubtda/issue/65123/890307>

Gurusinga, M. O., Ginting, R., & Sinulingga, S., (2025). Parallel Scheduling using Genetic Algorithm and Knowledge Based Approach. Jurnal

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

Sistem Teknik Industri, 27(2), 127-136.

<https://doi.org/10.32734/jsti.v27i2.19099>

Kauppinen, R., Lagstedt, A., & Lindstedt, J. P. (2023). Evaluating success in the digitalized thesis management process. In 2023 11th International conference on information and education technology (ICIET) (56–60). IEEE.

<https://doi.org/10.1109/ICIET56899.2023.10111267>

Katoch, S., Chauhan, S. S., & Kumar, V. (2020). A review on genetic algorithm: past, present, and future. Multimedia Tools and Applications, 80(5), 8091-8126.

<https://link.springer.com/article/10.1007/s11042-020-10139-6>

Khasawneh, Y. J. A., Jarrah, H. Y., Alsarayreh, R. S., & Khasawneh, M. A. S. (2023). The Role of Cloud Computing in Improving the Performance of School Principals. Eurasian Journal of Educational Research, 107(107), 110-125.

Khozaimi, A. (2023). Design and development of backend application for thesis management system using microservice architecture and restful api. Jurnal Ilmiah Kursor, 11(4), 179-186.

<https://kursorjournal.org/index.php/kursor/article/view/313>

Lyceum of the Philippines University-Cavite. (n.d.). <https://cavite.lpu.edu.ph/>.

<https://cavite.lpu.edu.ph/academics/undergraduate-colleges/>

Modern javascript frameworks and javascript's future as a fullstack programming language. Design of Single Chip Microcomputer Control System for Stepping

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

Motor, 1-5 (2023). <https://www.onlinescientificresearch.com/articles/modern-javascript-frameworks-and-javascripts-future-as-a-full-stack-programming-language.pdf>

Nico, N., Charibaldi, N., & Fauziah, Y. (2022). Comparison of Memetic Algorithm and Genetic Algorithm on Nurse Picket Scheduling at Public Health Center. *International Journal of Artificial Intelligence & Robotics (IJAIR)*, 4(1), 9-23. <https://doi.org/10.25139/ijair.v4i1.4323>

Pereira, J. A., Azanza, M., López, J. M., & Garmendia, X. (2023). Aplicación de la IA Generativa para Mejorar el Proceso de Elaboración y Evaluación de Trabajos de Fin de Grado (TFGs) - [Application of Generative AI to Improve the Process of Drafting and Evaluating Final Year Projects (TFGs)] . <https://doi.org/10.26754/cinaic.2023.0049>

Pigola, A. (2023). Artificial intelligence in academic research. *International Journal of Innovation*, 11(3), e25408. <https://periodicos.uninove.br/innovation/article/view/25408>

Pisica, A., Edu, T., Zaharia, R., & Zaharia, R. (2023). Implementing artificial intelligence in higher education: pros and cons from the perspectives of academics. *Societies*, 13(5), 118. <https://www.mdpi.com/2075-4698/13/5/118>

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

- Qasem, Y. A., Abdullah, R., Jusoh, Y. Y., Atan, R., & Asadi, S. (2019). Cloud computing adoption in higher education institutions: A systematic review. *Ieee access*, 7, 63722-63744.
- Ramírez-Adrados, A., Fernández-Elías, V. E., Fernández-Martínez, S., Martínez-Pascual, B., Gonzalez-de-Ramos, C., & Clemente-Suárez, V. J. (2022). The effect of studying a double degree in the psychophysiological stress response in the bachelor's thesis defense. *International Journal of Environmental Research and Public Health*, 19(3), 1207. <https://www.mdpi.com/1660-4601/19/3/1207>
- Rappos, E., Thiémar, E., Robert, S., & Hêche, J. (2022). A mixed-integer programming approach for solving university course timetabling problems. *Journal of Scheduling*, 25(4), 391-404. <https://doi.org/10.1007/s10951-021-00715-5>
- SEAL LLM Leaderboards: Expert-Driven Private Evaluations (2024). https://scale.com/leaderboard/adversarial_robustness
- Storey, V. A. (2023). AI Technology and Academic Writing: Knowing and Mastering the “Craft Skills”. *International Journal of Adult Education and Technology (IJAET)*, 14(1), 1-15. <https://doi.org/10.4018/IJAET.325795>
- Thavi, R., Jhaveri, R., Narwane, V., Gardas, B., & Jafari Navimipour, N. (2024). Role of cloud computing technology in the education sector. *Journal of Engineering, Design and Technology*, 22(1), 182-213.

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

Tindle, R., Pozzebon, K., Willis, R. L., & Moustafa, A. (2023, September 13).

Academic Misconduct and Generative Artificial Intelligence: University

Students' Intentions, Usage, and Perceptions [Preprint].

<https://doi.org/10.31234/osf.io/hwkgu>

Trang, N. T. Q. and Thi Thu, P. (2024). The role of ai in improving student

learning outcomes: evidence in vietnam. International Journal of

Multidisciplinary Research and Analysis, 07(06). <https://ijmra.in/v7i6/48.php>

Wang, L. (2023). An evaluation method of english composition automatic grading

based on genetic optimization algorithm and cnn model. International Journal of

Advanced Computer Science and Applications, 14(10).

<https://thesai.org/Publications/ViewPaper?Volume=14&Issue=10&Code=IJACS>

[A&SerialNo=7](https://thesai.org/Publications/ViewPaper?Volume=14&Issue=10&Code=IJACS)

Which LLM to Choose: 12 key aspects to consider building AI solutions. (2024,

July 1). [https://mindsdb.com/blog/which-llm-to-choose-12-key-aspects-to-](https://mindsdb.com/blog/which-llm-to-choose-12-key-aspects-to-consider-when-building-ai-solutions)

[consider-when-building-ai-solutions](https://mindsdb.com/blog/which-llm-to-choose-12-key-aspects-to-consider-when-building-ai-solutions)

Widodo, Y. B., Korwa, R. U., & Nuraini, R. (2023). Artificial intelligence based

decision support system for education management in higher education. Al-fikrah

Jurnal Manajemen Pendidikan, 11(2), 352.

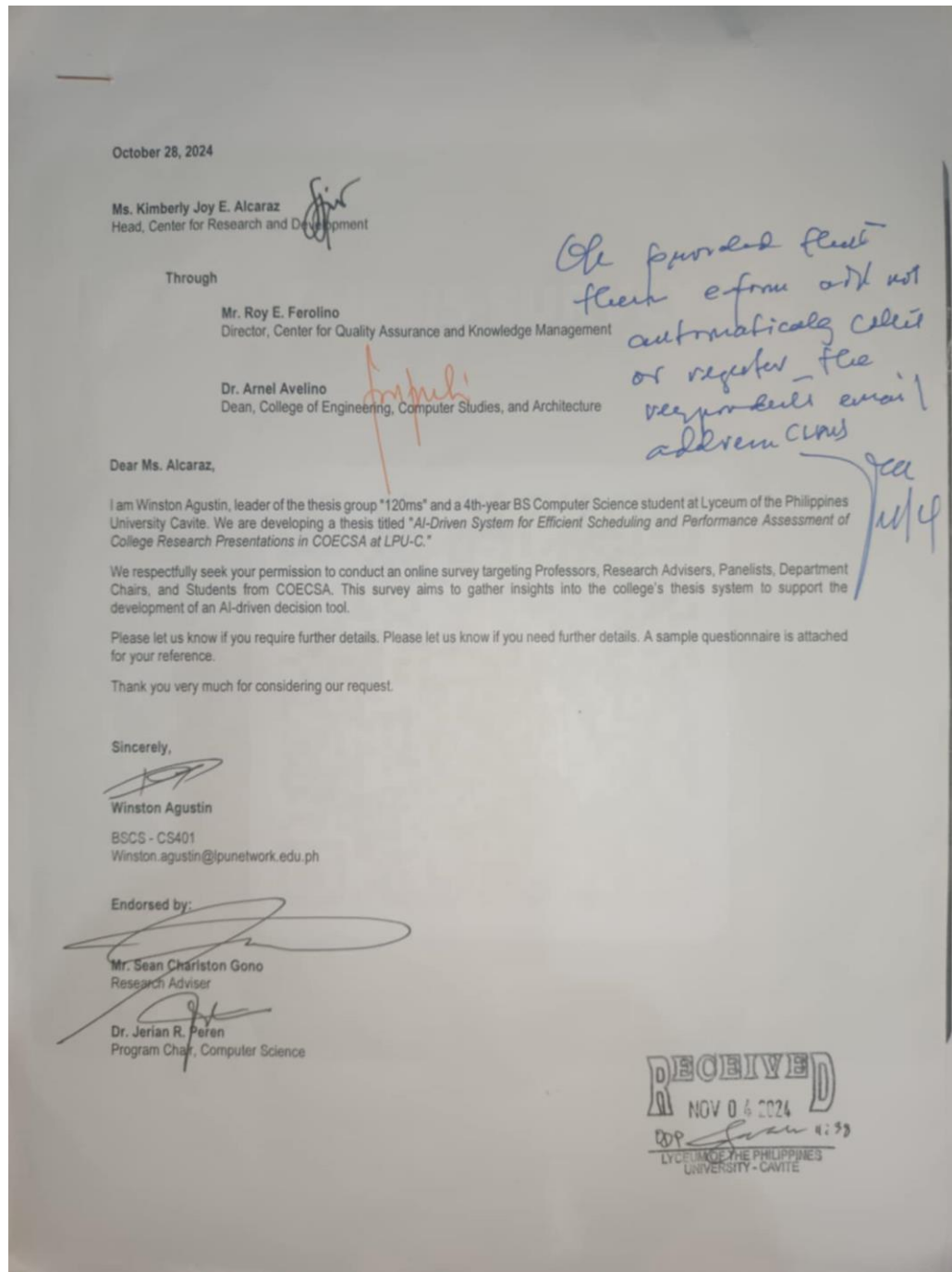
<https://doi.org/10.31958/jaf.v11i2.12120>

	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	William, F. (2024). Ai in academic writing: ally or foe?. International Journal of Research Publications, 148(1). https://www.ijrp.org/paper-detail/6273 Xin-She Yang (2019) Introduction to Algorithms for Data Mining and Machine Learning. https://www.sciencedirect.com/book/9780128172162/introduction-to-algorithms-for-data-mining-and-machine-learning Yopi Nugraha (2022). Web-Based Thesis Management Information System Design. https://jurnal.polgan.ac.id/index.php/sinkron/article/view/12244	
	COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE	140

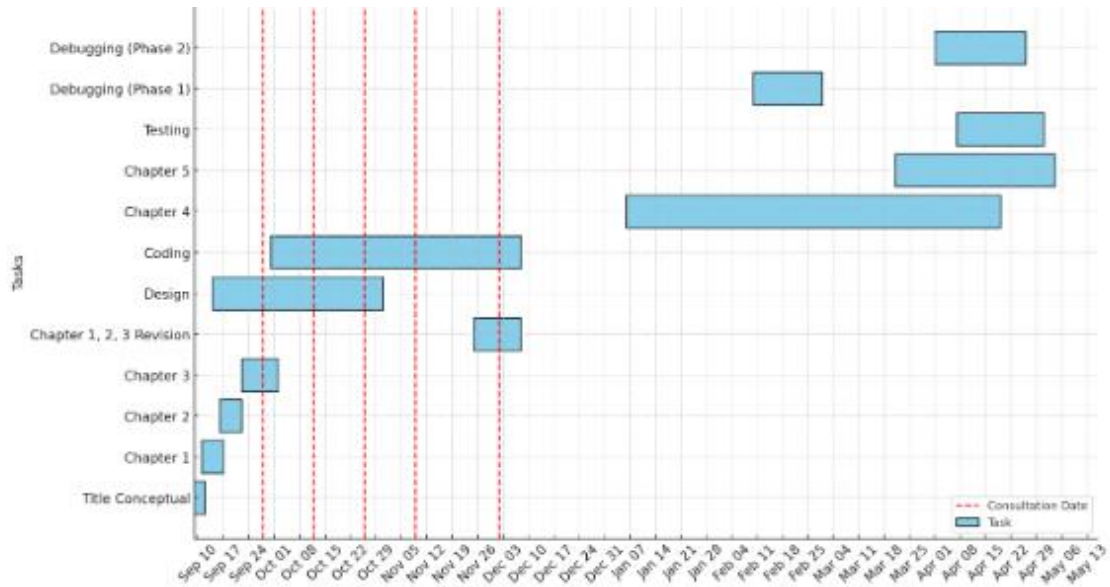
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APPENDICES

APPENDIX A LETTER OF REQUEST



**APPENDIX B
GANTT CHART**



APPENDIX C

WIREFRAME

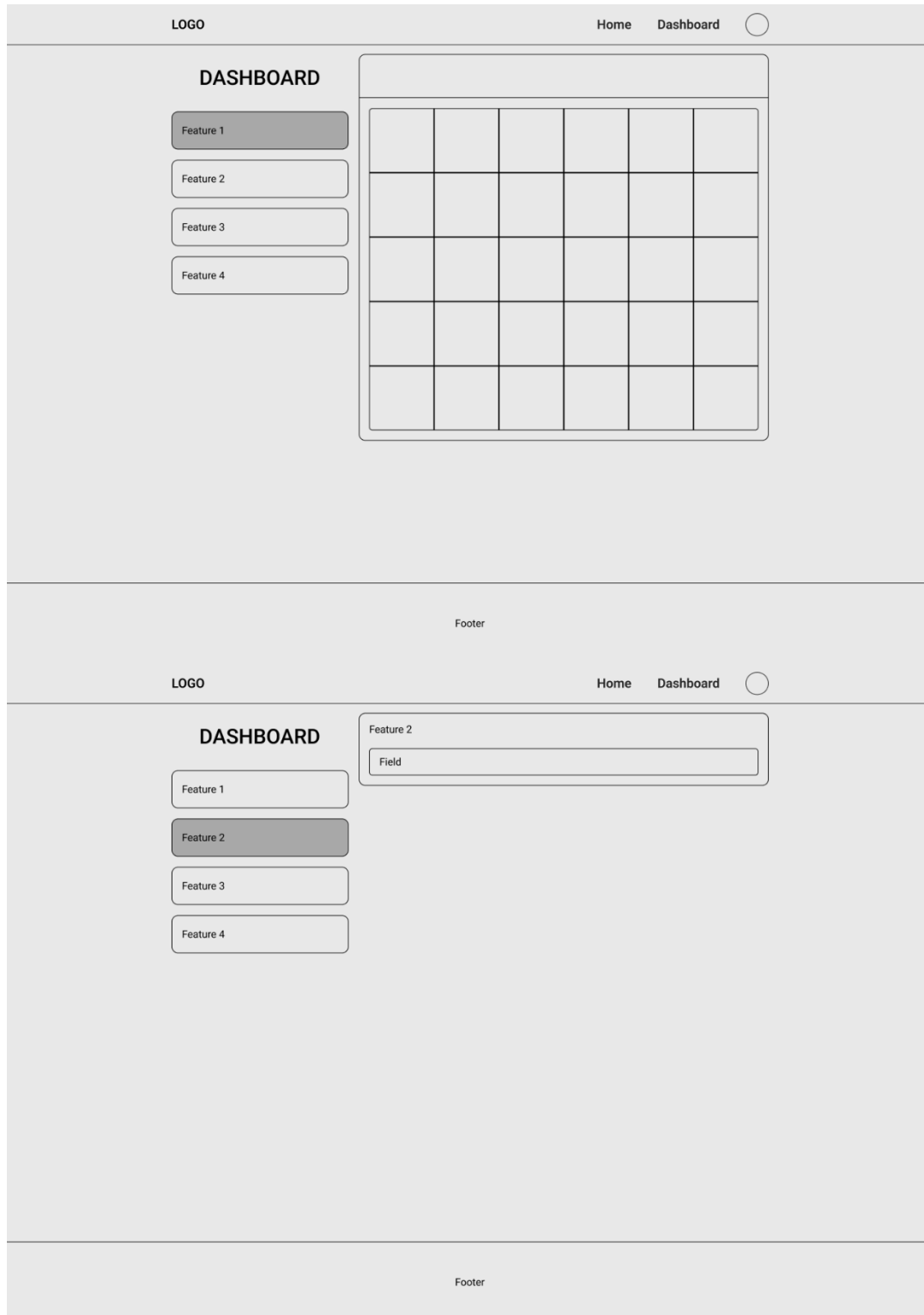
LOGO

Username

Password

[Forgot Password?](#)

Login



LOGO

HomeDashboard

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 3

Field

Field

Button

Footer

LOGO

HomeDashboard

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 4

Requirement
Details
Details
Details

Button

Requirement
Details
Details
Details

Button

Requirement
Details
Details
Details

Button

Requirement
Details
Details
Details

Button

Footer





LOGO

Home

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Requirements

Defense Sched

Footer

LOGO

Home

DASHBOARD

Feature 1

Feature 2

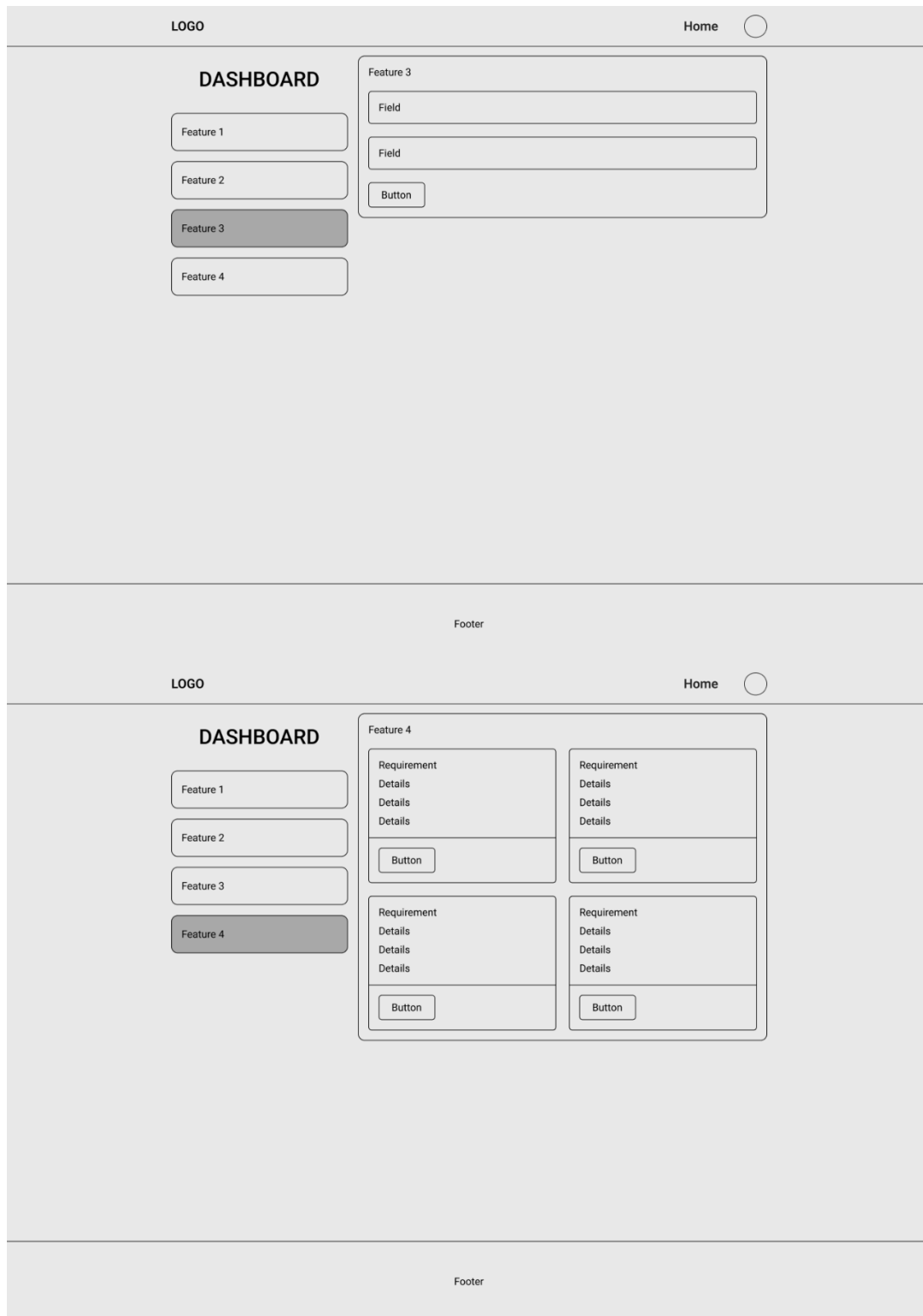
Feature 3

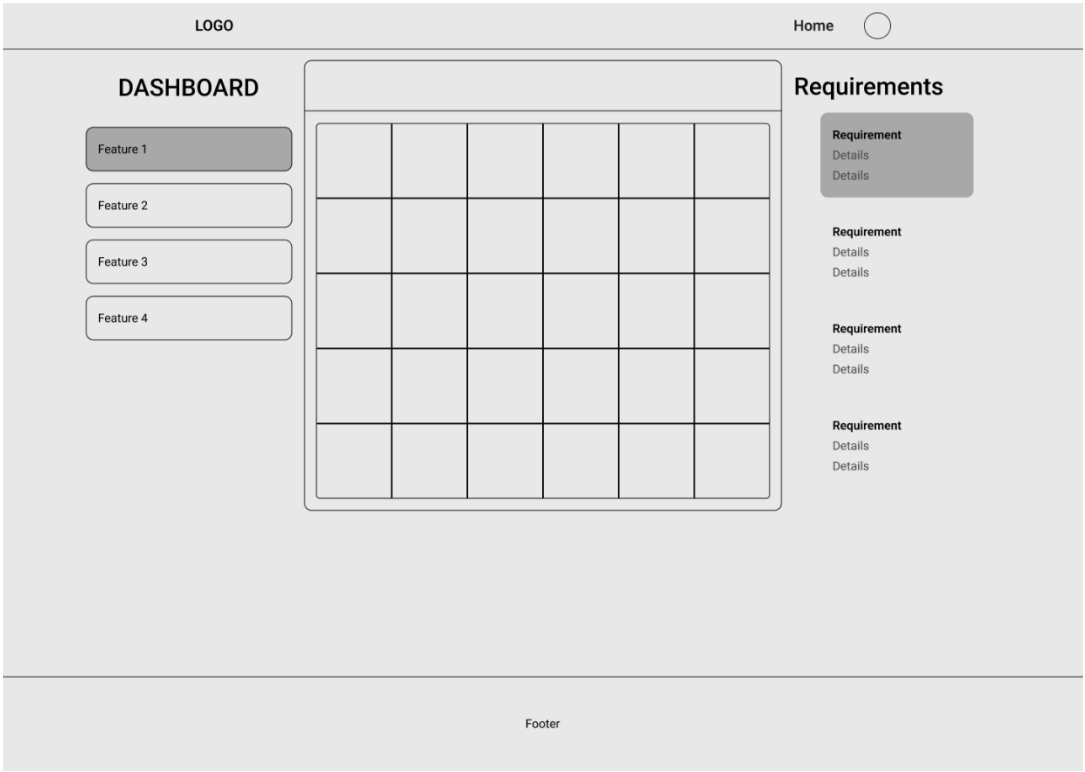
Feature 4

Feature 2

Field

Footer





**APPENDIX D
PSEUDOCODE**

BEGIN

ENABLE error reporting and configure logging
SET content type to JSON

TRY

IMPORT necessary database and helper functions

IF database connection is not valid THEN
THROW exception "Database connection error"

IF request method is POST THEN
FETCH teams and panelists from database
PARSE input data (duration, rooms, timeSlots, days) from POST request

VALIDATE duration is a positive integer
IF invalid THEN
THROW exception "Invalid duration"

FETCH user schedules from database

RUN genetic algorithm with parameters:
teams, panelists, rooms, timeSlots, days, userSchedules, population size, generations, mutation rate

IF saving schedule to database is successful THEN
PREPARE success response including:
- Initial population size
- Crossover and mutation counts
- Conflict counts and fitness scores per generation
- Population per generation

SAVE response to a JSON file
OUTPUT success response as JSON

ELSE
THROW exception "Failed to save schedule to database"

ELSE
THROW exception "Invalid request method"

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CATCH exceptions
 LOG error details
 OUTPUT error response as JSON

FUNCTION fetchTeams(pdo)
 EXECUTE SQL query to fetch teams with their advisers
 RETURN results

FUNCTION fetchPanelists(pdo)
 EXECUTE SQL query to fetch panelist IDs
 RETURN results

FUNCTION fetchUserSchedules(pdo)
 EXECUTE SQL query to fetch user schedules
 RETURN schedules organized by user ID

FUNCTION geneticAlgorithm(pdo, teams, panelists, rooms, timeSlots, days, userSchedules, populationSize, generations, mutationRate)
 INITIALIZE population using `createInitialPopulation`

FOR each generation
 CALCULATE fitness for each schedule in the population

 PERFORM selection on population
 REPLACE population with new schedules generated through crossover and mutation

SORT population by fitness DESCENDING
 RETURN the best schedule

FUNCTION createInitialPopulation(pdo, populationSize, teams, panelists, rooms, timeSlots, days)
 FOR each individual in population
 CREATE DefenseSchedule object with randomly assigned panelists, rooms, times, and days
 RETURN population

FUNCTION saveScheduleToDatabase(pdo, schedule)
 START database transaction
 FOR each defense in schedule
 INSERT defense details into the database
 COMMIT transaction
 RETURN true

FUNCTION DefenseSchedule(pdo, teams, panelists, rooms, timeSlots, days)
 INITIALIZE schedule with chromosomes for each team

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	ASSIGN random panelists, rooms, times, and days FUNCTION hasConflicts(pdo, defense, userSchedules, all_defenses) CHECK conflicts with user schedules and overlapping times RETURN true if conflicts exist, false otherwise END	
	COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE	153

**APPENDIX E
CERTIFICATION STATEMENT OF QUESTIONNAIRE STATEMENT AND TEST
CASES VALIDATION**

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APPENDIX F Test Cases

Project Name:	ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED EVALUATION FOR THE COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE						
Written By:	Agustin, Winston L.		Gerona, Jerald Ryan D.				
	Vicedo, Neil S.		Ilano, Ivan Kerwin R.				
Written Date:	Tuesday, April 22, 2025						
Test Case Description:	Functionality Test						
Testers:	(Test Respondent's Full Name)						
Test Conditions			Date:	April 26, 2025			
Total Test Case:	137		Rating:				
Pass:							
Fail:							
Test Case Scenario ID	Name of the Module Function	Test Case Scenario	Action	Actual Input	P A S S	F A I L	Comments/Suggestions

ADMIN						
AD#01	Admin Login	Admin enters registered Username and password	Successful login will navigate Admin to Admin Dashboard	Click Login		
AD#02		Admin enters unregistered Username and Password	A message will show that the "Username does not Exist"	Click Login		
AD#03		Admin enters only Username	A message will pop out saying "Please fill out this field"	Click Login		

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AD#04		Admin enters only password	A message will pop out saying "Please fill out this field"	Click Login			
AD#05	Admin Overview Preview	Admin Dashboard overview	Successful login, Admin will be redirected to Admin Dashboard on Overview Tab	Click Login			
AD#06		View of Team Completion Status	Successful login, Admin can access/view the Team Completion Status	Click Team Completion Status			
AD#07		View of Upcoming Defenses Schedule	Successful login, Admin can access/view the Upcoming Defense Schedule	Scroll down → Click an available group			
AD#08		View of Past Defense Schedules	Successful login, Admin can access/view the Past Defense Schedule	Scroll down → Click an available Group			
AD#09	Admin Users Tab	Admin Search Users	System searches for user	Input user → Click search button			
AD#10		Admin Sort Users	A dropdown list shows, which includes Default (Newest First), Default (Oldest first), Username (A-Z), Username (Z-A), First Name (A-Z), First Name (Z-A),	Click sort option beside search button			

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			Last Name (A-Z), and Last Name (Z-A)			
AD#11	Admin Adding Users	Select Add User	A window will pop up showing details of potential system user	Click Add User button		
AD#12		Admin leaves Username blank	The system saves the item but will have a blank username	Click Add User button		
AD#13		Admin leaves email blank	An error message will pop out "Invalid email format"	Click Add User button		
AD#14		Admin leaves password blank	The system saves the item but will not be eligible for login	Click Add User button		
AD#15		Admin leaves first name blank	The system saves the item but will have a blank first name	Click Add User button		
AD#16		Admin leaves last name blank	The system saves the item but will have a blank last name	Click Add User button		
AD#17		Admin leaves Program blank	The system saves the item but will have a blank Program	Click Add User button		
AD#18		Admin leaves User Type blank	The system automatically makes the added user as a student	Click Add User button		
AD#19	Admin Bulk Add students	Select Bulk Add Students	A window pops up showing details on bulk adding students	Click Bulk Add Students beside Add User button		

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AD#20		CSV File upload	Choose CSV File on Upload Method	Click Bulk Add Students → Click Submit button			
AD#21		Admin did not add CSV File	An error will pop up "No valid upload method selected, or data provided"	Click Bulk Add Students → Click CSV File → Click submit button			
AD#22		Admin pastes a CSV data	Choose Paste Text in Upload Method	Click Bulk Add Students → Click Paste Text			
AD#23		Admin did not paste a CSV data	An error will pop up "No valid upload method selected, or data provided"	Click Bulk Add Students → Click Paste Text → Click Submit			
AD#24		Admin manually enters ID, Name, Program	Choose Manual Form on Upload method	Click Bulk Add Students → Click Manual Form			
AD#25		Admin leaves ID blank	The item will be saved but will not show in the list	Click Bulk Add Students → Click Manual Form			
AD#26		Admin leaves Name blank	An error message will show	Click Bulk Add Students → Click Manual Form			
AD#27		Admin leaves Program blank	An error message will show	Click Bulk Add Students → Click Manual Form			

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AD#28	Admin Edit Users	Select Edit	A window pops up with the details of the user that can be edited	Click Edit button on the far right of a user			
AD#29	Admin Delete Users	Select Delete	A confirmation window will pop up	Click Delete button on the far right of a user			
AD#30	Admin Teams Tab	Admin Team Search	System searches for user	Click Teams tab ➔ Input Valid team ➔ Click search button			
AD#31		Admin Team Sort	A dropdown list shows, which includes Default (Newest First), Default (Oldest First), Team Name (A-Z), and Team Name (Z-A)	Click sort option beside search button			
AD#32	Admin Adding Teams	Select Add Team	A window pops up with details of adding a team	Click Add Team			
AD#33		Admin leaves Team Name blank	An error message will pop up	Click Add Team			
AD#34		Admin leaves Research Title blank	The system saves the item but will display a blank title	Click Add Team			
AD#35		Admin leaves Area of Expertise blank	The system saves the item but will display a blank area of expertise	Click Add Team			

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AD#36		Admin leaves Program blank	The system saves the item but will display a blank Program entry	Click Add Team			
AD#37		Admin leaves Team members blank	The system saves the item but will not display the team	Click Add Team			
AD#38	Admin Edit Team	Select Edit	A window pops up with the details of the team that can be edited	Click Edit button on the far right of a team			
AD#39	Admin Delete Team	Select Delete	A confirmation window will pop up	Click Delete button on the far right of a team			
AD#40	Admin Thesis Topics Tab	Get Latest Topics	The system will display the top 10 current trends based on the field of study chosen by the admin	Click Thesis Topics Tab → Select Field of Study → Click Get Latest Topics			
AD#41		Admin did not select a field of study	A pop-up shows "Please select a field"	→ Select Field of Study → Click Get Latest Topics			
AD#42		Admin adds new topic	A window pops up with the details of the topic	Select a Field of Study → Click Get Latest Topics → Scroll down and click add button			

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AD#43		Admin Search Thesis topic	System searches for thesis	Input valid topic → Click search button			
AD#44		Admin Filters Thesis Topic Categories	System shows a dropdown list and shows thesis topics by category, which include Architecture, Computer Science, Information Technology, Aeronautical Engineering, Civil Engineering, Computer Engineering, Electrical Engineering, Electronics Engineering, Industrial Engineering, and Mechanical Engineering	Click categories dropdown list			
AD#45		Admin Sort Thesis Topics	System shows a drop-down list and sorts thesis topics, which includes Default (Newest First), Default (Oldest First), Topic (A-Z), Topic (Z-A), Category (A-Z), and Category (Z-A)	Click Sort dropdown list			
AD#46	Admin Edit Thesis Topics	Select Edit	A window pops up with details about the topic	Click Edit button on the far right of a topic			

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AD#47	Admin Delete Thesis Topic	Select Delete	A confirmation window will pop up	Click Delete button on the far right of a topic			
AD#48	Admin Research Title Tab	Add Research Title	A window will pop up to add a Research Title	Click Add Research Title button			
AD#49		Admin leaves Team name blank	An error message will pop up	Click Add Research Title button			
AD#50		Admin leaves Title blank	The system saves the item but will not display a title	Click Add Research Title button			
AD#51		Admin Edit Research Title	A window will pop up with details of the research title	Click Edit button on the far right of research title			
AD#52		Admin Delete Research Title	A confirmation window will pop up	Click Delete button on the far right of research title			
AD#53	Admin Defense Schedule Tab	Admin Add Defense Schedule	A window will pop up with details of the Defense Schedule	Click Add Defense Schedule			
AD#54		Admin leaves Schedule Date blank	An error will pop up	Click Add Defense Schedule			
AD#55		Admin leaves Start time/ End time blank	An error will pop up	Click Add Defense Schedule			
AD#56		Admin leaves Room blank	The system will save the schedule but will not be shown in the system	Click Add Defense Schedule			

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AD#57		Admin leaves Team blank	An error will pop up	Click Add Defense Schedule			
AD#58		Admin leaves Panelists blank	An error will pop up	Click Add Defense Schedule			
AD#59		Admin Edit Defense Schedule	System shows a pop up for editing a Defense Schedule	Click Edit button on the far right on a schedule			
AD#60		Admin Delete Defense Schedule	A confirmation window will pop up	Click Delete button on the far right on a schedule			
AD#61		Admin Generate Defense Schedule	A window will pop up that will generate a Defense Schedule	Click Generate Defense Schedule button			
AD#62	Admin Rubrics Tab	Add Rubrics	A window will show and an Admin will input the Rubrics based on the programs	Click Add New Rubric Button			
AD#63		Admin leaves Rubric blank	The rubric will be saved but with N/A as details	Click Add New Rubric Button			
AD#64		Admin Edit Rubrics	A window pops up with details of an editable rubric	Click Edit button on the far right of a rubric			
AD#65		Admin Delete Rubrics	A confirmation window will pop up	Click Delete button on the far right of a rubric			
AD#66	Admin Evaluation Tab	Admin Evaluation View	The system will change the view	Click Change View button			

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AD#67	Admin Requirements Tab	Admin Add Requirement	A window will pop up with details of adding a requirement	Click Add Requirement button			
AD#68		Admin leaves Add Requirements blank	An item will be saved but will have no data	Click Add Requirement button			
AD#69		Admin Edit Requirements	A window will pop up with details that can be edited	Click Edit button on the far right of the requirement			
AD#70		Admin Delete Requirements	A confirmation window will pop up	Click Delete button on the far right of the requirement			
AD#71	Admin System Settings	Admin Add System Settings	A setting can be added by inserting the right Key, Value, And Description	Click Content Management Tab → Click Add Setting			
AD#72		Admin Edit System Setting	A window will pop up with editable details	Click Content Management Tab → Click Edit button on the far right of a setting			
AD#73	Admin Files	Access Admin Files	A new tab will open and will be able to see all college Research Repositories	Click Files tab			
AD#74		Admin Upload Files	Admin will upload file	Click Upload			

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AD#75		Admin leaves College blank	A warning will pop up "Please fill out this field"	Click Upload			
AD#76		Admin did not input a file	A warning will pop up "Please select a file"	Click Upload			
AD#77	Admin Guide	Admin Guide Access	The page will show a guide on Using the System	Click Guide			
US#01	User Profile	User Profile View	Shows the Username, Email Address, First Name, Last Name, Headline, Profile Details, Gender, Password Edit	Click the profile Icon banner in the Top right corner and Click Profile selection			
US#02		User Profile Edit	Edit your Picture, Username, Email address, First Name, Last Name, Headline, Profile Details, Gender And Password and It will show a "profile successfully updated"	Click the profile Icon banner in the Top right corner and Click Profile selection and Click Confirm Changes			
US#03		User Profile inputting a wrong Email Address	It will say" Please enter a part following"" is incomplete.	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Email Address and Click Confirm Changes			

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US#04		User Profile inputting a Wrong current Password	It will say "Incorrect Current Password"	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Password and Click Confirm Changes			
US#05		User Profile inputting a not same Password	It will say "Confirmed password does not match new password"	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Password and Click Confirm Changes			
US#06		Bio Overview	Shows the User Bio	Click the profile Icon banner in the Top right corner and Click your profile Icon in the left side			
US#7		User logout from system	User will be logged out from system	Click the profile Icon banner in the Top right corner and Click logout button			

ADVISER / PANELIST

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AP#01	Staff Login	Staff enters registered Username and password	Successful login will navigate Admin to Staff Dashboard	Click Login			
AP#02		Staff enters unregistered Username and Password	A message will show that the "Username does not Exist"	Click Login			
AP#03		Staff enters only Username	A message will pop out saying "Please fill out this field"	Click Login			
AP#04		Staff enters only password	A message will pop out saying "Please fill out this field"	Click Login			
AP#05	Staff Overview Preview	Staff Dashboard overview	If login was successful, the staff will be redirected to Staff Dashboard on Overview Tab	Click Login			
AP#06		Team Completion Status	view the Team Completion Status	Click Overview			
AP#07		Upcoming Defense Schedule	view the Upcoming Defense Schedule	Click Overview			
AP#08	Staff Calendar	Staff Calendar overview	Shows Month and Date and if there a upcoming Defense	Click Calendar			
AP#09		Date	Shows the week	Click a date			
AP#10		Date with a defense	Shows the Hours, Team Members, Adviser, Program, Research	Click a date with defense			

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			Paper PDF of the Papers, AI Analysis, and Score Sheet				
AP#11		Week	Shows the current date up to 7days and hours	Click week			
AP#12		Day	Show the whole hour day	Click Day			
AP#13		next month or previous month	Shows next month and previous month	Click the next month or previous month			
AP#14	Staff Requirement Status	Staff Requirements List	Shows the Requirements	Click Calendar tab			
AP#15		Staff click a requirement	Goes to the Requirement Checker Overview	Click Calendar tab and click a requirement			
AP#16	Staff Thesis Topics Decision	Staff Thesis Topics	Shows choose a filed	Click Thesis topic Decision			
AP#17			Shows the Topics and Description of the selected Field	Click Thesis topic Decision and select A field			
AP#18	Staff Research Title Acceptance Tool	Search a Title	The AI tool will give a analysis for the title	Click Research Title Acceptance , Fill up the list and then Check Title			
AP#19		Search a Title that have a similar title	A popup saying "The title is similar to an existing title". Do you still want to	Click Research Title Acceptance , Fill up the list and			

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			proceed with the analysis? If you Proceed it give a Analysis but if you don't proceed, It will give you a answer "The title is similar to an existing title. Consider revising it for uniqueness."	then Check Title			
AP#20		Enter without inputting	It will say "Since both the research title and the problem to solve are empty, I am unable to provide an analysis. Please provide the research title and the problem it aims to solve so that I can give you feedback."	Click Research Title Acceptance , Fill up the list and then Check Title			
AP#21	Staff Requirement Checker	Requirement Checker Overview	Shows the Team and Requirements that have been submitted and not submitted	Click Requirement Checker			
AP#22		Requirement Checker Team	Shows all of the team	Click Requirement Checker and The Team name			
AP#23		Requirement Checker Status	Changed the status of the Requirements to Pending, Submitted, Approved, Rejected	Click Requirement Checker and Status:			

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AP#24		Requirement Checker Feedback	Give a feedback to a requirement	Click Requirement Checker and Type at Feedback			
AP#25		Requirement Checker Feedback File upload	Upload a File that have a Feedback	Click Requirement Checker and Click Choose file			
AP#26		Requirement Checker Download File	Download a File	Click Requirement Checker and Click Download File			
AP#27		Requirement Checker View File	View the PDF file that have been Submitted	Click Requirement Checker and Click View File			
AP#28		Requirement Checker Update Requirements	Save the changed that have been made and It will have a Pop up message saying "Requirements updated successfully!"	Click Requirement Checker and Click Update Requirements			
AP#29	Staff Research Evaluation	Research Evaluation Overview	Shows the Research Evaluation Comments	Click Research Evaluation			
US#01	User Profile	User Profile View	Shows the Username, Email Address, First Name, Last Name, Headline, Profile Details, Gender, Password Edit	Click the profile Icon banner in the Top right corner and Click Profile selection			
US#02		User Profile Edit	Edit your Picture, Username, Email address, First Name, Last Name, Headline, Profile Details,	Click the profile Icon banner in the Top right corner and Click Profile selection			

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			Gender And Password and It will show a "profile successfully updated"	and Click Confirm Changes			
US#03		User Profile inputting a wrong Email Address	It will say "Please enter a part following"" is incomplete.	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Email Address and Click Confirm Changes			
US#04		User Profile inputting a Wrong current Password	It will say "Incorrect Current Password"	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Password and Click Confirm Changes			
US#05		User Profile inputting a not same Password	It will say "Confirmed password does not match new password"	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Password and Click Confirm Changes			
US#06		Bio Overview	Shows the User Bio	Click the profile Icon banner in the Top right corner and Click your profile			

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				Icon in the left side			
US#7		User logout from system	User will be logged out from system	Click the profile Icon banner in the Top right corner and Click logout button			

STUDENT							
ST#01	Student Login	Students enter registered Username and password	Successful login will navigate Student to Student Dashboard	Click Login			
ST#02		Student enters unregistered Username and Password	A message will show that the "Username does not Exist"	Click Login			
STD #03		Student enters only Username	A message will pop out saying "Please fill out this field"	Click Login			
ST#04		Student enters only password	A message will pop out saying "Please fill out this field"	Click Login			
ST#05	Student Overview Preview	Student Dashboard overview	If login was successful, the Student will be redirected to Student Dashboard on Overview Tab	Click Login			
ST#06		Team Completion Status	view the Team Completion Status	Click Overview			

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ST#07		Upcoming Defense Schedule	view the Upcoming Defense	Click Overview			
ST#08	Student Calendar	Student Calendar overview	Shows Month and Date and if there a upcoming Defense	Click Calendar			
ST#09		Date	Shows the week	Click a date			
ST#10		Date with a defense	Shows a Pop up Message saying "Event with Team: Room:"	Click a date with defense			
ST#11		Week	Shows the current date up to 7days and hours	Click week			
ST#12		Day	Show the whole hour day	Click Day			
ST#13		next month or previous month	Shows next month and previous month	Click the next month or previous month			
ST#14	Student Thesis Topic Decision	Thesis Topic Decision Overview	Shows a Choose a field	Click Thesis Topic Decision			
ST#15		Select a Field	Shows the Topic and the Description	Click Thesis Topic Decision and Select a Field			
ST#16	Student Requirement Status	Student Requirements	List Shows the Requirements	Click Calendar tab			
ST#17		Student click a requirement	Goes to the Requirement Checker Overview	Click Calendar tab and click a			

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				requirement			
ST#18	Student Title Acceptance	Search a Title	The AI tool will give a analysis for the title	Click Research Title Acceptance , Fill up the list and then Check Title			
ST#19		Search a Title that have a similar title	A popup saying "The title is similar to an existing title". Do you still want to proceed with the analysis? If you Proceed it give a Analysis but if you don't proceed, It will give you a answer "The title is similar to an existing title. Consider revising it for uniqueness."	Click Research Title Acceptance , Fill up the list and then Check Title			
ST#20		Enter without inputting	It will say "Since both the research title and the problem to solve are empty, I am unable to provide an analysis. Please provide the research title and the problem it aims to solve so that I can give you feedback."	click Research Title Acceptance , Fill up the list and then Check Title			

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				Email Address and Click Confirm Changes			
US#04		User Profile inputting a Wrong current Password	It will say "Incorrect Current Password"	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Password and Click Confirm Changes			
US#05		User Profile inputting a not same Password	It will say "Confirmed password does not match new password"	Click the profile Icon banner in the Top right corner and Click Profile selection and Edit Password and Click Confirm Changes			
US#06		Bio Overview	Shows the User Bio	Click the profile Icon banner in the Top right corner and Click your profile Icon in the left side			
US#7		User logout from system	User will be logged out from system	Click the profile Icon banner in the Top right corner and Click logout button			

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Project Name:	ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED EVALUATION FOR THE COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE					
Written By:	Agustin, Winston L.		Gerona, Jerald Ryan D.			
	Vicedo, Neil S.		Ilano, Ivan Kerwin R.			
Written Date:	Tuesday, April 22, 2025					
Test Case Description:	Compatibility Test					
Testers:	(Test Respondent's Full Name)					
Test Conditions			Date:			
Total Test Case:	5		Rating:			
Pass:						
Fail:						
Test Case Scenario ID	Test Case Scenario	Action	Expected Result	Pass	Fail	Comments/Suggestions
CT# 1	Check System Functionality on Google Chrome	Open browser and open the website	The website should be accessible and well functioned. The website should be accessible and well functioned. The website should be displayed and well functioned. The website should be accessible			
CT# 2	Check System Functionality on Mozilla Firefox					
CT# 3	Check System Functionality on Opera					
CT# 4	Check System Functionality on Microsoft Edge					
CT# 5	Run website on different resolution screens	Resize browser or open on a different device with $\geq 576\text{px}$				

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CT# 6		Resize browser or open on a different device with $\geq 768\text{px}$	and well functioned.			
CT# 7		Resize browser or open on a different device with $\geq 992\text{px}$				
CT# 8		Resize browser or open on a different device with $\geq 1200\text{px}$				
CT# 9	Open the website on Android mobile devices.	Open website on mobile devices and different browsers.				
CT# 10	Open the website on Apple mobile devices.					
CT# 11	Run AI evaluation on Google Chrome	Access AI tools in website and evaluate thesis paper.	The website should display AI results.			
CT# 12	Run AI evaluation on Mozilla Firefox					
CT# 13	Run AI evaluation on Opera					
CT# 14	Run AI evaluation on Microsoft Edge					

APPENDIX G
EVALUATION QUESTIONNAIRE

ATLAS POST-ASSESSMENT SURVEY

This short survey aims to gather your feedback on the performance and quality of the system we developed which is "ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED EVALUATION FOR THE COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE" as part of our thesis, based on the ISO/IEC 25010 software quality model.

Your responses will help us evaluate key quality characteristics such as usability, performance efficiency, reliability, and more.

We greatly appreciate your time and honest insights.

* Indicates required question

RESPONDENT'S PROFILE

Please fill out the following details to help us categorize the feedback accordingly.

1. **Name** (e.g., Dela Cruz, Juan)

2. **Age** *

3. **Gender** *

Mark only one oval.

☐ Male

☐ Female

☐ Other: _____

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4. Participant's Category *

Mark only one oval.

- ☐ IT Expert
☐ Admin
☐ End-User

5. Please specify specific position for IT Expert, Admin, Student, and End-user *

Example:

IT Expert - Software Engineer, Web Developer, etc.

Admin - Department Head

End-user - Adviser, Student

Student - COECSA, CFAD, CITHM, CAMS, CON, CLAE, CBA

SOFTWARE EVALUATION

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

User-Interface and User-Friendliness

6. Navigation *

The user can navigate the system with ease and minimal effort

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

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7. **Visual Design ***

The interface has a simplified and professional look. Which enhances the experience of the user.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

8. **Labeling and Icons ***

Buttons, icons, and labels are easy to understand and clearly represent their function.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

9. **Layout Consistency ***

The design across the system is consistent, which avoids confusion for the user.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

10. **Responsiveness ***

The system quickly responds to actions like clicking, typing, and navigating, providing a smooth experience for users.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

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BUG FREQUENCY AND ERROR TOLERANCE

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

11. **Bug-Free Operation ***

The system runs without frequent glitches, bugs, or crashes.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

12. **Stability ***

During use, no major system crashes or interruptions occurred while using the system.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

FUNCTIONAL SUITABILITY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

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13. Functional Completeness *

The system can perform tasks essential to the day-to-day operations of the Users: Admin, Panelists and students.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

14. Functional Correctness *

The system can keep accurate records of academic materials kept by the Users: Admin, Panelists and students.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

15. Functional Appropriateness *

The system allows the users to efficiently access the materials stored by the Users: Admin, Panelists and students.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

PERFORMANCE EFFICIENCY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

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16. **Time Behavior ***

The system performs its functions within an acceptable time frame.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

17. **Resource Utilization ***

The system uses the computer's hardware and software wisely, making sure everything works well without affecting performance.

Mark only one oval.

1	2	3	4
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18. **Capacity ***

The system can handle large number of users and theses with minimal effects to system functionality.

Mark only one oval.

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COMPATIBILITY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

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19. **Co-existence ***

The system efficiently carries out its functions while smoothly operating alongside other tools and resources used by Admins, Panelists, and Students.

Mark only one oval.

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☐	☐	☐	☐

20. **Interoperability ***

The system can export and import records reliably and within an acceptable time frame.

Mark only one oval.

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USABILITY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

21. **Appropriateness Recognizability ***

Users can quickly understand the system's features and easily see how it fits their needs.

Mark only one oval.

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22. **Learnability ***

The functionalities and features of the system are easy to learn.

Mark only one oval.

1 2 3 4

☐ ☐ ☐ ☐

23. **Operability ***

The functionalities and features of the system are easy to navigate and use.

Mark only one oval.

1 2 3 4

☐ ☐ ☐ ☐

24. **Error Protection ***

The system quickly protects users and their information from any mistakes during use.

Mark only one oval.

1 2 3 4

☐ ☐ ☐ ☐

25. **User-Interface Aesthetics ***

The system's design both pleases the eye and improves the experience of the user.

Mark only one oval.

1 2 3 4

☐ ☐ ☐ ☐

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26. **Accessibility ***

The system can be used by people with different characteristics and capabilities.

Mark only one oval.

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☐	☐	☐	☐

RELIABILITY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

27. **Maturity ***

The system can continue operating normally after repeated failures.

Mark only one oval.

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28. **Availability ***

The system is working and accessible when required to use.

Mark only one oval.

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29. **Fault Tolerance ***

The system can maintain consistent performance in the event of errors in the system.

Mark only one oval.

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☐	☐	☐	☐

30. **Recoverability ***

The system can quickly restore its performance and recover data immediately after a failure occurs.

Mark only one oval.

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SECURITY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

31. **Confidentiality ***

The system ensures that data is only accessible to those who are permitted to use it.

Example: Only approved advisers and panel members can view a student's full thesis file; random students or outsiders cannot open it.

Mark only one oval.

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32. Integrity *

The system prevents unauthorized access to or alteration of the data.

Example: Once a thesis is submitted, students cannot secretly change their papers. The system locks the document to keep its original form safe.

Mark only one oval.

1	2	3	4
☐	☐	☐	☐

33. Non-Repudiation *

The system can confirm that specific events occurred in its history.

Example: When a student submits their thesis or an adviser gives a grade, the system records the exact time and action nobody can later deny it happened.

Mark only one oval.

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☐	☐	☐	☐

34. Accountability *

The system can track the actions of users.

Example: If someone edits a student's thesis status or updates an evaluation, the system logs who did it and when, so actions are always traceable.

Mark only one oval.

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35. **Authenticity ***

The system can verify and confirm that an action was performed by an authorized user.

Example: When a panelist logs in to approve a research paper, the system checks their credentials to make sure it's really them not an impersonator.

Mark only one oval.

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☐	☐	☐	☐

Maintainability

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

36. **Modularity ***

System components are designed to be easily replaced without causing any issues

Mark only one oval.

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37. **Reusability ***

The data and components from the system can be shared and used across different systems.

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38. **Analyzability ***

The system can detect issues with the input data or identify the causes of failures.

Mark only one oval.

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39. **Modifiability ***

Authorized users can make changes to the system without significantly impacting on its overall functionality.

Mark only one oval.

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40. **Testability ***

The system can easily be tested to maintain quality standards.

Mark only one oval.

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PORTABILITY

Legend: 4 - Highly Acceptable 3 - Acceptable 2 - Fairly Acceptable 1 - Unacceptable

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41. **Adaptability ***

The system is capable of adaptation into different specified setups.

Mark only one oval.

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42. **Installability ***

The system can be successfully installed and/or uninstalled in a specified setup.

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☐	☐	☐	☐

43. **Replaceability ***

The data can be replaced by another specified software for the same purpose in the same setup as importing data from a third party application.

Mark only one oval.

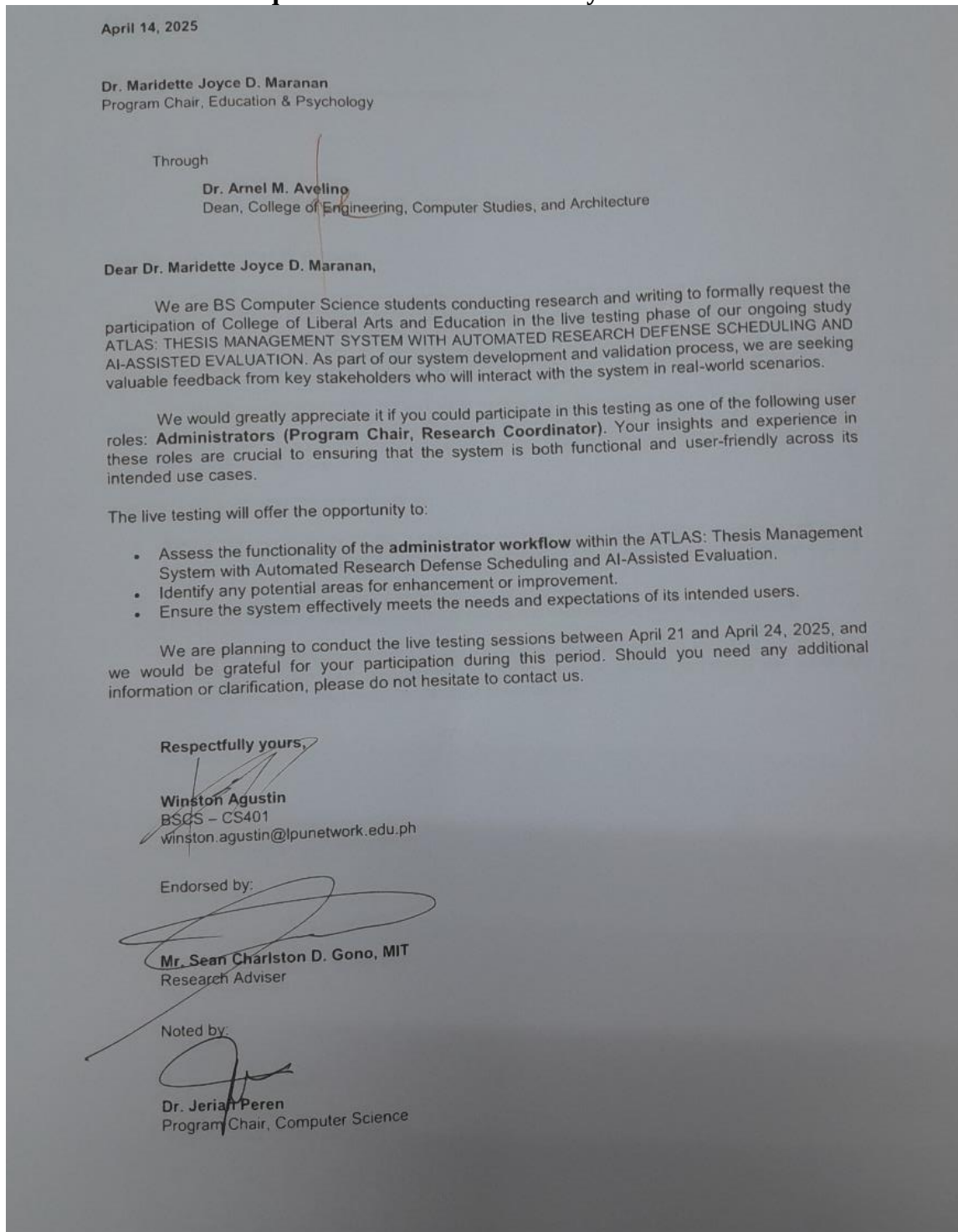
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SUGGESTIONS AND RECCOMENDATIONS

44. Suggestions and Recommendations

APPENDIX H

Request for the feedback of Key stakeholder



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April 14, 2025

Dr. Anjerick J. Topacio
Research Coordinator, College of Engineering, Computer Studies, and Architecture

Through

Dr. Arnel M. Avelino
Dean, College of Engineering, Computer Studies, and Architecture

Dear Dr. Anjerick J. Topacio,

We are BS Computer Science students conducting research and writing to formally request the participation of College of Engineering Computer Studies and Architecture in the live testing phase of our ongoing study ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED EVALUATION. As part of our system development and validation process, we are seeking valuable feedback from key stakeholders who will interact with the system in real-world scenarios.

We would greatly appreciate it if you could participate in this testing as one of the following user roles: **Administrator (Program Chair, Research Coordinator)**. Your insights and experience in these roles are crucial to ensuring that the system is both functional and user-friendly across its intended use cases.

The live testing will offer the opportunity to:

- Assess the functionality of the **administrator workflow** within the ATLAS: Thesis Management System with Automated Research Defense Scheduling and AI-Assisted Evaluation.
- Identify any potential areas for enhancement or improvement.
- Ensure the system effectively meets the needs and expectations of its intended users.

We are planning to conduct the live testing sessions between April 21 and April 24, 2025, and we would be grateful for your participation during this period. Should you need any additional information or clarification, please do not hesitate to contact us.

Respectfully yours,

Winston Agustin
BSQS – CS401
winston.agustin@lpunetwork.edu.ph

Endorsed by:

Mr. Sean Charlston D. Gono, MIT
Research Adviser

Noted by:

Dr. Jerian Peren
Program Chair, Computer Science

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April 14, 2025

Dr. Catherine Cababa
Dean, College of Allied Medical Science

Through

Dr. Arnel M. Avelino
Dean, College of Engineering, Computer Studies, and Architecture

Dear Dr. Catherine Cababa,

We are BS Computer Science students conducting research and writing to formally request the participation of College of Allied Medical Sciences in the live testing phase of our ongoing study ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED EVALUATION. As part of our system development and validation process, we are seeking valuable feedback from key stakeholders who will interact with the system in real-world scenarios.

We would greatly appreciate it if you could participate in this testing as one of the following user roles: **Administrator (Program Chair, Research Coordinator)**. Your insights and experience in these roles are crucial to ensuring that the system is both functional and user-friendly across its intended use cases.

The live testing will offer the opportunity to:

- Assess the functionality of the **administrator workflow** within the ATLAS: Thesis Management System with Automated Research Defense Scheduling and AI-Assisted Evaluation.
- Identify any potential areas for enhancement or improvement.
- Ensure the system effectively meets the needs and expectations of its intended users.

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Respectfully yours,

Winston Agustin
BSCS - CS401
winston.agustin@lpunetwork.edu.ph

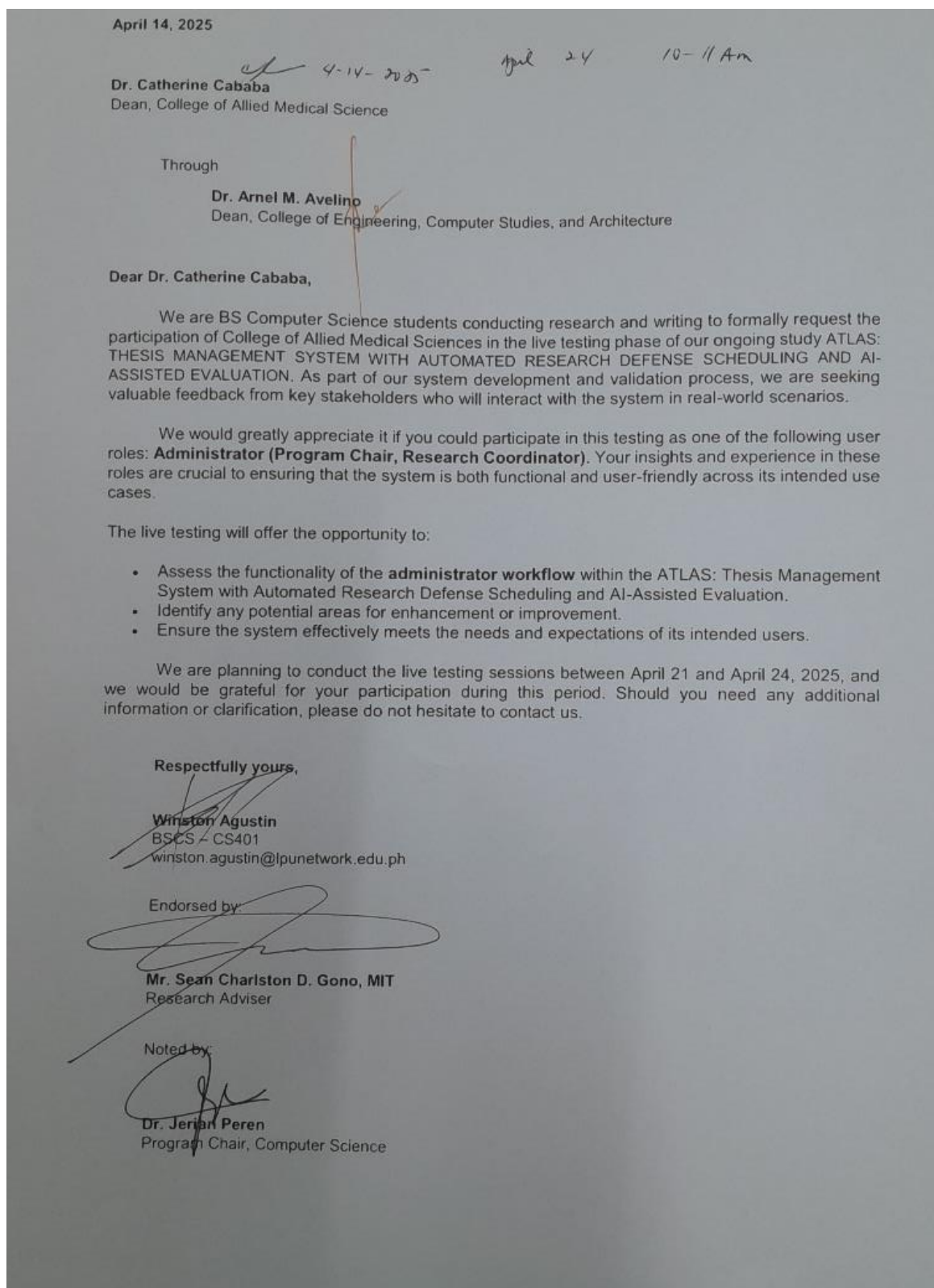
Endorsed by:

Mr. Sean Charlston D. Gono, MIT
Research Adviser

Noted by:

Dr. Jerian Peren
Program Chair, Computer Science

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April 14, 2025

Mr. Lucky Cedrick D. Guyamin
Research Coordinator, College of Business Administration

Through

Dr. Arnel M. Avelino
Dean, College of Engineering, Computer Studies, and Architecture

Dear Mr. Lucky Cedrick D. Guyamin,

We are BS Computer Science students conducting research and writing to formally request the participation of College of Business Administration in the live testing phase of our ongoing study ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED EVALUATION. As part of our system development and validation process, we are seeking valuable feedback from key stakeholders who will interact with the system in real-world scenarios.

We would greatly appreciate it if you could participate in this testing as one of the following user roles: **Administrators (Program Chair, Research Coordinator)** and **Panelists**. Your insights and experience in these roles are crucial to ensuring that the system is both functional and user-friendly across its intended use cases.

The live testing will offer the opportunity to:

- Assess the functionality of the **evaluation feature** and **administrator workflow** within the ATLAS: Thesis Management System with Automated Research Defense Scheduling and AI-Assisted Evaluation.
- Identify any potential areas for enhancement or improvement.
- Ensure the system effectively meets the needs and expectations of its intended users.

We are planning to conduct the live testing sessions between April 21 and April 24, 2025, and we would be grateful for your participation during this period. Should you need any additional information or clarification, please do not hesitate to contact us.

Respectfully yours,

Winston Agustin
BSCS – CS401
winston.agustin@lpunetwork.edu.ph

Endorsed by:

Mr. Sean Charlston D. Gono, MIT
Research Adviser

Noted by:

Dr. Jerian Feren
Program Chair, Computer Science

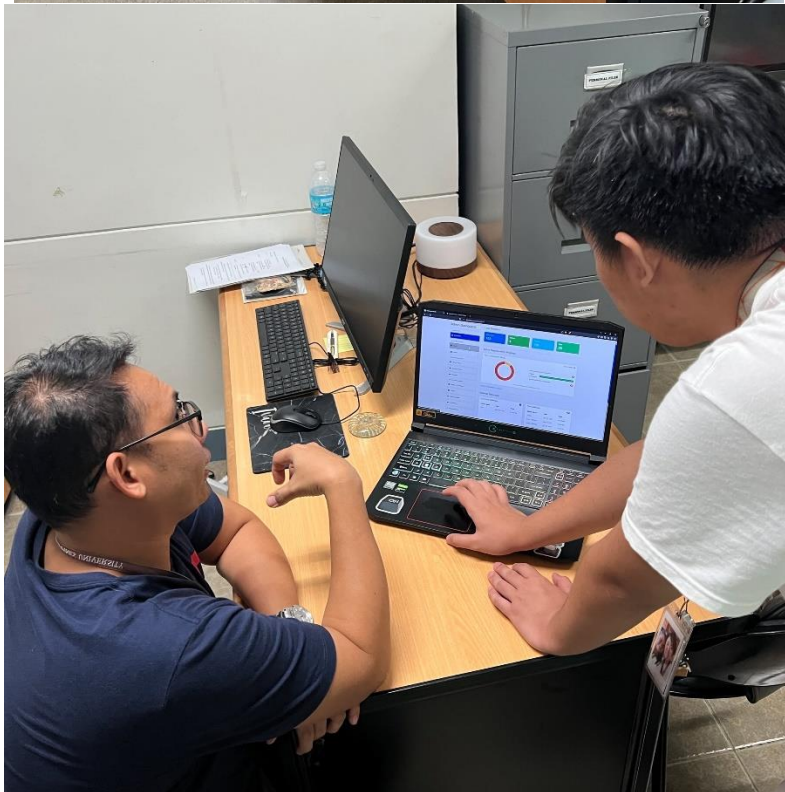
Approved:

Lucky Cedrick D. Guyamin

APPENDIX I
DOCUMENTATION IMAGES



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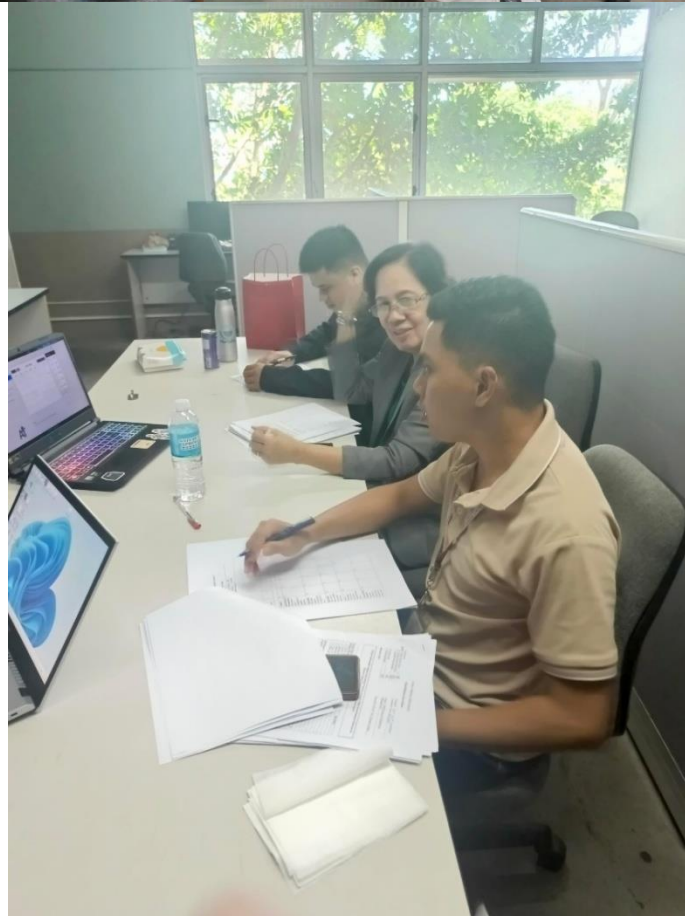
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